OECD ECONOMIC SURVEYS

1999

SPECIAL FEATURES

Labour and product markets

Enhancing environmentally sustainable growth

GERMANY

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1998-1999

GERMANY

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF GERMANY

THE LAND

Area (thousand sq. km)	357.0
Agricultural area (thousand sq. km)	195.4
Forests (thousand sq. km)	104.3

Major cities, June 1995 (thousand inhabitants)	
Berlin	3 470
Hamburg	1 707
Munich	1 241
Cologne	964
Frankfurt	651
Essen	616
Dortmund	600
Stuttgart	587
Düsseldorf	572
Bremen	549

THE PEOPLE

Population (thousands), end 1998	82 037
Number of inhabitants per sq. km	230
Net natural increase in population (thousands), 1998	–67
Net migration (thousands) 1998	47

Labour force (thousands), 1998	39 709
Employment (thousands), 1998	35 999
of which:	
Agriculture, forestry, fishing	972
Industry (including construction)	11 250
Private services	19 236

PRODUCTION

GDP, 1998 (billions of DM)	3 784
GDP per head, 1998 (US$)	26 224
Gross fixed investment 1998:	
Per cent of GDP	21.1
Per head (US$)	5 524

Origin of GDP, 1998 (per cent)	
Agriculture, forestry, fishing	1.2
Industry (including construction)	29.0
Services	69.9

THE GOVERNMENT

Public consumption, 1998, (per cent of GDP)	19.0
General government current revenue, 1998 (per cent of GDP)	46.2
Public debt, end 1998 (ratio to general government current revenue)	146.3

Composition of Parliament:	Seats
Social Democratic Party (SPD)	298
Christian Democratic Party (CDU)	198
Christian Social Union (CSU)	47
Greens	47
Free Democratic Party (FDP)	43
PSD	36

Last general election: September 1998
Next general election: 2002

FOREIGN TRADE

Exports of goods and services, 1998 (per cent of GDP)	28.9
Main exports, 1998 (per cent of total merchandise exports):	
Products of agriculture, forestry and fishing	0.9
Basic materials and semi-finished goods	21.6
Manufactured foods and tobacco	4.0
Other consumer goods	10.5
Investment goods	57.9
End products	5.1
Total	100

Imports of goods and services, 1998 (per cent of GDP)	27.2
Main imports, 1998 (per cent of total merchandise imports):	
Food,	8.9
Raw materials and semi-finished goods	12.1
Finished goods	70.7
of which:	
Primary products	11.4
End products	59.3
Other exports	8.3
Total	100

THE CURRENCY

Monetary unit: Deutschemark	
Currency units per euro, as from 1 January 1999:	1.95583

Currency units per per US$, average of daily figures:	
Year 1998	1.759
October 1999	1.827

Note: An international comparison of certain basic statistics is given in an Annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Germany by the Economic and Development Review Committee on 5 October 1999.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 10 November 1999.

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The previous Survey of Germany was issued in August 1998.

Assessment and recommendations

Overview of current policy issues

When the German economy was last reviewed in June 1998 there were signs that, despite slowing exports, growth would continue to be strong in 1999. In the event, business confidence relapsed and continued to worsen through 1998. Growth projections for 1999 have subsequently been reduced from 2½ per cent to 1½ per cent. This deceleration has occurred within a macroeconomic policy framework where monetary policy is now shaped by the requirements of the Euro area as a whole, and where fiscal policy is constrained by the National Stability Programme, which has the task of achieving a sustainable fiscal position in the long term. Against this background, the new government has set about establishing its own policy priorities with respect to structural reform, its approach to Germany's severe unemployment problem representing, in some respects, a departure from that recommended in the OECD *Jobs Strategy*. This is particularly so in the area of promoting labour market flexibility and improving work incentives in the tax/benefit system. The *Survey* begins with a review of short-term prospects and associated monetary conditions (Chapter I). Chapter II then reviews the fiscal consolidation programme from the perspective of the government's commitment to reduce the tax burden (especially for households). This, with the obligation to restructure family allowances, entails particular challenges for the containment of both health costs and public pensions. Chapter III discusses developments in structural policy, in the light of the government's aims to lower unemployment and modernise the economy. In this context, the commitment to environmentally-sustainable growth, including substantial reductions in greenhouse gas emissions and an eventual phasing-out of nuclear power plants is a major challenge. The issues

that arise in setting policies which are both economically efficient and environmental friendly are discussed in Chapter IV.

A new policy strategy has been evolving

The new policy strategy is still evolving. Four key elements were emphasised at the start. First, with the intention of stimulating employment and growth, demand was strengthened by tax cuts for households. To influence the supply side of the economy, the corporate tax burden was reduced for small and medium-sized firms and shifted to larger companies by a broadening of the corporate tax base. Second, to tackle youth- and long-term unemployment, active labour market measures have been given greater priority. Third, to address the problem of high labour costs, social security contributions are to be gradually lowered from 42.3 per cent of gross wages to under 40 per cent by the end of the legislative period in 2002. This process began in 1999 with a reduction in the pension contribution rate, paid for by a new energy tax. Fourth, talks between the social partners and the government (*Bundnis für Arbeit*) are aimed at developing a joint strategy for employment and growth, with wage developments "conducive to more employment" and increased resort to early retirement being on the agenda. The government moved immediately to implement its strategy in a number of areas, but in the meantime has needed to reassess some elements of its programme to provide for more clarity and ensure that it is consistent with its ultimate objectives. Fiscal strategy shifted to fiscal consolidation in mid-1999. After raising the tax burden on large enterprises, the government is now planning a wide-ranging reform of the business tax system, incorporating a net tax relief for the business sector. Most importantly, the goal of modernising the economy now includes a work-oriented social policy and greater self-reliance as key elements.

After a period of slowing growth and falling business confidence, the economy is now showing signs of recovery

The business climate, which had been slowly worsening during the summer, deteriorated markedly in September 1998 and continued to weaken until the beginning of the second quarter of 1999. The downturn in activity was related to the slowdown in exports and to the Russian crisis, but policy uncertainties related to the tax increases on business contributed to lower business confidence until well into 1999, as did the wage round, which started at the end of 1998 and resulted in settlements above 3 per cent. Consumer confidence remained rather resilient, but the economy slowed rapidly toward the end of 1998. With activity remaining sluggish in the first half of this year, growth for 1999 is likely to amount to some 1¼ per cent (on the new GDP basis). Despite high levels of active labour market programmes the unemployment rate has not continued the improvement seen in 1998. Prospects for an acceleration of activity are based on a projected rise in world demand and a gradual improvement in the investment climate. Domestic and foreign orders have been rising and the indicator of expected market conditions has been steadily improving since March 1999. Both export and domestic demand are projected to continue to strengthen in 2000 so that growth should accelerate to some 2¼ per cent. Changes in economic policy seem also to have contributed to an improved investment climate, even though details of business tax reform remain unsettled. Indeed, it is assumed that uncertainty about the business environment will gradually diminish as details of the business tax reform become clearer, otherwise the upturn in business investment could be slower than projected. Even so, the projected growth for 1999, 2000 and 2001 would not be sufficient for the unemployment rate (national accounts basis) to fall below 8¼ per cent.

Monetary conditions in the euro zone should continue to underpin activity

Monetary conditions were supportive of growth through 1998 and the first half of 1999 and this favourable stance should continue. In the first half of 1999, the euro effective exchange rate weakened, reversing the gains made at the end of 1998, so that international price competitiveness is now at levels seen at the start of 1998. In April the European Central Bank lowered its policy rate by 50 basis points to 2.5 per cent. Although bond rates have recovered from their historical lows at the start of the year they remain relatively

low, while short-term lending rates for companies and for consumers have continued to decline, supporting a rapid rate of credit expansion in the euro zone. Despite the recovery in raw material prices, the effects of declining telecommunications and electricity prices should help maintain virtual aggregate price stability: provided there is no further acceleration in wages, inflation should remain at an underlying rate of a little over 1 per cent. This would imply a relatively low real short-term interest rate of 1½ per cent. OECD projections are that policy rates could rise somewhat as activity picks up in the euro area, but monetary conditions should remain favourable for continued growth into 2000.

Fiscal policy initially slowed the pace of consolidation...

In comparison with the previous government's plans, the initial fiscal consolidation target for 2002 was some ½ percentage point less ambitious. The federal budget deficit for 1999 was planned to be only slightly smaller than the year before, even though net tax revenues were revised upwards. Offsetting increased tax revenues, federal spending was planned to increase by some 6 per cent, due in great measure to transfers to the pension system. The general government deficit was officially projected to stabilise at some 2 per cent of GDP in 1999, with both current receipts and disbursements rising as a ratio of GDP. Even at the time the budget strategy was completed, the risks of substantial slippage in 2000 and beyond were significant, due particularly to a Constitutional Court ruling which raised family allowances and to government promises to reduce the tax burden on industry from 2001 onwards.

... but this policy has been reversed and more ambitious fiscal targets have been set

Faced with these increased commitments, and to restore future room for fiscal manoeuvre, especially given high public debt interest payments, the government introduced an ambitious consolidation package in June, amounting in financial terms to DM 30 billion in 2000 (0.8 per cent of 1998 GDP). A comprehensive fiscal plan was introduced in which the general government deficit target was lowered by ½ per cent of GDP in both 2000 and 2001 to 1 per cent. There has also been a new commitment to further significant consolidation after 2002, the aim being to balance the federal budget in 2006. This programme needs to be implemented as a priority, since, on OECD estimates, a ½ per cent budget

deficit is the minimum needed to prevent breaching the 3 per cent limit in the event of a series of adverse economic developments. Equally important, the Minister of Economics has recently proposed a significant reduction in the public expenditure/GDP ratio to 40 per cent in the longer run, and ways of achieving this overall objective need to be developed. Meanwhile, attaining the existing consolidation plan for expenditures will require that the lower levels of government carefully reassess their social security expenditures and policies. Clear guidelines about what the government intends by way of modernising the social security system are also necessary; the nature and extent of the consolidation will be importantly influenced by decisions with respect to both social security and tax reform.

Personal taxes have been cut, and a comprehensive business tax reform is being planned for 2001...

Personal taxes were cut at the beginning of 1999, the net benefits to households from the *Tax Relief Law* being estimated at more than DM 20 billion once all changes are in force; but for 1999 and 2000 the relief is fully compensated by higher business taxation via a broadening of the tax base. Moreover, while the reform to the personal income tax is an important step towards a more efficient tax structure, the changes to the business tax regime have so far been essentially partial and *ad hoc*. A committee of experts was thus appointed in December 1998 to develop a more comprehensive approach to business tax reform, more consistent with the government's investment and growth objectives. On the basis of the report, presented in April, the government has decided on the main guidelines of reform to enterprise taxation, to be implemented in 2001. Taxes on retained profits will be reduced to 25 per cent for both incorporated and unincorporated businesses. Distributed company income will also be taxed at the 25 per cent. But domestic taxpayers will get only incomplete relief for company tax paid on the dividends they receive, compared with full imputation at present. This partial double taxation will imply a significantly higher combined marginal tax on distributions (43 per cent for those paying income tax at the highest income tax rate) than the 25 per cent on retained earnings. If all elements of the tax reform package are implemented, statutory corporate tax rates will fall to below 40 per cent, including local taxes and the solidarity

surcharge. In conjunction with a systematic broadening of the profits tax base, there will be a tax relief to business of DM 8 billion from this phase of the reform.

... which will make the tax system more neutral but still leave personal tax rates high

The proposed reform represents a long overdue move towards a more neutral tax system, with a more internationally-competitive corporate tax rate. The bias in the tax accounting system which presently favours large capital intensive companies, and those able to form tax-free reserves, should be significantly reduced, so that the system will be much more neutral in that respect. Large enterprises will see their tax liability increase, while for the smaller businesses it will decline. The tax base will be substantially broader and should resemble tax accounting standards in force in many countries. The lower overall corporation tax rate places Germany in a median position among OECD countries, putting it in a better position to attract international capital. However, the reform could still imply distortion in the market for capital. In particular, the preferential treatment of retained earnings, while intended to foster investment and employment, is likely to hamper the reallocation of funds via the domestic capital market. The discrimination could lead shareholders to withdraw their earnings via capital gains, and to prevent this it is proposed to reduce the share-ownership level above which such capital gains are fully taxed from 10 per cent to 1 per cent. While this increase in capital gains tax would have the advantage of placing incorporated and non-incorporated enterprises on a more equal footing, the incentive effects would appear to be at variance with the government's goal of encouraging market-based equity investment and attracting more venture capital. Moreover, the current distortion, whereby entrepreneurial income is taxed at lower rates than labour income, dividends, and certain types of self-employment income, will become more pronounced. What is needed is a tax system which, taking account of the need to maintain a rate of profits tax which does not deter international investors, is more even-handed with respect to types of domestic activity. For this to be achieved, the higher rates of personal income tax need to be unified and reduced. This reinforces the need for tightly controlling budget outlays.

Short-term measures to control pension outlays do not obviate the need for fundamental pension reform

Germany is facing both short and long-run budget challenges with respect to the pension system, which demand urgent action. The most immediate difficulty has arisen from the fact that pensions are indexed to net wages. Technical changes to the way child allowances are paid and tax cuts for households have had the effect of driving up net wages which would have given rise to a substantial but anomalous rise in pensions in 2000 and 2001 with a substantial budgetary impact (unless offset by a rise in contribution rates). The government now proposes to index pensions to inflation for these two years before returning to a system based on net wages. This would appear justified as an expedient, but at the same time the government has cancelled the introduction of a demographic life-expectancy adjustment to the annual pension-adjustment formula. It has also reversed reforms to the invalidity pension schemes. The contribution base has been widened to casual jobs and some of the self-employed, which will raise claims in the future. From a longer term perspective the government's aim is to reduce and stabilise the pension contribution rate at around 19 per cent up to 2010. The receipts from the new energy tax are being used to this end, but for the government's objective to be achieved a broader approach to pension reform is needed. A fundamental pension reform has been promised for 2001, and a number of reform ideas have been put forward by members of the government, including a compulsory supplemental funded component to the pay-as-you-go system. This idea needs to be developed. In addition, as recommended in previous *Surveys*, invalidity pensions need to be decoupled from the labour market situation and all forms of early retirement should be related to an appropriate actuarial discount.

Recent measures to control health costs do not constitute a viable structural reform for the longer term

The objective of stabilising the rate of social security contributions is also closely related to decisions on reform of the health system, the latest stage of which has been formulated subject to the overriding constraint of not increasing contribution rates. There is a global budget limit, and expenditures by the health funds, both individually and collectively, are not permitted to grow faster than the wage base used for calculating contributions. Since rationing of health care services is seen as unacceptable, any expansion

in medical services due to increased entitlements, technological development or an ageing population is meant to be financed by higher efficiency on the part of health suppliers. A positive list of authorised medicines will be introduced to hold pharmaceutical costs in check, co-payments having been reduced. However, at this stage it appears that a sustainable medium-term growth path for health spending might not yet have been achieved:

- It is doubtful whether a fixed contribution rate (and hence a budget ceiling) represents an appropriate target given the well established tendency for health expenditures to rise faster than incomes. In the absence of price incentives such as co-payments, achieving the budget objective might well ultimately require some form of queuing, which would amount to a rationing of health services. The 1999 pharmaceutical budget is already in danger of overrunning in some regions.
- Sanctions do not appear to be strong. Scrutiny of aberrant funds by other health funds may be weak where all funds are subject to the same pressure; and the fact that all health funds must deal collectively with health suppliers means that there is little room for manoeuvre for individual health funds.
- Experience elsewhere with a positive list for medicines shows that they do not produce the hoped-for savings. A negative list would be more appropriate. And there is a more general problem with binding pharmaceutical budgets in that reducing other types of medical costs might require increased use of medicines.
- The powers of the health funds to reduce excess capacity among hospitals is limited because the Länder maintain final responsibility for any restructuring. Moves to more closely integrate the ambulatory and stationary sectors appear to be formal rather than substantial.
- The ability of individual health funds to compete is highly restricted. The funds have greater scope to negotiate with groups of health suppliers without the need for approval by the doctors' association; but the fact that all health funds need to negotiate collectively reduces the flexibility of this mechanism.

Although the system might temporarily deliver stable contribution rates, it lacks a medium term perspective. The special study of the German health system included in the 1997 OECD *Economic Survey of Germany* concluded that there are arguments for both a universal health fund and competing health funds. But the present system is neither one nor the other. Funds compete for members but cannot compete on the basis of services or costs. They should be given powers to deal directly with suppliers and the possibility to vary contribution rates. In sum, the new reform has failed to build on the consensus achieved during the last ten years and could prove to have only a short run effect in terms of expenditure control. The aim should be to build a system which is consistent and sustainable over the medium term.

The strategy to reduce unemployment is a departure from previous policies and a new consensus has yet to emerge

The projected GDP growth for 1999 and 2000 would still leave a relatively large output gap and would do little to raise employment or to lower the unemployment rate, which is currently over 8 per cent on a national accounts basis. Those involved in Federal Labour Office programmes, including early retirement on account of unemployment, amount to over 4 per cent of the labour force. To correct labour-market imbalances, the OECD *Jobs Strategy* has advocated the adoption of a broad range of structural policies, with respect to both product and labour markets. Recognising that each country should pursue the path which was most appropriate to its situation, specific recommendations have been formulated for Germany, based particularly on the need for greater wage flexibility, reductions in non-wage labour costs and less intrusive labour-market regulation. Reforming the tax and transfer system to reduce labour costs remains a clear objective, even if there are differences as to the means of achieving it with the emphasis being on redistributing the overall tax and social security burden. But in some respects the current policies diverge from the OECD approach, particularly in the area of promoting labour market flexibility, and reducing the work disincentives built into the tax and benefit systems. Indeed, labour market flexibility has been reduced by tightening regulations governing the definition of self employment for social security purposes, and by more restrictive job protection provisions

for small firms. The *Bündnis für Arbeit* is meant to develop a new consensus over a wide range of topics bearing on the labour market, including fostering part-time employment by older workers and reducing overtime, more flexible opportunities for early retirement, modernising the vocational training system and creating more employment opportunities for low qualified persons. However, apart from a declaration by the social partners that productivity gains should be used, in part, for employment creation, there has been little in the way of agreement so far.

The wage bargaining system has been moving in the direction of more flexibility and this now needs to be supported by policy measures

The German wage bargaining system has been evolving rapidly, *de facto* if not *de jure*, in the direction of greater flexibility. Although the branch-wide wage agreement remains dominant, the tendency toward more decentralised wage bargaining has continued, especially in the new Länder, where an increasing number of enterprises have quit the employers' association and so are not bound by their agreements. Non-observance of branch wage agreements is also growing, with companies making agreements with their workforces to secure employment and cut costs. Many of these "agreements" made with the works councils might not be in line with legal provisions. As a result the government is considering increasing the powers of the social partners to challenge agreements in the courts. In addition, it has compulsorily extended wage agreements to non-contracting parties in the construction sector. Proposals to strengthen the powers of the social partners to challenge local wage agreements need to be resisted, although there is a need for some legal clarification. Indeed, as argued in the 1998 *Survey*, the authorities could support plant-level agreements by strengthening the role of works councils. Even with greater wage flexibility, the low-wage sector would be a problem, and the operation of the tax and benefit systems in this segment of the labour market needs to be reviewed. Attention has focused on across-the-board wage subsidies. But policy here needs to be seen as a supplement, rather than as a substitute, to a greater spread of the wage distribution, which would create employment opportunities for the less qualified.

Active labour market measures have to be more carefully defined to help solve the unemployment problem

An important element of the government's labour-market strategy is a greater emphasis on active labour market measures (ALMP). Resources devoted to ALMP had been significantly increased by the previous government and this policy has been maintained, while eligibility criteria have been significantly widened. Persons who have been unemployed for six months are now eligible and those older than 56 years can be supported, in effect, until retirement. Greater emphasis has been given to job-creation measures (*Arbeitsbeschaffungsmassnahmen*) in the public sector. At the same time, 100 000 short-term training places are being made available for youths who have not been successful in entering normal training programmes. The original idea was to give greater emphasis to ALMP in comparison with passive benefits, but unemployment benefits have in fact been enhanced, in part by reversing earlier reforms. There is, however, little evidence from the experience in the new Länder, that large-scale job-creation measures can be effective in solving the employment problem. They should be confined to high-risk groups and should not be used as a social policy instrument to reduce the labour supply, particularly in the new Länder. Carefully-chosen training is to be preferred to job creation projects, since this appears to be more effective in supporting re-entry to normal employment. For all programmes, monitoring of their effectiveness is crucial, as is a clear sense of policy direction.

Product markets have become more competitive but pressures to protect vested interests need to be resisted and subsidies reduced

Product markets have generally become much more competitive and efficient, as privatisation, deregulation and pressure from the competition authorities has introduced competition into the utilities sector. The benefits of liberalisation and competition have been evident in falling prices in both electricity and in telecommunications, showing that the welfare gains from further efforts to encourage competition may be large. There is a need to move forward in a number of areas including regional support and subsidies:

– Despite welcome proposals for a new agreement between industry and the electricity sector which would lower significantly the charge for transmitting electricity over long distances, it will remain necessary for the authorities to ensure non-discriminatory access by all potential competitors to the transmission network. Calls

to protect small community generators need to be resisted, the more so as the already-dominant generators are often shareholders in community generation plants. The local generators do need to be given room to restructure and amalgamate, which they are often prohibited from doing by local and Länder laws.

- In the telecommunications area, the impressive results from opening the market for long-distance calls using the fixed network point to the potential benefits from encouraging competition in the local net. In opening the local network to competitors, there should be no need to protect the interests of the former monopolist. Rather, the regulator, in setting access prices, should encourage competition by setting them at levels which could be achieved by efficient and competitive operation. To avoid a potential conflict of interest, the government should sell its remaining shares in Telekom.
- Subsidies need to be reduced and eventually eliminated in a number of sectors, particularly banking and mining. Regional aid needs to be restricted and, with respect to the new Länder, a start needs to be made to developing a more appropriate framework to replace the current system, which expires in 2004.
- Shop opening hours need to be liberalised as the distortions and contradictions in the present system are becoming increasingly apparent.

The efficiency gains from further market liberalisation in the above areas could make an important contribution both to consumer welfare and international competitiveness.

The regulatory approach to better environmental performance does not seem to have been associated with high abatement costs so far...

The German public has been highly sensitive to environmental concerns, leading to many policy initiatives in this area since the early 1980s, based primarily on a regulatory approach. Regulations have been developed for quite specific sources of pollution, specifying emissions and technologies to be used. Water quality has improved dramatically and emissions of major pollutants have been sharply reduced. The improvement in environmental performance has been particularly marked in the new Länder during the 1990s with the modernisation of the capital stock, although most of this improvement has been due to the initial col-

lapse of industrial activity. Cost minimisation has been an important criterion for policy but until the recent introduction of an energy tax, economic instruments had scarcely been utilised. Implementation has been through detailed regulation and, more recently, an increasing resort to voluntary agreements with industry, which have been justified by the principle of co-operation. A cost-effective policy would aim to prevent resource misallocation by equalising marginal abatement costs across emission sources. In theory, a fully-informed regulatory agency could design measures so as to achieve such an efficient outcome, but the necessary information is rarely available, so that the cost differentials between polluters are difficult to take sufficiently into account. In the German case, however, regulations have been implemented in close consultation with polluters and have often been so differentiated that marked variations in marginal abatement costs across sectors and sources might have been avoided. Such an outcome would have been facilitated by the fact that, on the available evidence, marginal abatement costs have apparently been quite low.

... but making further progress may be more costly, calling for increased use of economic instruments

The situation may now have changed significantly and the period of low abatement costs would seem to be over for emissions such as greenhouse gases. A continued reduction in emissions is still observable for the country as a whole, but progress is slower than in the 1980s and is often due to developments in the new Länder. Western German emission levels have either been constant or increasing. The voluntary agreement with industry to reduce specific carbon dioxide emissions by 20 per cent resulted in rapid reductions at first, but progress has since slowed markedly. With Germany having a domestic commitment to reducing carbon dioxide emissions by 25 per cent by 2005, and an international commitment to cutting greenhouse gas emissions by 21 per cent by 2008-12, both in comparison with 1990, more attention will have to be devoted to minimising the costs of achieving environmental objectives. Under such circumstances, regulations would have to be adapted to many different sources and co-ordinated to ensure abatement costs did not differ substantially, while voluntary agreements would be increasingly difficult to negotiate without the authorities giving harmful financial incentives.

Greater emphasis will therefore need to be given to policies affecting broad sources of emissions, notably household energy consumption and transport, with greater use of economic instruments to achieve an equalisation of abatement costs. Economic instruments are not a complete substitute for regulation; indeed, there is often a strong complementarity. However, attention will also have to be given to ensuring that the regulatory system is compatible with a greater use of economic instruments where these are appropriate. As a first step, this will require a codification of environmental legislation. A start has been made on this project (creating an U*mweltgesetzbuch*), but the process needs to be accelerated.

Clear and consistent objectives and efficient markets would facilitate the attainment of environmental objectives...

In moving ahead towards a more cost-efficient environmental policy, clear and consistent objectives are required across all public programmes, since economic instruments can only be effective in markets where there are proper economic incentives. At present, individual markets are often distorted by regulation, government intervention and conflicting policy aims. In spite of recent progress in reducing output-related support, agriculture is still an example of a distorted market leading to negative environmental affects, as are subsidies for coal production. Urban sprawl which, *inter alia*, impacts on land use, biodiversity and transport-associated pollution is in part encouraged by the pattern of public support for homebuilding. Rent laws fail to give sufficient incentive to tenants and proprietors to save energy, because they prevent cost recoupment. At a minimum, if the housing market is not liberalised and changes to the existing legal framework minimised, normal rent comparisons should be defined to include heating costs. The electricity market has been highly regulated up to the present and although it has held energy prices high, this has scarcely been for the right reasons and innovation and energy conservation have been discouraged. Increasing competition places generating companies under pressure to select low-cost energy sources and to make sure that the conversion of fuel to electricity is energy-efficient. In this respect, and more generally, improved efficiency resulting from competition should, be welcomed from an environmental perspective.

… which requires a less discriminatory approach to ecological taxation

A prime example of the problems involved in grafting environmental incentives onto existing regulatory structures and conflicting economic objectives is the energy tax which came into force in April 1999. The aim of the tax reform is to combine ecological and employment objectives; it will be followed by further tax increases. The energy tax is to be welcomed as a move towards the more systematic use of economic instruments. However, major factors conditioning the shape of the energy taxes have been the desire of the authorities to maintain support for the domestic coal industry, to avoid giving nuclear power a competitive advantage (an objective is to phase out nuclear power which presently accounts for one third of electricity generation capacity) and maintain industrial competitiveness, in the absence of an EU-wide energy tax. It must also conform to the EU regulations covering the liberalised energy market, which rules out taxes on imported electricity. The result has been that the energy tax is levied on energy output rather than input of primary energy sources, which are more related to environmental outcomes. Industry is taxed at a lower rate than households. If the goal of meeting the Kyoto Protocol is to be met in an efficient way, the best solution would be to impose a tax or tradeable permit on carbon dioxide emissions rather than to tax electricity. Coal needs to be treated equally with other fuels if environmental objectives are to be achieved efficiently and electricity generators must be given incentives to use an efficient combination of fuels, which the current tax does not provide. *Ceteris paribus*, this development would reduce the attractiveness of coal, including imported coal, to the advantage of gas. This would go some way to compensating for the induced rise in CO_2 and in other emissions arising from a phase-out of nuclear power, which should not be under-estimated. Most fundamentally, attempts to meet environment aims at the same time as sectoral employment and competitiveness objectives through the same instrument risks falling short on all counts. In this context, the selective nature of the current ecological tax will limit the environmental dividend arising from a shift in the incidence of taxation from labour to the "environment". Achieving the full "double dividend", in terms of employment and environmental performance,

depends on a more broadly-based future use of economic instruments.

Summary

In sum, the new government has set itself the ambitious tasks of lowering unemployment, modernising the economy and the social system, and securing the long term viability of the budget and the health and pension systems. Ecological goals have been given equal prominence in order to ensure the environmental sustainability of economic development. In some fields there has been progress. However, the fiscal package needs to be fully implemented to put public finances on a sustainable path and to create a tax regime that is more business friendly. These policies should be underpinned by structural reforms that strengthen future growth prospects. Such policies can benefit both macroeconomic performance and future fiscal outcomes. While short-term growth prospects are already improving, unemployment remains a major problem. With respect to its strategy for reducing unemployment, the government is seeking to obtain consensus, *inter alia* on an employment-friendly wage policy, via round-table talks with the social partners. It is important that a consistent set of policy instruments emerge that establish clear links between policies and ultimate policy goals. For Germany to attain the employment, growth and environmental aims commensurate with its key position in the European economy, requires not only favourable macroeconomic conditions, including aggregate wage developments, but a policy emphasis which more effectively enhances labour-market flexibility, as well as structural reforms that strengthen individual initiative, economic choice and the role of competition. Since structural and macroeconomic policies tend to have synergies which make them mutually reinforcing, achieving a more flexible and dynamic use of resources will help to assure progress towards the country's social, budgetary, environmental and economic goals.

I. Slowdown and recovery: trends, prospects and monetary conditions

Overview

The current expansion, which began as long ago as 1994, has been characterised by intermittent spells of growth and slowdown. At the time of the last review, in mid 1998, the economy appeared to be gaining strength after the 1995-96 slowdown, supported by a broadening upswing in domestic demand and underpinned by rising employment and buoyant consumer and business sentiment. However, exports slowed more quickly than expected during the summer of 1998 and business confidence deteriorated markedly in August and September. The business climate continued to worsen in the following months, due in part to uncertainties surrounding policy initiatives of the new government, including an initial tax reform which shifted the tax burden from households to businesses, and a reversal of some labour-market policies. Consumer demand continued to recover throughout the year and appears to have remained robust, underpinned by increases in real disposable income. Nevertheless, growth slowed markedly in the second half of 1998 and early 1999, leading to a downward revision in official projections for 1999 to around 1½ per cent (on an old national accounts basis: see Box 1). A pick-up was projected for the second half and forward-looking indicators now suggest more favourable prospects for foreign demand. Despite a much stronger second half, growth is likely to amount only to some 1¼ per cent in 1999 on a new national accounts basis. There should be a gradual recovery in investment in machinery and equipment and a stabilisation of construction activity after a protracted decline extending over the last four years. Monetary policy in the euro area is supportive and the real exchange rate favourable. Growth is projected to pick up to around 2¼ per cent in 2000 and to 2½ per cent in 2001. But despite large increases in the number of people on active labour market programmes, there is likely to be only modest progress in reducing unemployment rates this year and in 2000/2001.

Box 1. Revised national accounts

Germany released a first set of revised national accounts figures in April 1999 based on the new 1995 European System of Accounts (ESA 95). The revised accounts are available from 1991 onwards, with earlier revisions planned only for 2005. Moreover, the Federal Statistical Office will only publish data for Germany overall. The switch to the new system entails major revisions to the level of national income, which originate both in revised accounting concepts and more up-to-date and broader statistical data. Prices have also been rebased from 1991 to 1995.

The most important conceptual change affects investment, which has been broadened to encompass non-material items such as expenditures on computer software and copyrights. Further conceptual changes comprise, *inter alia*, an extension of depreciation to include public infrastructure and a widening of the basis for estimating imputed rents in household consumption. The statistical base has been revised mainly by taking into consideration new census information. The revision has increased the level of real GDP, but growth rates since 1995 have been reduced by ½ percentage point a year on average. The level effect mainly results from the broadening of the investment and consumption concepts. A major factor underlying the reduction of GDP growth rates is a downward revision of imputed rents. Table shows major demand aggregates before and after the revision.

Demand and output in 1998 before and after the adoption of ESA 95

	GDP		Private consumption	Government consumption	Gross fixed investment[1]	Exports	Imports	Change in stock-building[2]
	Level, DM billion at current prices	Growth rate (real)	Growth rate (real)					
Prior to revision, at 1991 prices	3 758.1	2.8	1.9	0.6	1.6	5.4	6.6	0.3
After revision, at 1995 prices	3 784.2	2.2	2.3	0.5	1.4	7.0	8.5	0.7

1. After revision now including intangible investment.
2. Contribution to change in real GDP (percentage of real GDP in previous year).
Source: OECD.

The level of employment has also been revised for 1991 to 1998 to follow the International Labour Office definition; no statistics are currently available for 1999. Jobs involving less than 15 hours per week have not been covered in the past, but they are now included. Revised employment in 1998 totals 36 million persons, 2 million more than previously estimated. Employees account for 32 million (an increase of 1.7 million), and 4 million persons are self-employed or supporting family members (an increase of 0.3 million). Estimates of the unemployed are also derived from the labour force survey. In 1998, the unemployment rate on a national accounts or labour force basis was 1.8 percentage points lower than the monthly unemployment rate, which is based on registered unemployed.

Faltering growth with renewed signs of recovery

Following the traditional pattern, since the 1993 recession growth has been driven to a great extent by net exports. Domestic demand has actually been slower than usual to develop in this cycle. The pattern changed in 1998, with the growth contribution from net exports turning negative while domestic sources – notably investment in machinery and equipment, private consumption and stockbuilding – became the main engine of demand expansion (Table 1 and Figure 1). Real GDP grew by 2.2 per cent in 1998, 0.7 percentage points more than in 1997 (Box 1). However, the annual average conceals a rapid slowing of growth in the course of the year, which carried over into 1999.

Table 1. **Demand and output**[1]

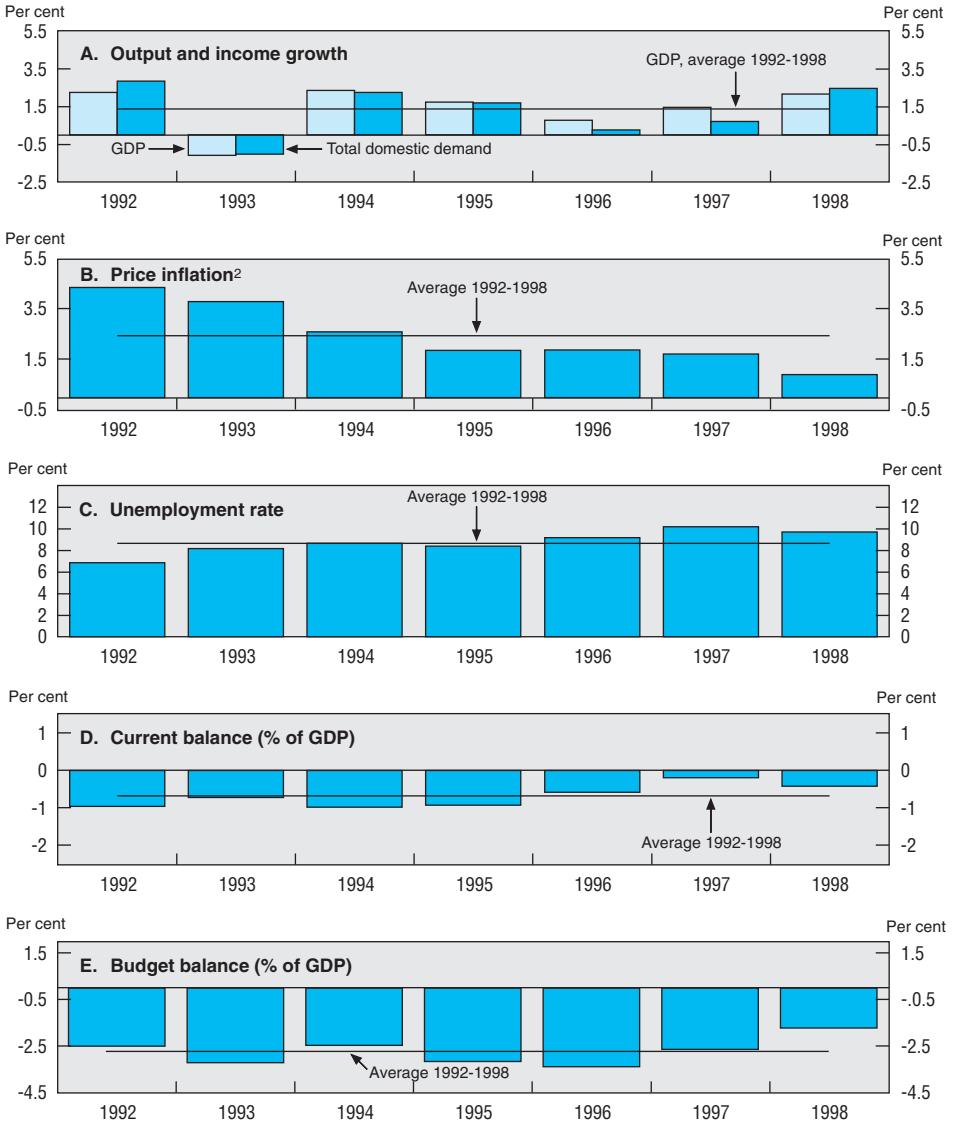
Percentage change from previous period, 1995 prices

	1994	1995	1996	1997	1998	1999	2000	2001
Private consumption	1.0	2.1	0.8	0.7	2.3	1.7	2.2	2.4
Government consumption	2.4	1.5	2.1	–1.1	0.5	0.7	0.5	0.5
Total fixed investment	4.0	–0.7	–1.1	0.5	1.4	3.2	2.8	3.6
Construction	6.9	–1.8	–2.9	–1.4	–3.9	–0.1	0.7	1.0
Machinery and equipment	–1.0	1.5	2.1	3.7	9.9	7.8	5.5	6.8
Stockbuilding[2]	0.3	0.4	–0.4	0.4	0.7	–0.1	–0.0	–0.0
Total domestic demand	**2.2**	**1.7**	**0.3**	**0.7**	**2.5**	**1.7**	**2.0**	**2.3**
Net exports[2]	0.1	0.1	0.5	0.8	–0.3	–0.4	0.4	0.3
Exports of goods and services	7.6	5.7	5.1	10.9	7.0	1.7	6.1	6.3
Imports of goods and services	7.3	5.6	3.2	8.3	8.5	3.1	4.9	5.7
GDP at market prices	**2.3**	**1.7**	**0.8**	**1.5**	**2.2**	**1.3**	**2.3**	**2.5**
Memorandum items:								
Total employment[3]	–0.3	–0.1	–0.7	–0.8	0.4	0.2	0.3	0.5
Household savings rate[4]	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Relative ULC in manufacturing	1.3	6.8	–3.7	–7.3	–0.6	n.a.	n.a.	n.a.
Export market growth in manufacturing	11.4	10.1	6.7	10.5	8.0	4.7	6.9	6.5
GDP deflator	2.5	2.0	1.0	0.8	1.0	1.2	1.3	1.3
Private consumption deflator	2.6	1.9	1.9	1.7	0.9	0.7	1.4	1.4
Unemployment (in millions)[5]	3.3	3.2	3.5	3.9	3.7	3.6	3.5	3.3
Unemployment (per cent of labour force)	8.3	8.1	8.8	9.8	9.3	9.0	8.7	8.3
Current account (per cent of GDP)	–1.1	–0.8	–0.2	–0.1	–0.2	–0.0	0.1	0.5

1. Based on the European System of Accounts in 1995 prices.
2. Contributions to changes in GDP (as a percentage of real GDP in the previous period).
3. According to the labour force survey. Now includes jobs with less than fifteen hours per week.
4. Household accounts according to the new system of national accounts are currently not available.
5. Unemployed according to the labour force survey; ILO definition.

Source: OECD.

Figure 1. **Macroeconomic performance**[1]



1. Based on the new European System of Accounts, at 1995 prices.
2. Growth in private consumption deflator.

Source: OECD.

Exports slowed rapidly during 1998...

Export growth declined dramatically in 1998 and even became negative in seasonally-adjusted terms in the second semester. This development mainly reflected the substantial decline of foreign demand originating in the crisis areas of Asia, Latin America and Russia (Table 2). The decline was sharper in Germany than in most other European countries, due not only to the area composition of

Table 2. **Trade by region**

Percentage change from previous year

	1992	1993	1994	1995	1996	1997	1998	Share 1998	Share 1998 II	Share 1999 I
Exports to[1]										
EU	0.7	–13.2	9.3	8.2	3.1	8.7	9.5	56.4	55.8	57.9
EFTA	–4.7	–6.1	10.6	6.8	–3.1	6.3	8.8	5.4	5.6	5.4
Central and Eastern Europe	–5.7	9.9	12.8	14.4	17.1	28.8	11.1	10.8	10.7	9.4
of which: Russia	..	..	–4.5	–4.0	11.0	43.4	–13.5	1.6	1.3	0.9
North America[2]	0.1	7.6	16.0	0.3	9.7	30.3	15.5	10.0	10.1	10.5
Japan	–11.3	4.9	14.0	6.3	12.5	–2.7	–11.1	1.9	1.9	2.0
Asia[3]	4.3	15.4	23.2	13.0	5.3	8.4	–24.9	3.8	3.6	3.5
Central and South America	5.8	6.7	10.6	8.3	4.7	28.5	19.1	3.0	3.3	2.7
Other	8.7	10.5	0.7	3.1	4.2	12.4	2.3	8.8	9.1	8.4
Total	0.5	–5.8	10.1	7.7	4.9	12.6	7.5	100.0	100.0	100.0
Imports from[1]										
EU	–1.0	–16.3	8.2	9.2	3.5	9.3	6.6	55.3	54.5	54.7
EFTA	1.9	–4.8	11.1	5.6	1.8	10.1	1.4	5.2	5.3	5.2
Central and Eastern Europe	1.7	–1.3	21.8	16.8	6.7	20.4	13.3	10.0	10.5	10.5
of which: Russia	..	..	20.1	2.3	13.4	11.9	–14.9	1.7	1.7	1.6
North America[2]	0.0	–5.3	9.3	4.4	7.4	18.1	13.0	8.7	8.7	9.1
Japan	–5.7	–9.1	–0.5	3.7	–5.1	9.8	6.4	4.9	4.8	4.8
Asia[3]	–5.9	4.3	8.5	8.0	–4.9	13.0	–0.2	4.8	4.9	4.6
Central and South America	–5.6	–15.3	14.7	1.3	0.8	15.5	4.7	2.0	2.0	1.9
Other	–2.3	13.2	5.1	–7.7	8.4	17.1	3.8	9.0	9.2	9.1
Total	–1.4	–10.1	8.7	7.0	3.4	12.0	6.8	100.0	100.0	100.0

1. F.o.b., including adjustment.
2. United States and Canada.
3. Dynamic Asian economies.

Source: Deutsche Bundesbank, Zahlungsbilanzstatistik.

Table 3. **The current account of the balance of payments**

DM billion

	1991	1992	1993	1994	1995	1996	1997	1998
Goods	31.6	43.6	68.1	82.7	93.2	107.4	125.1	139.3
Exports (f.o.b.) [1]	668.1	671.4	632.7	696.4	749.9	786.5	885.9	952.0
Imports (f.o.b.) [1]	636.6	627.9	564.5	613.7	656.7	679.1	760.8	812.7
Services	–35.3	–47.4	–54.7	–64.1	–65.0	–66.2	–71.7	–77.3
of which: Tourism	–34.4	–39.5	–43.0	–49.3	–49.0	–50.5	–51.7	–53.7
Factor income	33.1	34.0	27.4	4.9	0.3	1.7	–3.0	–16.1
of which: Investment income	29.8	32.1	26.6	4.5	1.7	3.5	–1.3	–14.3
Transfers	–57.8	–51.2	–55.7	–59.9	–55.7	–51.3	–52.8	–53.2
of which:								
Net contribution to the EU	–21.9	–25.3	–27.3	–31.7	–30.0	–27.6	–28.5	–30.4
Other public transfers	–24.1	–12.1	–14.1	–12.9	–10.3	–7.7	–8.3	–7.0
Current account	–28.4	–21.1	–14.9	–36.5	–27.2	–8.4	–2.4	–7.4
(As a per cent of GDP)	–1.0	–0.7	–0.5	–1.1	–0.8	–0.2	–0.1	–0.2

1. Special trade, according to the foreign trade statistics, including supplementary trade items; imports exclude freight and maritime transport insurance costs, which are included under Services.

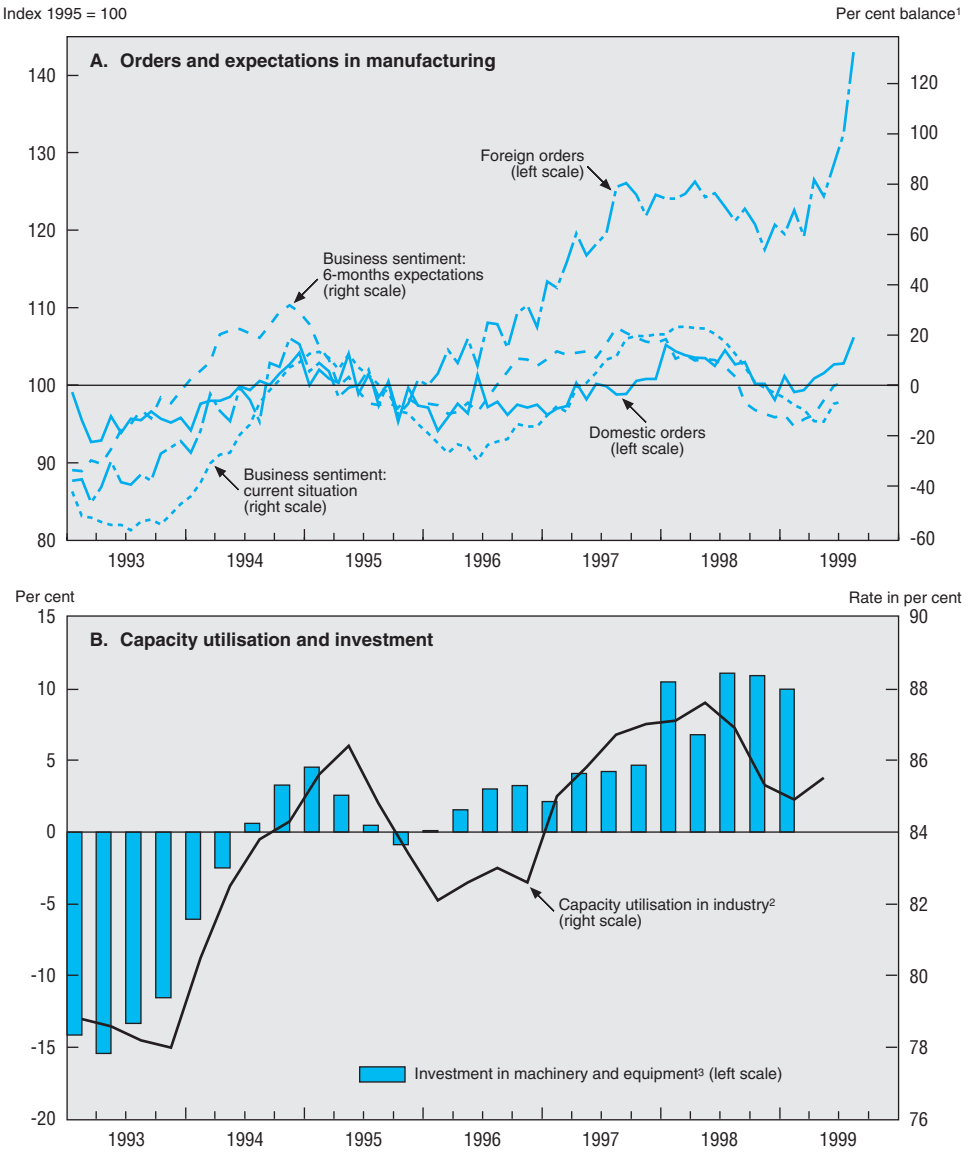
Source: Deutsche Bundesbank, Zahlungsbilanzstatistik.

German trade but also to the specialisation in machinery and intermediate industrial goods, demand for which was cut quite strongly in the crisis countries. In addition, the direct impact of the slowdown in exports on total economic growth was more pronounced in Germany, given that exports of goods and services account for almost 30 per cent of nominal GDP, which is well above the OECD average. The trade surplus still increased in 1998, because of the improvement in the terms of trade. However, following a continuous improvement since 1995, the current account deficit (–0.2 per cent of GDP) widened somewhat because of larger deficits on services and factor income from abroad (Table 3). Exports and imports started to grow again in the first quarter of 1999, as did incoming foreign orders in manufacturing (Figure 2). While exports to South America and to Russia continued to decline, those to the dynamic Asian countries and Japan began to improve. Exports to North America remained particularly buoyant.

... and were only partially offset by stronger domestic demand...

Declining exports contributed to a cooling of the business climate in manufacturing in the second half of 1998 and, as in the past, investment plans were revised downward (Figure 2, panel B). Having accelerated steadily from 1995, the growth of investment in machinery and equipment (including intangibles) reached an unusually high rate of 13 per cent in the first semester 1998 (seasonally-adjusted annual rate), but thereafter it decelerated. High

Figure 2. **The business climate**



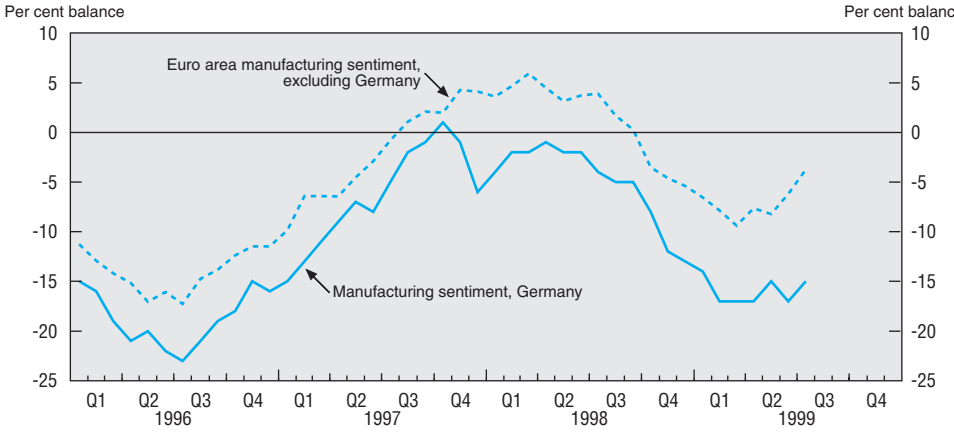
1. Balance of positive-negative responses. A level of 0 corresponds to normality.
2. Refers to western Germany.
3. At 1995 prices. Percentage change over previous year.

Source: Ifo Institut; Deutsche Bundesbank; OECD.

stockbuilding contributed to the robustness of total domestic demand in 1998, but toward the end of the year the Ifo index of stock adequacy pointed to an overhang developing, which contributed to the slowing in inventory accumulation in the first half of 1999.

The deterioration in the business climate in Germany was more pronounced after the Russian crisis in August 1998 than in other EU countries, and was partly related to domestic policy shifts which followed the election of the new government in September (see chapter III below) (Figure 3). Uncertainty about the policy framework increased further during the discussions about tax reform, which commenced almost immediately.[1] The tax burden on business activity was increased during the first phase of reform to compensate for net tax reductions for households. Household confidence, by contrast, remained quite buoyant in 1998, helping to underpin activity. Personal consumption was supported by real disposable income growth of 1.9 per cent following a decline of 0.1 per cent in 1997,[2] attributable to somewhat higher wage growth, rising employment and falling inflation. While consumption slowed in the first half of 1999, household income was further supported by the tax reductions that came into effect at the start of 1999 and which were targeted at low-income groups. Cuts in pension contribution rates financed by increases in energy taxes also had a positive short-term impact, although the tax falls to a great extent on households.

Figure 3. **Business sentiment in the euro area**[1]



1. European industrial confidence indicator in manufacturing. Euro area aggregate is weighted by GDP share in the euro area, excluding Germany. Scale refers to balance of positive and negative responses both expressed as a percentage of total replies.

Source: European Commission; OECD.

... but activity has begun to pick up

In addition to the recovery in export activity, incoming domestic orders in manufacturing began to increase from the spring, their volume in the second quarter exceeding that in the first quarter by 7½ per cent at an annual rate. The Ifo index of business expectations in manufacturing also started to improve in March with relatively large gains from May onwards. From mid year, assessments of current business conditions also became more positive. Manufacturing production rose continually between May, and August, as production plans began to be revised upwards. Rising orders and improving expectations also suggest that the recession in construction has begun to level off.

Employment recovered in 1998 but unemployment remains high

After declining consistently from 1993 to 1997, employment increased in 1998 by an average of 135 000 on a national accounts basis (Table 4 and Box 1). In an accounting sense, some 60 per cent of the additional working places were due to subsidised jobs: job creation schemes along with training and retraining programmes were expanded rapidly from the first quarter of 1998, reversing the preceding policy of a phased reduction in active labour market programmes (Figure 4, panel C). At a sectoral level, the increase in employment was entirely attributable to greater employment in services. Employment in the general service sector grew by 0.8 per cent, approaching its secular growth path after a weak expansion in 1997. Financing, leasing and enterprise-related services contributed most to employment growth (3.6 per cent). By contrast, employment in construction declined again, reflecting the further deterioration in construction activity in both east and west. Employment in industry continued its long-term decline, although at a reduced rate.

Monthly unemployment rates declined from the spring to the end of 1998 by ½ percentage point in the west and 1 percentage point in the east (seasonally-adjusted), aided by the increase in active labour market measures. But they remained high at 9 per cent in the west and 18 per cent in the east (national measure using registered unemployed). The decline in seasonally adjusted unemployment rates ceased in the spring of 1999, a result both of an end to the expansion of subsidised employment and training schemes and of stagnation in normal job creation (Figure 4). Nevertheless from a longer-term, developments in 1998 were rather encouraging. While the participation rate continued to fall, the overall employment rate (employment divided by the workforce) rose only for the second time since 1990 (Figure 5, panel A). At the same time, the configuration of private and public sector employment creation was such that, for the first time for two decades, private sector employment rose while government employment fell (panel B).

Table 4. **The German labour market**

Thousands of persons

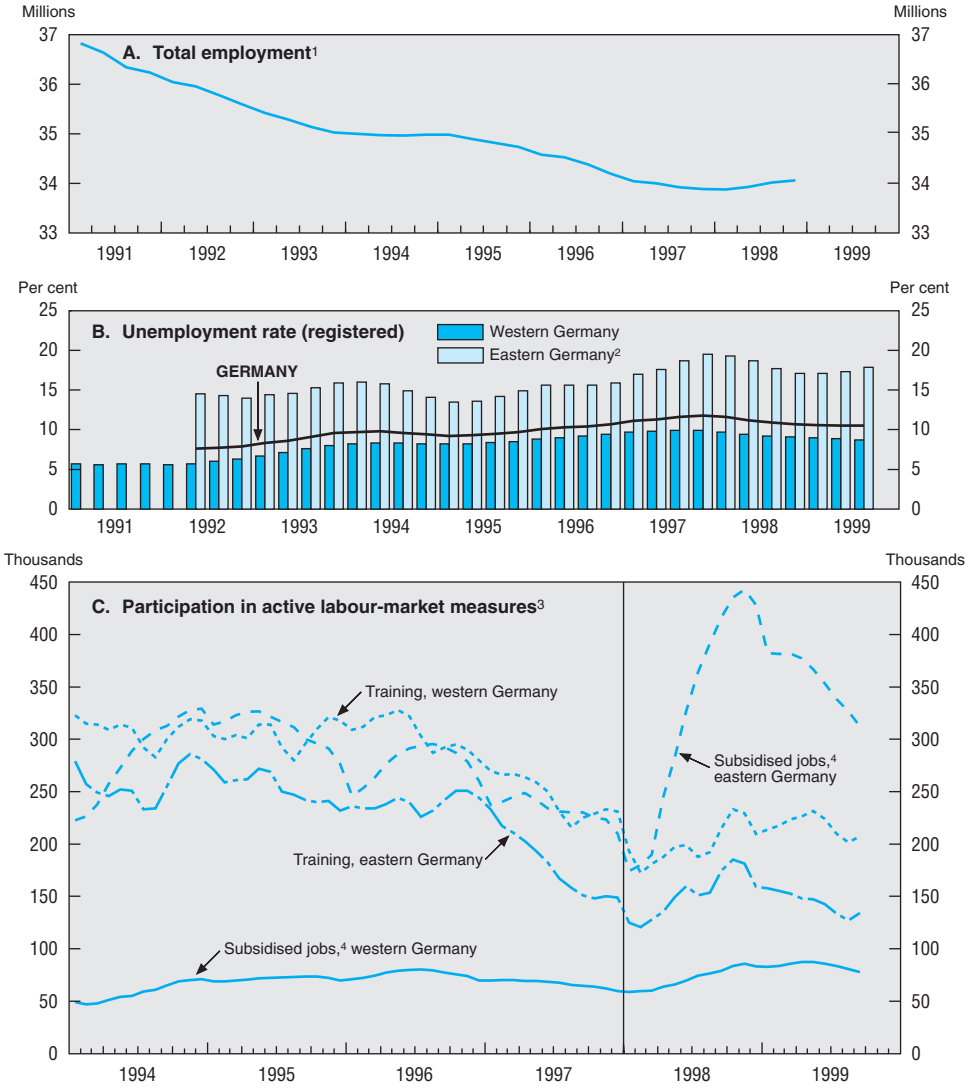
	Germany		Western Germany		New Länder (eastern Germany)	
	1997	1998	1997	1998	1997	1998
	Change from previous year[1]					
Labour force	195	–70	..	..	..	..
of which:						
Unemployed[2]	339	–382	..	..	..	..
Commuters[3]	7	3	..	..	..	..
Employed (domestic)	–137	315	..	..	..	..
Memorandum item:						
Registered unemployed[4]	406	–369	138	–202	269	–167
Under-utilised labour	–228	215	–34	27	–194	188
of which:						
Short-time workers[5]	–30	–7	–23	–3	–7	–5
Subsidised jobs (ABM)[6]	–69	230	–12	21	–58	209
Retraining courses[7]	–94	–36	–57	–46	–37	10
Early retirement programmes[8]	–33	27	58	54	–92	–26
	Annual average					
Labour force	39 704	39 645	..	..	..	..
of which:						
Unemployed[2]	3 907	3 710	..	..	..	..
Commuters[3]	62	64	..	..	..	..
Employed (domestic)	35 859	35 999	..	..	..	..
Memorandum item:						
Registered unemployed[4]	4 384	4 279	3 021	2 904	1 364	1 375
Under-utilised labour	1 956	1 901	1 028	982	928	919
of which:						
Short-time workers[5]	80	53	55	36	26	17
Subsidised jobs (ABM)[6]	302	385	68	71	235	314
Retraining courses[7]	761	650	453	367	308	282
Early retirement programmes[8]	812	814	452	508	359	306
	Unemployment rate (per cent)					
Registered unemployment[4]	11.4	11.1	9.8	9.4	18.1	18.2
Registered unemployment and under-utilised labour[9]	15.4	15.0	..	..	..	..

Note: Employment on a national accounts basis has been revised upwards (see Box 1) but is no longer available for east and west Germany separately.

1. Fourth quarter to fourth quarter; according to national account definitions.
2. Source: national accounts, ILO definitions.
3. Balance of commuters: persons commuting between east Germany and west Germany are included.
4. Source: Federal Labour Office (*Bundesanstalt für Arbeit*).
5. Programmes subsidised by the Federal Labour Office; number of persons calculated as equivalent to unemployment.
6. Public employment programmes subsidised by the Federal Labour Office.
7. Full-time participants in further education programmes, retraining courses, and German language courses subsidised by the Federal Labour Office.
8. Programmes subsidised by the Federal Government, by the Federal Labour Office and by the Public Pension Systems. Excludes invalidity pensions.
9. Registered unemployment and estimates of under-utilised labour as a percentage of the labour force.

Source: Sachverständigenrat zur Begutachtung der gesamtwirtschaftlichen Entwicklung.

Figure 4. **Employment, unemployment and active labour-market measures**

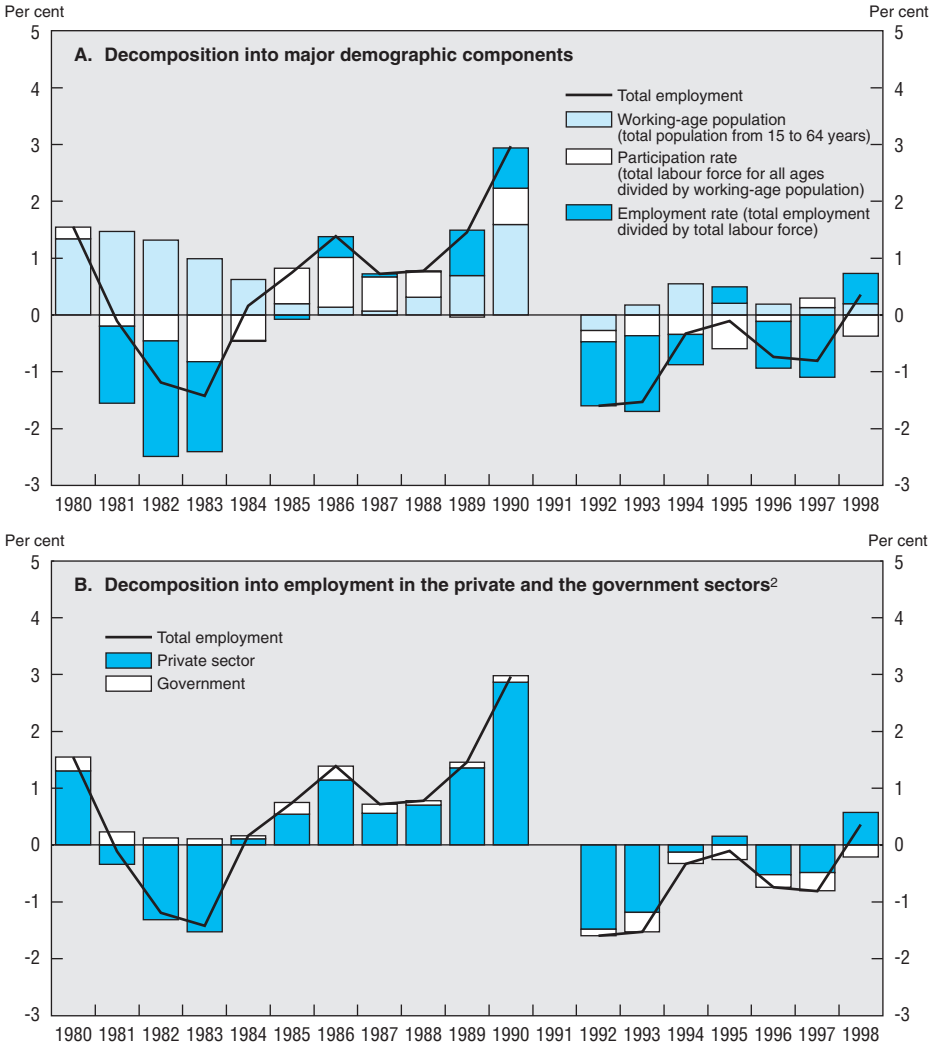


1. Revised series following the adoption of the European System of Accounts. Data are not yet available for 1999.
2. Data not available before Q2 1992.
3. Not seasonally adjusted. The vertical line indicates a change in the definition of training support programmes in 1998.
4. Employment Creation Measures (ABM) and Structural Adjustment Measures (SAM).

Source: Deutsche Bundesbank; Federal Labour Office; OECD.

Figure 5. **Decomposition of change in total employment**[1]

Percentage change



1. Prior to 1991 figures relate to western Germany and are on the old national accounts basis. After 1991 figures refer to total Germany and are based on the new system of national accounts.
2. The decomposition of employment growth has been approximated by a weighted sum of employment growth in the private and the government sectors, with the weights being the share of each type of employment in total employment, recalculated each year. Employment in Treuhand companies in eastern Germany is classified as the private sector.

Source: OECD.

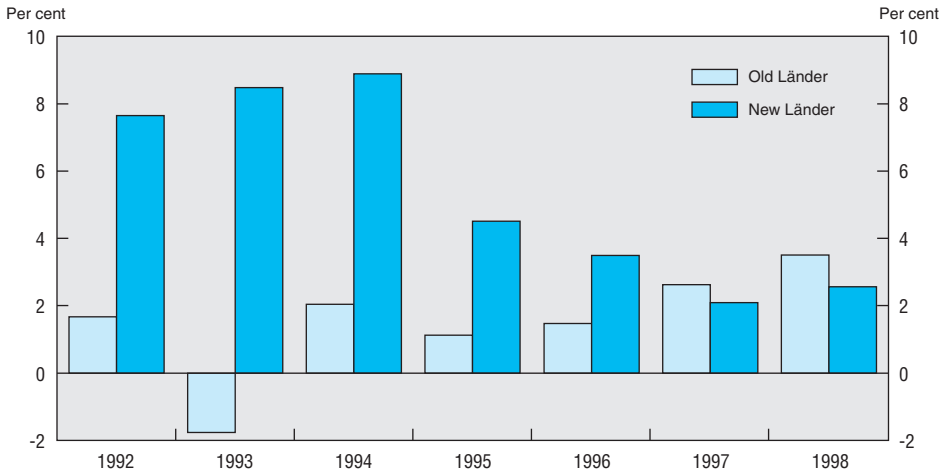
There were also significant developments in the composition of unemployment, partly due to policy measures. The share of the long-term unemployed (with duration of one year or more) declined until August 1999, year-on-year, by 1 percentage point to 36.7 per cent in the west and by 2½ percentage points to 31.4 per cent in the east . The above-average decline in long-term unemployment largely reflects the expansion of work provision schemes, which target the long-term unemployed as a problem group. Similarly, special training programmes designed to reduce youth unemployment were implemented in January 1999 and probably contributed to the decline in unemployment among younger people (aged between 15 and 25 years) by 1.4 percentage points, year-on-year, to 8.9 per cent by September 1999 in the west and by 0.7 percentage points to 17.6 per cent in the east. A more comprehensive measure of labour under-utilisation would take into account persons participating in active labour market schemes or being in early retirement and related schemes designed to reduce labour supply. These would add nearly 4 per cent to the usual definition of unemployment (Table 4).

Convergence has stalled, but restructuring is continuing in the new Länder

Although there has not been further progress in convergence over the last two-and-a-half years, the restructuring of the new Länder has been proceeding quite rapidly. Strong growth in the new Länder during the years following reunification has been followed, since 1997, by slower growth rates than in the west. In 1998, GDP – as measured by the old system of national accounts (Box 1) – expanded by 2 per cent, which was 0.8 percentage points less than in the west (Figure 6). To a great extent, this weakening is attributable to the recession in construction in response to the over-capacity built up in the first half of the 1990s, and economic activity in the new Länder has been characterised by a marked dichotomy in the development of manufacturing on the one hand and by construction on the other. Between 1995 and 1998 production in manufacturing increased by 28½ per cent, while construction output fell by 15 per cent. Growth in services, which had fallen back to around 3 per cent in 1997, recovered to almost 5 per cent in 1998. Business-oriented services have increasingly gained in importance in recent years, employment in this branch increasing by 10 per cent between 1995 and 1997.

Following its previous rapid growth, manufacturing production in the new Länder declined in the final quarter of 1998, mainly due to weaker domestic demand. The drop in world trade is likely to have had an indirect effect via orders from western firms. But the direct impact of the crisis in Asia and elsewhere appears to have materialised later than in the west and, given the lower share of exports in eastern German GDP, adverse developments in international trade have had a smaller impact on overall economic activity. In 1998, less than one fifth

Figure 6. **Real GDP growth in the old and new Länder**[1]



1. Based on national accounts at 1991 prices, prior to the adoption of the European System of Accounts (ESA).
Source: Statistisches Bundesamt.

of the revenues of eastern German companies originated from exports as opposed to more than one third in western German enterprises. However, there is also a secular trend for eastern German enterprises to become more export-oriented, and this development continued despite the contraction in world trade. All of the five industrial branches which grew at double-digit rates between 1995 and 1998 can be classified as export-oriented.[3] One estimate suggests that manufacturing export growth in the new Länder totalled some 15 per cent in 1998, which would be about twice as high as the percentage increase in goods exports of Germany overall.[4] The export expansion has been associated with a redirection of trade towards western Europe and markets overseas. Nevertheless, companies in the new Länder still need to overcome domestic structural weaknesses to expand exports. Thus there is some evidence that success in exports has been associated with significant price cuts in order to establish market position.

The recession in construction became more pronounced in 1998. Construction accounts for 13 per cent of value added in the new Länder, which is three times the level in the west. The creation of over-capacity in both industrial and residential construction during the first half of the 1990s was fostered by a high rate of subsidisation, and the decline in construction activity over the past year and a half has been associated with the tightening of financial aid for the

construction of rental accommodation. Despite a marked reduction of over-capacity, the recession continued into 1999 although prospects have improved: incoming orders declined at a slower rate in the first quarter and increased markedly in the second quarter of 1999. Building permits appear to have stabilised, and business surveys indicate that the optimism in the sector is recovering.

Tariff wages in the east continued to increase more rapidly than in the west in 1998 (2.6 per cent in the east and 2.0 per cent in the west, at the time of contract implementation). This has brought tariff wages to 90.5 per cent of the level in the west, with the spread ranging from 66 per cent in the clothing industry to 100 per cent in construction. However, with wage agreements often undercutting negotiated tariff rates (see Chapter III), the average ratio of wages actually paid in the east relative to those paid in the west is almost 15 percentage points lower than the tariff-wage ratio.

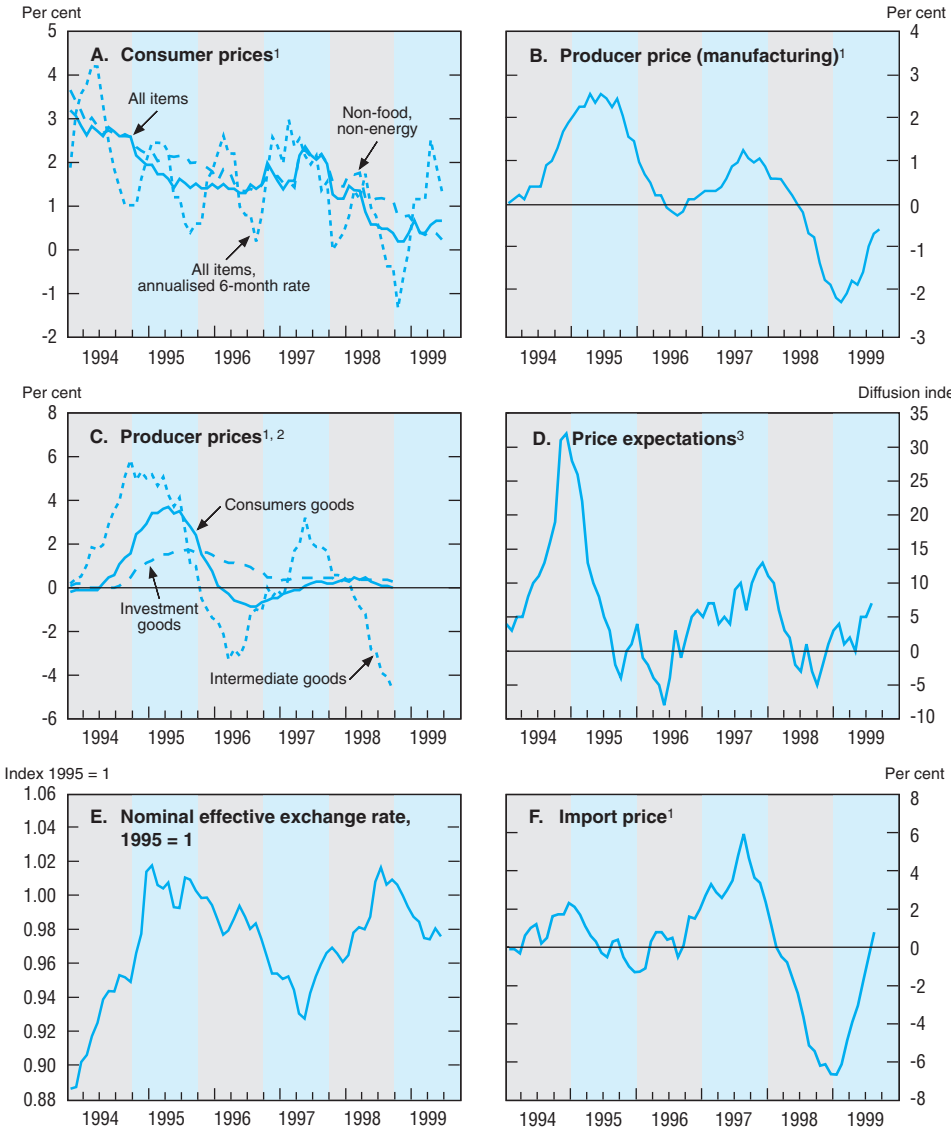
Monetary conditions, inflation and competitiveness

Inflation is subdued and real wages have accelerated...

The inflation environment remains favourable. The consumer price index rose by 1.0 per cent in 1998 (1.9 per cent in 1997), despite tax increases in April 1998 and again in April 1999, and in the middle of 1999 consumer prices were rising at a rate of about ½ per cent (Figure 7). One-off factors have been important in driving down the headline rate of inflation but the underlying rate of consumer price inflation is probably only around 1 per cent. The collapse in raw material prices, which was partly reversed in spring 1999, has led to weak or declining producer prices and has also helped to hold down consumer prices. Substantial falls in electricity and telecommunications prices have also had a marked impact on price developments. While future declines in telecommunications charges should be more moderate, electricity prices for households are expected to fall appreciably.

Wages set by tariff agreements accelerated from 1.5 per cent in 1997 to 1.9 per cent in 1998 and in 1999 the increase will be closer to 3 per cent, led by the metal industry in the west. However, once account is taken of reductions in other wage costs (in some cases the thirteenth month's pay has declined), actual compensation has been growing more slowly than negotiated wages. Pension contribution rates by employers were reduced in April 1999 by some 4 percentage points so that actual gross wage costs are increasing by less than the rise in negotiated wages.

Figure 7. **The inflation environment**



1. Percentage change over 12 months.
2. Series in the course of revision.
3. Diffusion index. Balance of higher and lower expectations. Seasonally adjusted

Source: Deutsche Bundesbank, *Monatsberichte;* OECD, *Main Economic Indicators.*

... but sustained productivity growth means competitiveness remains sound

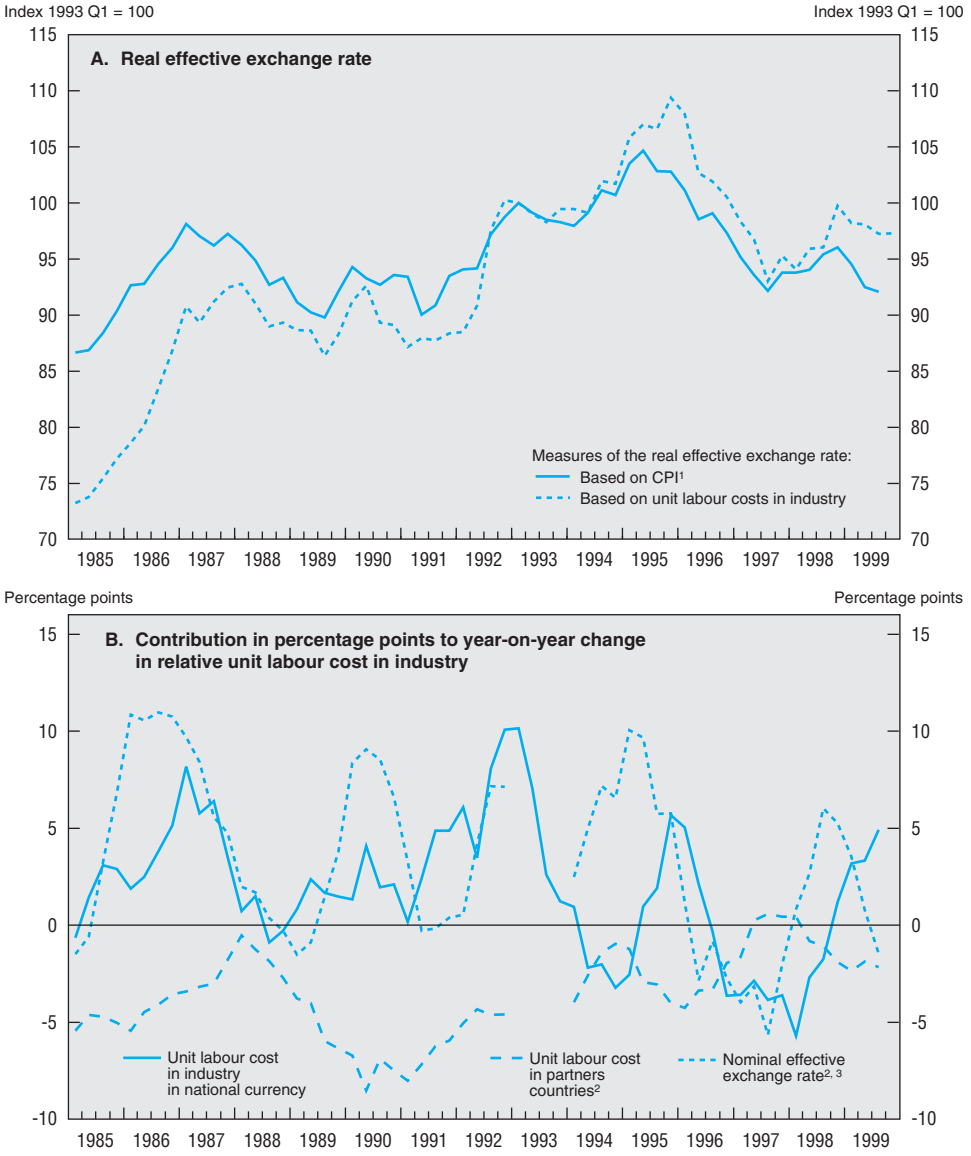
A notable feature of developments in Germany is the continuing decline in relative unit labour costs in industry, due to sustained productivity growth (Figure 8). With output growth slowing rapidly at the end of 1998, labour productivity stagnated and unit labour costs rose. However, this slowdown is likely to have been only temporary so that despite the high wage round for 1999, international competitiveness should remain sound. In the run up to the introduction of the euro, the nominal effective exchange rate of the Deutschemark appreciated and German exports lost competitiveness. However, during the first half of 1999, the euro depreciated in effective terms and the real effective exchange rate returned to the level which characterised the economy during the export boom of 1997 (Figure 8, panel A).

Monetary conditions are favourable for growth

Monetary policy has provided a supportive framework for growth in Germany over the review period. During 1998, policy-controlled interest rates in Germany were kept low and unchanged until December 1998 when, in a co-ordinated action with other euro zone banks and in agreement with the ECB, they were lowered by 30 basis points to 3 per cent. On 1 January 1999 responsibility for setting monetary policy shifted to the European Central Bank (ECB), the Deutsche Bundesbank participating in the formulation of monetary policy for the euro zone as a member of the European System of Central Banks, while also continuing to implement policy in Germany.[5] German money market rates had remained roughly stable throughout 1998, but between December 1998 and May 1999 short-term euro rates declined by some 80 basis points (Figure 9). In April, the ECB lowered the interest rate on two-week repurchase operations, its main refinancing instrument, by 50 basis points to 2.5 per cent. The reduction in policy rates was seen to be consistent with both monetary stability and an appropriate response to conjunctural risks.

Capital market rates (as measured by 9-10 year Bunds) continued their downward trend in 1998 but this was partly reversed during the first half of 1999. The yield on 10-year government bonds fell to a low of 3.7 per cent in December 1998, having declined by 160 basis points during the year. A major factor behind this movement was the general "flight to quality" in the wake of turbulence in Asia and elsewhere; but with prices nearly stable or even declining, inflation expectations were probably being revised down as well. The decline in long term rates came to a halt in the first months of 1999, and by mid-October they were around 5.3 per cent, exceeding the January level by nearly a full 1½ percentage points (Figure 9). The stabilisation of international financial markets, gradually improving prospects for faster growth in Germany and the euro area, and a widening interest rate differential to the US may all have played a role in the increase. Real interest rates, as measured by subtracting current inflation rates, have also risen though they remain very low

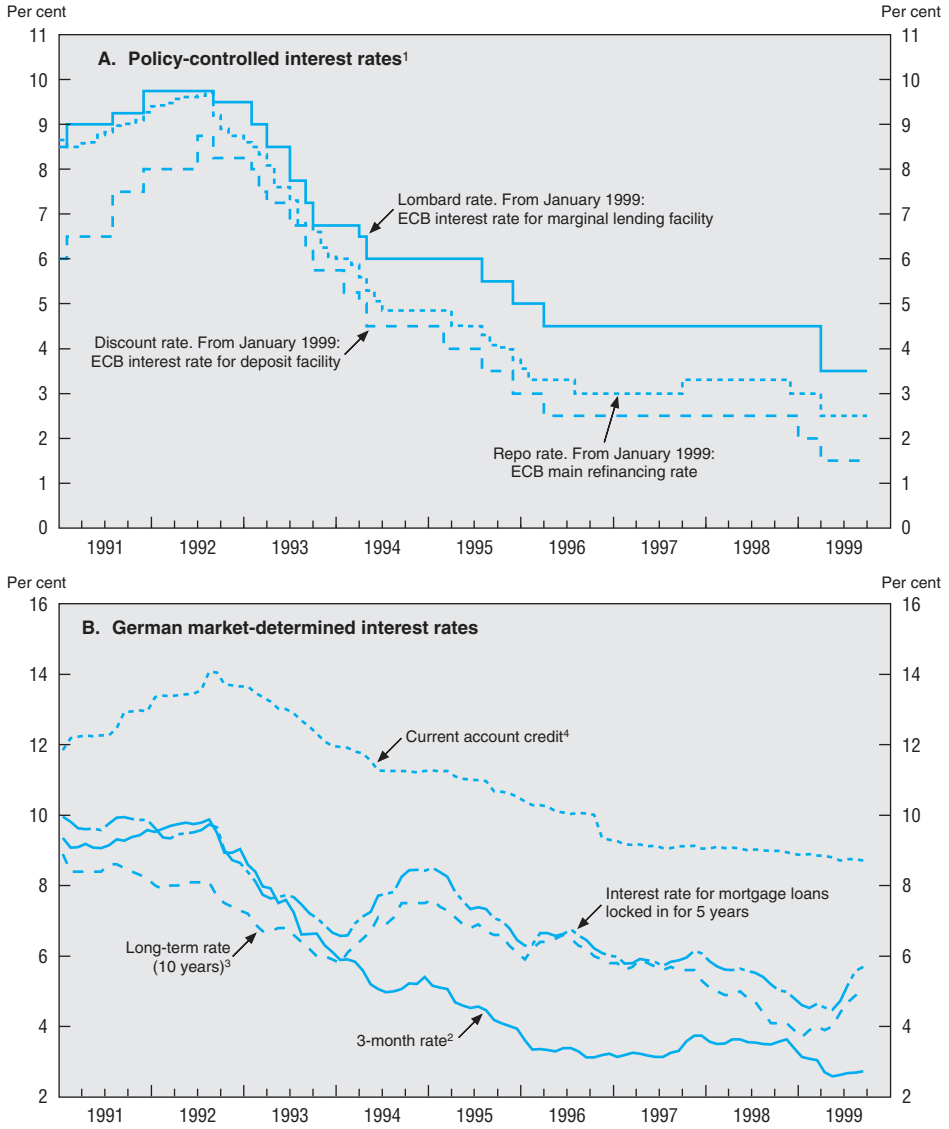
Figure 8. **International competitiveness**



1. Weighted real external value of the DM against currencies of 40 industrial countries.
2. Due to the introduction of 4 new countries in 1993 Q1 the decomposition is misleading in 1993 and has not been included.
3. Weighted external value of the DM against currencies of 40 industrial countries.

Source: OECD.

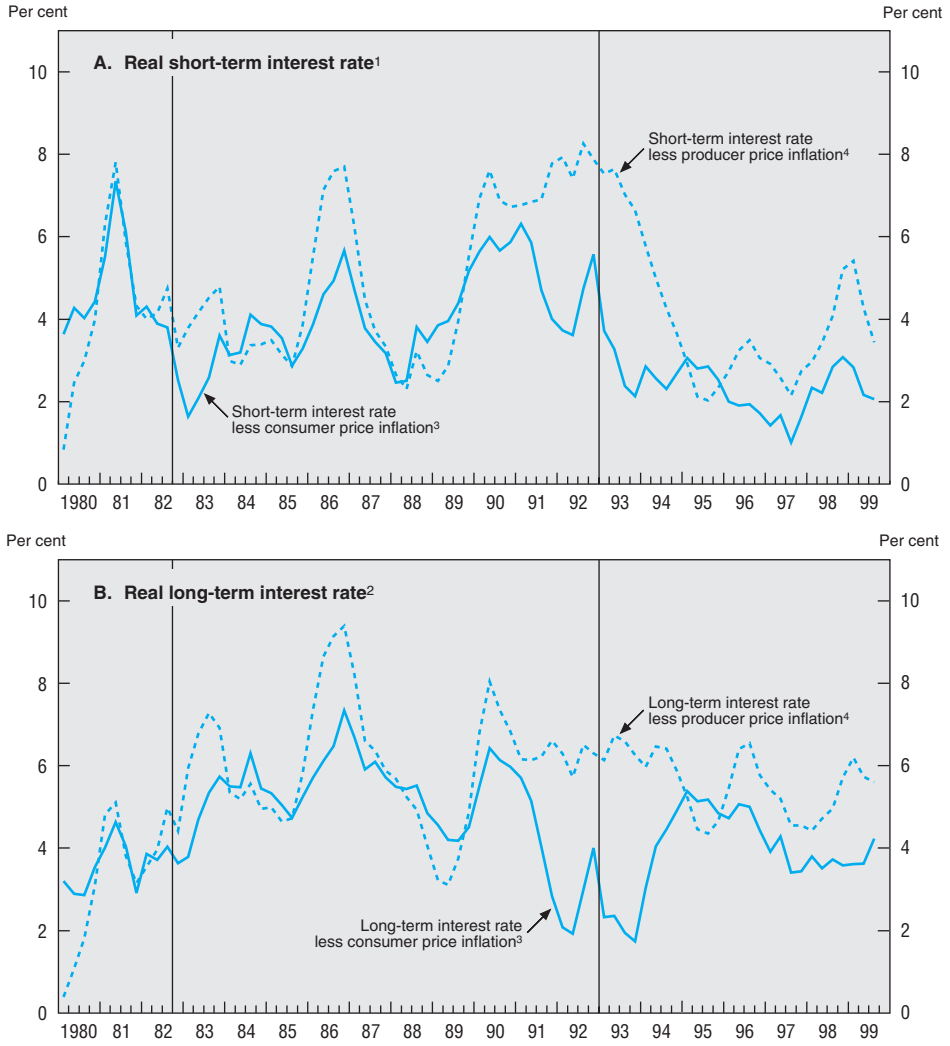
Figure 9. **Interest-rate developments**



1. End of period.
2. Prior to January 1999: Fibor, thereafter: Euribor.
3. Yield on government bonds outstanding.
4. Credit of less than DM 1 million.

Source: Deutsche Bundesbank.

Figure 10. **Real interest rates**
Quarterly average in per cent

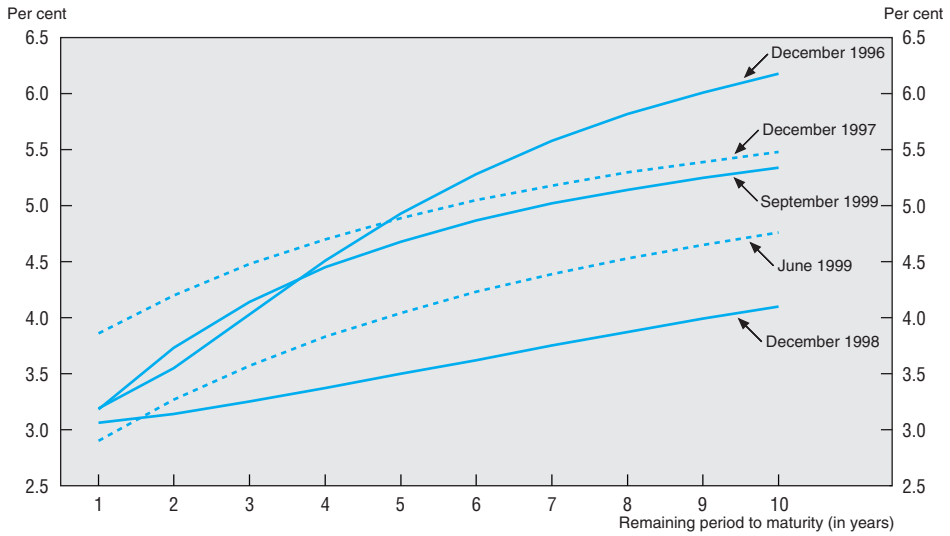


The vertical lines indicate the throughs.

1. 3-month rate. Prior to January 1999: Fibor, thereafter: Euribor.
2. Until 1990 7-15 year German public sector bonds, from 1990 yields on listed federal securities with residual maturity of 9 to 10 years.
3. As measured by the 4-quarter percentage change in the consumer price index excluding food and energy.
4. As measured by the 4-quarter percentage change in the producer price index.

Source: Deutsche Bundesbank; OECD, *Main Economic Indicators*.

Figure 11. **Yield curves**



Source: Deutsche Bundesbank.

(Figure 10).[6] The yield curve, having shifted down and flattened somewhat in 1998 has thus steepened this year, and this may be interpreted as reflecting the expansionary stance of monetary policy in the euro zone (Figure 11).

Overall, the stance of monetary policy which has emerged from the requirements of the euro zone can be seen as coinciding with the conjunctural needs of the German economy. Short-term interest rates for consumer credits and lending to enterprises in the euro zone have continued to decline, supporting a rapid rate of credit expansion and Germany has closely followed these trends. Credit growth in the German banking system has been quite rapid. With actual and expected inflation low in the euro area, monetary conditions are assumed to remain favourable throughout the remainder of 1999 and into 2000. Some increase in policy rates by the ECB are assumed to take place in line with the general increase in activity projected for the euro area.[7] However, these increases should not be great.

Short term prospects and risks

Growth should gain momentum...

Against the background of supportive monetary policy, a neutral fiscal policy (see below), growing real household income and a pick-up in exports, the

short-term prospects for the German economy are for somewhat faster growth. By mid 1999, exports appeared to be recovering, with incoming foreign orders in manufacturing during the second quarter of 1999 substantially exceeding those in the first quarter (Figure 2). Correspondingly, export expectations in the manufacturing sector have improved substantially since May. Export market growth is projected to accelerate in line with the upswing in the EU and improving conditions in the dynamic Asian economies and Japan. Demand from North America should remain strong. Competitiveness is also projected to underpin export growth. The effective euro exchange rate is assumed to remain at the favourable level reached by the end of the first semester of 1999 and, with wage growth continuing to be offset by productivity rises, the real effective exchange rate (in terms of relative unit labour costs) should remain competitive.

Export-led growth is also projected to be supported by domestic demand. With respect to private consumption, real household disposable income is being raised by tax and contribution cuts which are enacted for 2000 and 2001 while inflation should remain low. Investment in machinery and equipment is projected to improve slowly as exports improve and the policy environment becomes clearer. Although tax relief for enterprises has only been announced for 2001 – and does not fully offset tax increases in 1999 and in 2000 – declining uncertainty about the path of reform and improving prospects for some lowering of the tax burden should have a positive impact on investment intentions. This appears to be reflected in recent surveys. In the construction sector the worst seems to be over. While the contraction of construction activity is continuing, this will be at a slower rate in eastern Germany, where over-capacity is still substantial, and in the west the decline in residential and business-near construction appears to have bottomed out.

Overall, with exports and domestic demand picking up during the last half of 1999, growth for the year is projected to amount to some 1¼ per cent which would not be sufficient to lead to an increase in employment. The recovery of activity is projected to continue into 2000, GDP growth amounting to some 2¼ per cent, and to strengthen to a growth rate of around 2½ per cent in 2001. This will imply some narrowing of the output gap to around ½ per cent of potential GDP, which will nevertheless remain higher than most other euro zone economies. The unemployment rate is projected to decline only modestly, remaining at around 8¼ per cent in 2001 (on a national accounts basis). With productivity growth remaining robust, any fall will be partly due to demographic factors causing a trend decline in the labour force.

... with risks broadly balanced

The projection is contingent on a continuing recovery of export market growth and assumes unchanged nominal exchange rates. The former will depend

to an important degree on continuing strength in the US economy, as well as further progress in stabilising the dynamic Asian economies. Most importantly, it is assumed that uncertainty about the business environment will gradually wear of f as details of the government's reform strategy are settled. This holds for important features of the business tax reform as well as for policies in the fields of labour market and social reform. Renewed policy uncertainty would threaten the upturn in business confidence and investment would then be slower than projected. This might eventually lower consumer confidence, which has so far been resilient. With respect to the upside risks, the euro zone economies could recover more rapidly than anticipated so that growth in Germany might prove stronger.

II. Issues in fiscal policy

Overview of fiscal policy objectives

The new coalition government, which came to office in October 1998, has retained the budget stabilisation priorities of its predecessor but has adopted a programme which differs substantially as to the short-term means and distributional aims of fiscal policy. Reflecting its view that more needed to be done to stimulate demand by households, the rate of budget deficit reduction was at first slowed, but the priority given to budget consolidation has been reasserted in the June fiscal consolidation package. A fundamental reform of the personal and the corporate tax systems, which has been on the policy agenda for several years, is now under way, involving lower statutory tax rates, a broader tax base, a shift towards an ecological tax and a lower net tax burden for households; but in the short-term, company taxation has been increased and a reduction from the new higher level is only planned with the implementation of the proposed general reform of business taxation which will come into effect in 2001. The emphasis on reducing the tax burden on labour is evident in the commitment to lowering social security contributions from 42.3 per cent to under 40 per cent of the wage base by the end of the legislative period in 2002. This is to be accomplished in great part by lowering the pension contribution rate for both employers and employees and increasing transfers to the pension system out of the federal budget. But savings measures introduced by the previous government, designed to make the system demographically sustainable, have been withdrawn as socially unacceptable and the elements of a longer-term reform package are currently being elaborated. The government has also moved in the area of health sector reforms, retracting some measures introduced by the previous government, while emphasising the need to maintain stable health insurance contribution rates and maintaining a "solidarity" approach over the "insurance" approach to health care.

The Chapter begins by considering the prospects for the consolidation process in the light of the projected upswing in economic activity and the new demands which will weigh heavily on the budget, notably a scheduled re-design of the family support system. The three major structural policy issues which will

shape the stance of fiscal policy in the medium term – tax reform, reform of the pension system and the reorganisation of health care management – are highlighted in the subsequent sections of the Chapter.

Progress in fiscal consolidation

The 1998 and 1999 budgets: mixed progress towards consolidation

The general government deficit fell to 1.7 per cent of GDP in 1998 (2 per cent on the basis of the old system of national accounts), nearly 1 percentage point lower than in 1997 and ½ percentage point lower than initially projected (Table 5). The federal deficit came in as targeted, with higher transfers to the pension system than originally planned being matched by lower outlays in other fields and recovering tax receipts (Table 6). The observed tendency over the last few years for tax revenues to come in substantially lower than forecast appears to have been arrested in 1998, supported by a recovery in profits.[8] From the spending side, the better-than-expected deficit outcome was principally due to restraint at the community level,[9] reflecting efforts by the communities to reduce their outlays by relying on outsourcing and private project funding. Gross debt stabilised at around 61 per cent of GDP (Maastricht treaty definition), helped by privatisation receipts of DM 20 billion (Table 7).

The 1999 federal budget, which passed parliament in May 1999, incorporates a slightly lower deficit than the 1998 budgetary outcome (Table 6). Spending is set to increase by some 6 per cent, largely due to higher budgetary transfers to the pension system, for the purpose of lowering contribution rates. This in turn is being financed by new energy taxes (see below). With these transfers and additional provisions for the Post Office netted out, federal spending is set to increase by under 2 per cent. The government's priority to bring down unemployment is reflected in higher outlays for active labour market measures, and funding for universities is also being stepped up. While tax legislation implemented in 1999 is roughly revenue neutral, incoming data on tax receipts for the first half of 1999 support a relatively buoyant revenue outlook (see Box 2). Receipts from privatisations – mainly from selling Telekom shares – are scheduled to reduce the federal debt by DM 27 billion.

With the Länder also benefiting from higher tax receipts, the overall tax ratio will rise in 1999, contrary to the government's intention of lowering the tax/GDP ratio (including social security contributions) to 40½ per cent by 2002 (from 42½ in 1998). However, with spending less restrained than last year, the OECD projection is that the general government deficit will remain roughly unchanged in terms of GDP at 1.6 per cent. The structural deficit will be cut by 0.3 percentage points to 0.8 per cent due to additional receipts generated by the 1998 VAT

Table 5. **Public sector financial balances**

DM billion

	1993	1994	1995[1]	1996	1997	1998	1999[2]
General government (Maastricht basis)	**–103.4**	**–83.5**	**–111.2**	**–121.2**	**–96.5**	**–64.5**	**–63**
(Per cent of GDP)	(–3.2)	(–2.5)	(–3.2)	(–3.4)	(–2.6)	(–1.7)	(–1½)
of which:							
Territorial authorities[3]	**–137.8**	**–116.3**	**–109.4**	**–120.3**	**–94.8**	**–56.0**	**–73½**
Federal government	–66.9	–50.6	–50.5	–78.5	–63.4	–56.6	–53.6
State government west[4]	–22.5	–24.7	–28.6	–32.3	–24.4	–18.5	–18
Local government west	–8.9	–5.9	–12.4	–5.2	–4.2	5.7	2½
State government east[4]	–19.9	–19.9	–16.4	–15.0	–12.4	–9.6	–8
Local government east	–4.4	–4.8	–1.9	–2.7	–1.6	–0.8	–2
German Unity Fund	–13.5	–3.0	2.3	12.6	13.1	23.8	5
Other funds[5]	–1.7	–7.5	–1.8				
National accounts adjustments[6]	**28.7**	**29.3**	**13.1**	**10.9**	**–5.2**	**–16.3**	**–5½**
Social security	5.7	3.5	–14.9	–11.8	3.5	7.8	16
Memorandum items:							
Treuhandanstalt[7]	–38.1	–34.4	..	..	..	..	
Public enterprises[8]	–20.5	–20.2	..	..	..	..	
Extended public sector	**–162.0**	**–138.1**	**–111.2**	**–121.2**	**–96.5**	**–64.5**	**–63**
(Per cent of GDP)	(–5.0)	(–4.1)	(–3.2)	(–3.4)	(–2.6)	(–1.7)	(–1½)
(Cyclically-adjusted, per cent of potential GDP)	(–4.5)	(–3.8)	(–2.9)	(–2.6)	(–1.8)	(–1.1)	(–0.8)

1. 1995 figures do not include (as a capital transfer item) the take-over by the federal government of debts of the Treuhand (DM 204.6 billion) and of the East German Housing Fund (DM 30 billion). Inclusion of these items would imply a general government deficit in 1995 of the order of 9 to 10 per cent of GDP.
2. Government estimates.
3. Until 1997 including hospitals. The sum of the individual items differs for some years from the summary entry for the territorial authorities due to discrepancies in statistical reporting.
4. West including Berlin-West; east including Berlin-East.
5. Credit Fund, Economic Recovery Programme Fund, Financial Equalisation Fund, Compensation Fund, Railway Fund, Coal Fund and Inherited Debt Fund.
6. Including lending operations, privatisation receipts, timing adjustments, and, until 1995, profits paid by the Bundesbank into a debt sinking fund.
7. Wound up at the end of 1994. The successor organisation is financed directly from the federal budget.
8. Until 1995 included Post/Telecom and railway companies in the west and in the east; Post and Telecoms were incorporated in 1995 pending privatisation while subsidies are now paid directly to the railways by both the state and federal governments.

Source: BMF Finanzbericht; Deutsche Bundesbank, Monthly report; submission from the Ministry of Finance; OECD projections.

increase (which becomes fully effective only in 1999) and a recovery of the income tax base due to the gradual phasing out of previous allowances. The Länder deficit should stabilise, but community balances could diminish as outlays for personnel are rising and revenues from asset sales appear to be declining.[10] With the health insurance system producing a deficit of several billion in the first half

Table 6. **The Federal budget**

DM billion

	1995 Outcome	1996 Outcome	1997 Outcome	1998 Budget	1998 Outcome	1999 Budget	2000 Budget
Expenditure	**464.7**	**455.6**	**441.9**	**456.8**[1]	**456.9**	**485.7**[1]	**478.2**
(Per cent of GDP)	(13.2)	(12.7)	(12.1)	(12.1)	(12.1)	(12.5)	(11.9)
of which:							
Consumption	90.7	92.0	90.6	92.1	90.7	94.7	95.6
Interest payments	49.7	50.9	53.4	56.5	56.2	81.5	78.6
Investment	12.3	12.1	12.2	13.7	13.5	14.1	14.0
Transfers and lending	311.9	300.6	285.8	295.5	296.5	296.5	295.9
of which:							
To other administrations	88.5	92.5	79.7	79.8	78.8	48.6	47.8
Others	223.4	208.1	206.1	215.7	217.7	247.9	248.1
Revenues	**414.1**	**377.0**	**378.4**	**400.3**	**400.3**	**432.1**	**428.7**
(Per cent of GDP)	(11.8)	(10.5)	(10.3)	(10.6)	(10.6)	(11.1)	(10.7)
of which:							
Taxes	366.1	338.6	331.1	331.8	341.5	371.7	387.9
Other	48.0	38.4	47.4	68.6	58.8	60.5	40.8
Financial balance	**–50.5**	**–78.5**	**–63.4**	**–56.5**	**–56.6**	**–53.6**	**49½**
(Per cent of GDP)	(–1.4)	(–2.2)	(–1.7)	(–1.5)	(–1.5)	(–1.4)	(–1.2)

1. Total expenditure is smaller than the sum of the individual spending items by DM 1.0 billion in 1998, DM 1.1 billion in 1999 and DM 5.8 billion in 2000. This is due to overall expenditure restrictions not having been allocated to an expenditure category.

Source: Ministry of Finance; OECD.

Table 7. **Public debt by government level**[1]

DM billion

	1993	1994	1995	1996	1997	1998	1999[3]
Federal government	685.3	712.5	754.3	833.2	899.1	954½	1 387
Länder west	391.3	409.5	438.7	468.8	498.1	519.4	539
Länder east	37.3	51.2	64.6	78.6	88.2	96.3	103
Communities west	134.1	136.3	139.4	141.6	140.6	135.8	133½
Communities east	18.3	23.3	26.3	27.6	29.2	30.1	32
Special funds of the Federal Government	217.1	291.1	530.8	522.0	515.8	500.0	119
Total – territorial authorities[4]	**1 499.2**	**1 645.1**	**1 976.1**	**2 093.6**	**2 191.3**	**2 256.7**	**2 334½**
(Per cent of GDP)[2]	(46.3)	(48.5)	(56.1)	(58.4)	(59.8)	(59.6)	(60)
Maastricht definition	**1 518.9**	**1 672.0**	**2 006.4**	**2 140.6**	**2 230.0**	**2 295.7**	**2 369**
(Per cent of GDP)[2]	(46.9)	(49.3)	(57.0)	(59.7)	(60.8)	(60.7)	(61)
Financial liabilities	**1 585.4**	**1 670.5**	**2 082.5**	**2 219.2**	**2 304.3**	**2 393.9**	**..**
(Per cent of GDP)[2]	(49.0)	(49.2)	(59.1)	(61.9)	(62.8)	(63.3)	..

1. The Ministry of Finance calculation does not correspond to that published by the Bundesbank and which is reported in the *Economic Outlook* (financial liabilities). There are differences in valuation methods and in netting intra-government holdings.
2. GDP according to ESA 95. The debt ratios differ from those notified by the German authorities which are still based on the old GDP figures.
3. Government projections. In 1999 the federal government assumed the debt of the Inherited Debt Fund, the Coal Fund and the Railways Fund.
4. Including indebtedness of municipal special-purpose associations and municipal hospitals, and of smaller funds not listed in the table.

Source: Ministry of Finance; Deutsche Bundesbank; OECD.

Box 2. Assumptions underlying the OECD fiscal projections

Tax revenues in 1999 will benefit from an overhang effect of the VAT increase by 1 percentage point in April 1998. Since the reduction in the pension contribution rate by 0.8 percentage points is matched by increases in energy taxes, the two items are assumed to be deficit-neutral. The income tax measures and new regulation concerning minor employment and apparent self-employment implemented in 1999 and 2000 are projected to generate additional revenues of DM 1½ billion in 1999 and DM 2 billion in 2000. The higher benefit allowances mandated by the Constitutional Court have been incorporated, implying revenues foregone of DM 3.8 billion in 2000. No provisions have been made for the company tax reform, which will only become effective in 2001, since the programme has not yet been submitted to Parliament. To cover additional outlays associated with the conflict in the Kosovo, budgetary provisions have been made of DM 0.75 billion.

Both public sector wage growth and health care spending will accelerate in 1999, but the policy to restrict current expenditures will limit growth in real government consumption to around 1 per cent in 1999. In 2000 growth in government consumption is assumed to decelerate to 0.7 per cent, due to the government's savings package. Since unemployment will fall slightly in 1999 and to a larger degree in 2000, income replacement benefits for the unemployed are projected to remain roughly stable in 1999 and to decline in 2000. But outlays on active labour market measures will increase in 1999 and are assumed to remain roughly stable in 2000.

of the year, the surplus in the social security system is likely to deteriorate. More generous health care benefits and the rescinding of co-payment increases introduced by the previous government have contributed to this outcome. The general government debt ratio is projected to decline due to relatively high revenues from privatisations (Table 8).[11]

The 2000 Budget: consolidation efforts renewed

The budget plan for 2000 projects a reduction in the federal deficit to DM 49½ billion in 2000, from DM 53½ billion in 1999 (Table 6). In national accounts terms (Maastricht basis), the general government deficit is officially projected to fall from 1.6 per cent this year to 1.2 per cent in 2000,[12] which is also the OECD projection. This would suffice to maintain the debt-to-GDP ratio on a downward path according to OECD calculations (Table 8, and Box 2). Federal government spending is scheduled to decline by 1½ per cent in nominal terms, while the new tax measures will not generate additional net revenues in 2000.

To establish the deficit on the above declining path in the face of severe threatened slippage, the government was forced to introduce a multi-year savings

Table 8. **Appropriation account for general government**

Including social security

	1997	1998	1999	2000	2001
	DM billion				
Current receipts[1]	1 667.6	1 724.7	1 784.6	1 834.1	1 891.2
of which:					
Direct taxes	409.1	434.3	461.4	476.7	488.7
Social security contributions	719.7	726.1	736.1	752.2	775.5
Indirect taxes	418.5	438.6	463.2	484.0	505.9
Sales and property income	114.2	117.3	115.5	112.7	112.7
Current disbursements[1]	1 667.6	1 688.7	1 742.9	1 772.1	1 814.4
of which:					
Government consumption	714.2	719.4	738.0	749.3	766.4
Property income payable	134.2	135.2	138.0	140.7	154.5
Subsidies	67.1	69.4	71.6	72.2	73.3
Social security outlays to households	695.6	703.5	726.0	740.0	748.5
Other current transfers	56.5	61.3	69.3	70.0	71.8
Gross investment	70.2	66.9	67.4	68.6	70.2
Net lending	–96.5	–64.5	–63.1	–50.0	–36.8
	Per cent of nominal GDP				
Memorandum items:					
Net lending, Maastricht basis	–2.6	–1.7	–1.6	–1.2	–0.9
Cyclically-adjusted net lending (potential GDP)	–1.8	–1.1	–0.8	–0.6	–0.6
Financial liabilities	62.8	63.3	62.6	61.7	60.2

1. The grouping shown here is not that utilised by the Federal Statistical Office. The data is subject to revision.
Source: Statistisches Bundesamt for raw data; OECD.

package in June 1999. Savings of DM 30 billion (0.8 per cent of 1998 GDP) are foreseen relative to the notional baseline in 2000, consisting almost entirely of spending cuts. These are incorporated into the budget figures above. The need for a savings package arose from the fact that privatisation receipts are scheduled to decline rapidly after 1999,[13] while a ruling of the Constitutional Court that family allowances should be restructured will cost DM 3.8 billion in 2000 as the first step of the chosen scheme comes into operation in that year (see Table 9 for projected costs). The government's decision to cut net business taxes beginning in 2001 by some DM 8 billion (see below) was also a factor. The package is a significant step both towards fiscal consolidation and the creation of a safety margin with respect to the Maastricht deficit limits. But the ultimate effect of the expenditure cuts may be significantly smaller than the magnitudes suggested. Some reductions in federal transfers shift the budgetary burden from the central

government to the social security system, which will require further cuts in social spending if social security contributions are not to rise. Similarly, some measures, such as the adjustment to housing benefits, imply a higher budgetary burden for the communities, although if the proposal to index civil service salaries to inflation for the next two years is implemented there could be a significant budgetary benefit for lower levels of government.

Medium term fiscal objectives

Germany's Stability Programme, presented to the EU in January 1999, foresaw a reduction in the general government deficit (old basis of national accounts) in equal steps to 1 per cent of GDP in 2002, while the consolidation programme outlined in summer 1999 aims to reach a deficit/GDP ratio of ½ per cent in 2003. The debt/GDP ratio is to be reduced to 58½ per cent in 2003, by which year the savings generated by the 1999 fiscal package will increase to DM 49.3 billion. However, some items still need to be agreed by the government, while, as noted, implementation will require more permanent fiscal measures at the level of the Länder and social security funds. The government has not ruled out the need for additional cuts. Indeed, recently-implemented tax legislation and planned future tax cuts combined are estimated to imply significant revenues foregone (Table 9). Hence, without the fiscal consolidation package, the structural deficit could increase substantially after 2000. While the package marks substantial progress in consolidation policy, it needs to be combined with structural reform to identify economically important priorities for savings. There is still significant scope for further cuts in subsidies and tax expenditures, which combined have been estimated to total some 8 per cent of GDP. More effective control of social spending and measures to improve public sector efficiency are also required.

This ½ per cent deficit target appears to be the minimum that would be required to prevent breaching the 3 per cent deficit limit in the event of adverse economic developments. Based on estimates of the sensitivity of the general government deficit with respect to economic shocks, the OECD has simulated the probability of breaching the 3 per cent deficit limit over a pre-specified period of time under the assumption that public spending and tax parameters remain unchanged.[14] Generally, the likelihood of exceeding the 3 per cent limit increases with both the size of the deficit at the beginning of the period and the length of the period considered, since over a longer time span there is an increased probability of a sequence of unfavourable events hitting the economy. The results indicate that for a one-year horizon a deficit-to-GDP ratio of 1½ per cent would suffice to keep the 3 per cent deficit limit at a confidence level of 90 per cent. However, in order to avoid breaching the threshold with 90 per cent confidence over a five year period, the deficit should be confined to ½ per cent of GDP.

Table 9. **Financial effects of the tax reform programme**

Including child benefits
– = tax relief; + = tax burden

	Date of implementation	Revenues expected in year (financial base)				
		1999	2000	2001	2002	2003
		DM (billions)				
Tax reform law 1998: legislated	January 1999					
Tax relief: valuation of pension reserves		–0.73	–1.62	–6.48		
Tax relief law 1999: legislated[1]	January 1999					
Tax relief		–7.77	–7.11	–7.13	–7.16	–7.23
Reduction of entrance income tax rate to 23.9 per cent		–1.37	–1.31	–1.33	–1.36	–1.43
Increase in child benefits		–6.40	–5.80	–5.80	–5.80	–5.80
Tax relief law 1999/2000/2002 legislated[1]						
Step 1999	January 1999					
Tax relief		–2.30	–3.99	–5.20	–5.39	–5.55
Reduction of top income tax rate for business income to 45 per cent		–0.93	–1.12	–1.38	–1.50	–1.60
Reduction of corporate tax rate for retained profits to 40 per cent		–1.37	–2.87	–3.82	–3.89	–3.95
Step 2000	January 2000					
Tax relief			–13.55	–13.69	–14.31	–14.65
Increase in basic tax allowance and reduction of entrance income tax rate to 22.9 per cent			–4.68	–4.44	–4.50	–4.54
Reduction of top income tax rate for non-business income to 51 per cent			–1.49	–1.63	–1.84	–2.00
Reduction of top income tax rate for business income to 43 per cent			–1.10	–1.34	–1.55	–1.66
Reduction of intermediate tariff			–6.28	–6.28	–6.42	–6.45
Step 2002	January 2002					
Tax relief					–28.15	–28.07
Increase in basic tax allowance and reduction of entrance income tax rate to 19.9 per cent					–7.08	–6.85
Reduction of top income tax rate for non-business income to 48.5 per cent					–2.17	–2.32
Reduction of intermediate tariff					–18.90	–18.90
Financing measures		+10.80	+24.28	+31.96	+36.43	+32.10
Net tax relief from the Tax Relief Laws of 1999 and 1999/2000/2001[2]		+0.73	–0.37	+5.94	–18.58	–23.40

Table 9. **Financial effects of the tax reform programme** *(cont.)*

Including child benefits
– = tax relief; + = tax burden

	Date of implementation	Revenues expected in year (financial base)				
		1999	2000	2001	2002	2003
		DM (billions)				
Revised taxation of Minor Employment and Apparent Self-Employment: legislated[1]	April 1999					
Income tax revenues foregone on minor employment		–1.37	–2.11	–2.11	–2.11	–2.11
Additional revenues of the social security system from minor employment and apparent self-employment		+3.25	+4.85	+4.85	+4.90	+4.90
Net revenues from the revised taxation of minor employment and apparent self-employment		+1.88	+2.74	+2.74	+2.79	+2.79
Business Tax Reform planned	January 2001					
Tax relief preliminary				–22.39	–30.74	–32.66
Corporations:						
Reduction of tax rate for retained profits to 25 per cent				–11.26	–14.37	–15.57
Reduction of tax rate for distributed profits to 25 per cent				–4.22	–4.53	–3.19
Replacement of imputation system by "half-revenues" system				+0.09	+1.16	+1.60
Transition provisions				–3.00	–2.00	–0.50
Partnerships:						
Preferential treatment of retained profits at 25 per cent tax rate[3]				–4.00	–11.00	–15.00
Financing measures				+9.12	+17.96	+24.49
Abolition of rate reduction for entrepreneurial income				+4.48	+3.66	+4.69
Reduction of declining-balance depreciation rate to 20 per cent				+3.98	+12.58	+17.19
Others				+0.66	+1.72	+2.61
Net tax relief from the Business Tax Reform				–13.27	–12.78	–8.17
Further refinancing tax measures planned:			+1.07	+3.18	+5.89	+8.69
Updating of effective life of equipment for depreciation tables			+0.88	+2.27	+4.78	+7.36
Others			+0.19	+0.91	+1.11	+1.33
Family Tax Reform planned						
Step 2000	January 2000					
Tax relief including child benefits			–3.80	–5.34	–5.53	–5.72
Financing measures				+0.17	+0.19	+0.22

Table 9. **Financial effects of the tax reform programme** *(cont.)*

Including child benefits
– = tax relief; + = tax burden

	Date of implementation	Revenues expected in year (financial base)				
		1999	2000	2001	2002	2003
		DM (billions)				
Step 2002	January 2002					
Tax relief					–0.72	–3.42
Financing measures					+1.23	+2.52
Net tax relief from the Family Tax Reform			–3.80	–5.17	–4.84	–6.40
Net tax relief from all income tax reforms combined[2]						
Including additional social security revenues from new employment taxation		+1.16	–1.98	–13.06	–27.51	–26.49
Memorandum item:						
Ecological tax reform and reduction in pension contributions: legislated[1]	April 1999					
Step 1999						
Revenues from energy-taxes		+8.40	+12.34	+12.33	+12.31	+12.30
Reduction in pension contributions by 0.8 percentage points		–8.40	–12.34	–12.33	–12.31	–12.30
Steps 2000-2003	January 2000-2003					
Revenues from energy-taxes			+5.10	+10.50	+15.80	+21.20
Reduction in pension contributions			–5.10	–10.50	–15.80	–21.20
Net revenues from ecological tax reform and reduction in pension contributions		0.00	0.00	0.00	0.00	0.00
Net tax relief from all tax reforms combined[2]		+1.16	–1.98	–13.06	–27.51	–26.49

1. For 2003 estimates by the Institut für Wirtschaftsforschung Halle.
2. For 2003 the sum involves estimates by the Institut für Wirtschaftsforschung Halle.
3. The figures are based on the reform option which foresees that half of distributed profits enter the personal income tax base.

Source: Ministry of Finance; Insititut für Wirtschaftsforschung Halle; OECD.

Tax reform

Developing an overall strategy

While fundamental reform of the tax system has been on the policy agenda for several years, agreement has been hampered by differences as to aims and means. Distributional issues, including the redistribution of taxes between federal and state levels, have proved an important source of major disagreement, the reform proposal put forward in 1997 being rejected by the second chamber of parliament in part on these grounds. Proposals have often stressed the need to promote investment and thereby jobs, either directly or through improving the international competitiveness of Germany as a location for investment, and defining objectives in these specific terms has led to conflicting judgements about the precise measures to be adopted. The OECD view has been that fundamental reform is required to create a less discriminatory system, which would be more even-handed in creating a favourable growth environment for all types of activity. As has been outlined in previous *Surveys*, the German tax code is characterised by relatively high statutory tax rates and numerous tax breaks and allowances that make the system complex and opaque, while favouring certain sectors and activities which are, *inter alia*, capital- and inventory intensive and which involve rapid turnover of equipment.

The government has now embarked on a wide-ranging reform project involving the introduction of an energy tax, which will be used to reduce labour taxes, and a reform of income and business taxation, aimed at reducing statutory tax rates, broadening the tax base and simplifying the tax codes. The first part of the tax reform, the "Tax Relief Laws", came into operation in January 1999 (although some parts will only come into effect in January 2000 and 2002: see Box 3 and Table 9) and addressed the government's immediate objective of raising household disposable income. It involved a cut in statutory tax rates for wage and non-wage income, increased tax allowances for families and compensated for lost revenues by a large increase in tax payments by enterprises (especially large firms) via a broadening of the tax base. The ultimate effect will be a substantial lowering of the net tax burden for households, an impact accentuated by changes to the family support system, where there will be a further major revision based on a recent judgement of the Constitutional Court. The Court ruled that the present family support system is unconstitutional in that married parents are disadvantaged in comparison with single parents and expenses for raising children are not sufficiently taken into consideration. The support system therefore needs to be reformed in two steps in 2000 and 2002.

There has been, and remains, some uncertainty as to the government's objectives with respect to the ultimate tax burden on the business sector. The original proposals were not well received by the business sector and needed

Box 3. The major components of tax reform

Major features of the government's tax reform package are:

I. Tax Relief Law – legislated

The first phase of the Tax Relief Law is in operation since January 1999, and the remainder of the package was enacted by parliament in March 1999.

- The bottom statutory tax rates are being reduced from 25.9 per cent in 1998 to: 23.9 per cent (January 1999); 22.9 per cent (January 2000) and 19.9 per cent (January 2002);
- For non-entrepreneurial income (notably wages, dividends and income from certain types of self-employment such as physicians and lawyers) the top statutory tax rate is being decreased from 53 per cent in 1998 to 51 per cent (January 2000) and 48.5 per cent (January 2002);
- For entrepreneurial income the top statutory tax rate is lowered from 47 per cent in 1998 to 45 per cent (January 1999) and 43 per cent (January 2000)
- Child allowances have been increased. The basic tax-free allowance (*Grundfreibetrag*) will be increased in 2000 and 2002.
- For incorporated companies the statutory rate for retained profits has been decreased from 45 per cent in 1998 to 40 per cent in January 1999.
- A number of measures have been implemented in 1999 to broaden the tax base of business income including changes in the ways inventories are valued, discounting of reserves against future liabilities, and in the way tax free reserves are accumulated in the insurance and energy sectors. In the last two sectors tax free reserves will need to be unwound in coming years.

Due to tax raising measures the Tax Relief law is projected to be broadly fiscally neutral in 1999 and 2000. For 2001, the government projects net revenues of some DM 6 billion, but for 2002 there is an estimated net tax relief of DM 18.6 billion.

II. Ecological tax reform – legislated

New taxes on energy consumption, implemented in April 1999 (DM 8½ billion revenues in 1999) are used to lower pension contribution rates by 0.8 percentage points. Further increases in energy taxes and reductions in contribution rates until 2002 have been announced.

III. Revision of family taxation – partly decided in June 1999

The reform will be implemented in two steps in 2000 and 2002. Benefits and tax credits for child minding will by raised in 2000 (DM 3.8 billion in 2000), and further increases will be legislated in 2001.

IV. Business tax reform – elements decided in June 1999

Incorporated companies:

- Reduction of the tax rate to 25 per cent for both retained profits (from 40 per cent) and distributed profits (from 30 per cent)
- At present the corporate tax system is fully integrated with the personal income tax: domestic shareholders receive a tax credit for the tax on distributed profits paid by a company so that profits are effectively taxed at the personal income tax rate. Because the tax credit is not available for non-residents who are not

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liable to pay income tax in Germany, this system has been criticised for treating residents and non-residents in an unequal way. The government plans to remedy this shortcoming by abolishing the imputation system and treating companies and shareholders as entirely separate entities. On current plans profits will be taxed at 25 per cent on the company level. As regards distributed profits, 50 per cent will enter the shareholder's personal income tax base and will be taxed at their personal tax rate (with personal marginal tax rates between 19.9 and 48.5 per cent in 2002).

Non-incorporated partnerships:

- Taxation of retained profits at a tax rate of 25 per cent.
- Several options are under consideration for treating distributed income including application of personal income tax rates on 50 per cent of distributed profits ("half-base system") and full taxation.

Financing measures comprise *inter alia*:

- Lowering the threshold for full taxation of capital gains from selling stakes in incorporated companies to a share of 1 per cent or more in the company's capital, from 10 per cent or more
- Abolition of special depreciation allowances
- Reduction of the maximum declining-balance depreciation rate for investment in machinery and equipment to 20 per cent (from 30 per cent); reduction of the linear depreciation rate for buildings to 3 per cent (from 4 per cent).
- Further measures to broaden the profit tax base by changing the tax accounting system.

The planned net tax relief for business totals DM 8 billion for the first full year in operation. Several details of the reform still have to be decided. On current plans, the reform will become effective in January 2001.

technical corrections. To prepare a more comprehensive business tax reform the government set up a reform commission in December 1998 to make proposals for the design of a tax system in which income from all business activity – both incorporated and partnerships – is taxed at a uniform tax rate of at most 35 per cent. The commission was also asked to reconsider Germany's imputation system for corporate taxes. Based on the Commission's recommendations, made in April 1999, the government has decided the following key features of the enterprise tax reform scheduled to be implemented in 2001 (Box 2): *i*) taxes on retained profits will be reduced to a uniform 25 per cent for both incorporated business and partnerships (net of the local business trades tax which at present totals another 10 per cent of profits on average); *ii*) for corporations, the tax rates for distributed and retained profits will be unified at 25 per cent and the imputation system will be replaced by a dividend relief system in which 50 per cent of distributed profits will be subject to personal income tax; and *iii*) the profit tax

base will be further broadened, in particular, the top depreciation rate for investment in machinery and equipment will be lowered. The draft law is currently being prepared and will be presented for discussion at the end of 1999. The reform was to take place in 2000 but due to the difficult nature of the exercise it has now been delayed until 2001. At a later stage, the local business trade tax will be replaced by a less distortionary instrument which will have to be negotiated with the Länder and communities.[15]

In considering business tax reform, the main issues which have arisen relate to the following areas:

- *The taxable profit base*. The effective tax rate (see Box 4) is considerably lower than suggested by high nominal rates due to relatively generous depreciation and tax allowances and a wide scope for creating untaxed reserves. The issue is how to change these in a systematic manner especially when they appear to be related to either investment or other goals such as making provisions for the decommissioning of nuclear reactors.
- *Discrimination against cross-border profit flows*. At present, domestic shareholders can set corporation tax *against* their income tax liabilities (*i.e.* a full imputation system applies to the taxation of domestic profits distributed to resident shareholders). This system formally discriminates against both foreign shareholders in German corporations and resident shareholders in foreign corporations, and may also be contrary to the EU's directive on freedom of establishment. Issues arise as to how the system can be more friendly to foreign investors, and more consistent with tax harmonisation within the EU, by lowering the statutory corporation tax rate while limiting any increase in the taxation on domestic shareholders.
- *Discrimination between retained and distributed profits*. A higher rate of profits tax is currently applied to retained corporate *earnings*, although there are generous provisions for accumulating tax-free reserves. If the imputation system is reformed, issues arise as whether any resultant tax preference for retained earnings will stimulate saving and investment, and/or the extent to which entrepreneurs will be allowed to translate their earnings into capital gains, given that statutory income tax rates on distributions and self-employment income will remain high for revenue purposes.
- *Treatment of unincorporated enterprises*. Germany is characterised by a large number of unincorporated enterprises, which are taxed according to the personal income tax system and in many cases they are under the threshold for paying local business tax (*Gewerbesteuer*). Since 1994, income from unincorporated enterprises has been taxed at lower rates than capital and labour income. Issues arise as to whether there should be equal tax treatment to distributed and retained profits for corporations and non-incorporated businesses.

Box 4. Measuring tax rates of German corporations

There are three commonly-used measures of corporate tax rates: *statutory tax rates*, *average effective tax rates* and *marginal effective tax rates*. Statutory rates apply to the taxable profit base, after allowances and depreciation adjustments; average effective tax rates measure the tax liability relative to statistically-recorded profits; marginal effective tax rates measure the effects of the main corporate tax parameters (and sometimes income tax) on a representative firm's target rate of return on an additional unit of investment.

In 1998, *statutory corporate tax rates* in EU countries ranged from 28 per cent (Finland and Sweden) to 56.3 per cent for retained profits in Germany (including the business trades tax and the "solidarity" tax surcharge). Statutory rates are likely to be relevant for international investment decisions, but they typically overstate the true tax burden for the domestic owners of companies. This is particularly true for Germany where companies face comparatively generous depreciation and tax allowances and have a wide scope to create untaxed reserves.

International comparisons of *average effective tax rates* are complicated because there exist several different concepts and methodological approaches. Moreover, even where they provide insights into the tax rate on some definition of operational profit, the results will be sensitive to various assumptions made, and the economic structure of a firm considered as "representative". With these caveats in mind, a recent study based on firm-specific data in the manufacturing sector between 1990 and 1996, commissioned by the Dutch Ministry of Finance,[1] found that taxes on profits were on average highest in Germany (at over 38 per cent). The difference between Germany and other countries is much smaller than suggested by a comparison of statutory rates of the corporate income tax. However, earnings figures taken from annual accounts reflect differences in each country's national accounting practices as well as statistical and methodological discrepancies in the corporate balance sheet data. Annual accounts of incorporated companies in the old Länder reporting to the Deutsche Bundesbank show an average tax rate of about 46 per cent in recent years.[2] In another study, average tax rates for a medium-sized corporation were simulated for a "representative" hypothetical firm, based on both relevant provisions in the national tax codes and calibrated parameter values for various firm characteristics such as assets, financing structure, and employment.[3] Among the five countries considered for 1999, France has the highest estimated tax rate, followed by Germany.

Macroeconomic measures of the average effective tax rate, which express revenues collected as a percentage of the operating surplus of either the corporate sector or – in the case of Germany – the entire business sector as measured in the national accounts, give a different picture, with Austria and Germany showing the lowest tax rates (7 and 14 per cent respectively). But this ratio may be a highly unreliable indicator of the effective tax rate on profits for several reasons. In particular, there is a mismatch between the numerator and the denominator in that operating surplus is measured gross of interest on debt and includes profits from tax-free entities such as the Deutsche Bundesbank. Similarly, in countries that tax resident companies on their foreign source income an additional mismatch is introduced as operating surplus includes only domestic profits. Recent studies of average effective tax rates on a *forward-looking basis*, referring to 1998, suggest that for *new investment*, measured on a

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project basis, German and French firms in the manufacturing sector are exposed to the highest rates (over 45 per cent).[4]

Similarly, m*arginal effective tax rates* are estimated to be highest in France (over 41 per cent), followed by Germany (37 per cent) and Spain (33 per cent).[5] Again, the results of these studies are tied to a set of assumptions which are highly stylised and may not reflect taxes actually paid.[6]

1. Buijink W., B. Janssen and Y. Schols, University of Maastricht (MARC), "Corporate effective tax rates in the European Union", April 1999 (commissioned by the Ministry of Finance of the Netherlands).
2. Deutsche Bundesbank, "Ertragslage und Finanzierungsverhältnisse westdeutscher Unternehmen im Jahr 1997", *Monthly Report October* 1998; and November issues of the *Monthly Reports* in previous years. The figure shown in the reports (an average rate of 35 per cent) include unincorporated companies, which are taxed on the basis of the income tax system.
3. Zentrum für Europäische Wirtschaftsforschung, Statement to the Parliamentary Finance Committee of the Deutscher Bundestag, Bonn 21st June 1999.
4. Price Waterhouse Coopers, "Study of potential of effective corporate tax rates in Europe", 1998.
5. Baker & McKenzie, "Survey of the effective tax burden in the European Union", Amsterdam 1999 (commissioned by the Ministry of Finance of the Netherlands).
6. An assessment of international comparisons of company tax burdens is also contained in: Bundesministerium der Finanzen, *Steuerbelastung deutscher Unternehmen – nationaler und internationaler Vergleich*, Bonn, March 1999.

This section highlights important features of these projects; key elements are depicted in Box 3. The ecological tax reform – introducing taxes on energy consumption by households and enterprises – whose first step is in operation since April 1999 is considered in Chapter IV below.

The first phase of tax reform

In its initial phase, the tax relief produced by the legislated Tax Relief Law is concentrated on households, for whom the starting rate for income tax will be cut from 25.9 per cent in 1998 to 19.9 per cent in 2002. The corresponding cut in the top statutory tax rate on non-entrepreneurial income will be from 53 per cent to 48.5 per cent, while that on entrepreneurial income will decline from 47 per cent to 43 per cent in 2000. It has been estimated that the net benefits from this law accruing to private households from lower personal tax rates and higher tax allowances for children total some DM 3 billion in 1999 and more than DM 20 billion once all changes are in force.

For 1999 and 2000 the tax relief for private households is fully compensated by higher taxes on business activity. The latter originate mainly from more restrictive rules for assessing the level of profits – notably with respect to offsetting losses against profits, the valuation of assets and the creation and valuation of tax-free reserves – and to a lesser extent from reductions in business tax expenditures (Table 9). Although some tax expenditures in the income tax code were dealt with – notably, reduced taxation on redundancy payments, bonuses for employees, and some provisions for agriculture – other important concessions identified by a tax reform commission in 1997 have been preserved. Examples are the tax relief for travel expenses to the working place and the preferential treatment of night and overtime work. There is thus still scope for improving the allocative efficiency of the tax system, as well as creating room for further rate reductions, by cutting back tax expenditures.

The measures to widen the corporate and business tax base were in many respects *ad hoc* and did not represent a systematic attempt to create a new coherent tax accounting system. Indeed some further problems might have been created. For example, the new law generally restricts the possibilities to offset losses against profits across different income categories. While the economic rationale for these restrictions might be debated, they will complicate tax administration. In particular, partnerships are forbidden to offset losses against gains across income categories if the partnership has taken on the losses for the purpose of reducing the income tax liability. The latter is assumed to be the case if the post-tax return on capital is more than twice as high as the pre-tax return. Given the fact that the impact of the loss take-over on capital return varies with the tax situation of the taxpayer, the "double return rule" will be difficult to verify and may lead to debates and tax evasion.

Evaluation of the company tax reform proposals

The full details of the company tax reform still need to be fixed over the next months, and the government is in the process of evaluating the impact of the planned measures on company profits. Overall, the targeted net tax relief to enterprises from this particular business tax reform totals DM 8 billion in the first full year. Tax relief in later years will depend essentially on the exact details of the tax reform and there is no official target. Preliminary figures released by the government suggest that the net relief from the business tax reform tapers off over the medium-term planning horizon extending until 2003 (Table 9). However, the sectoral and company size distribution of the tax burden should change significantly.

If all elements of the proposed tax reform package are enacted, statutory tax rates in Germany will be considerably lower and several tax exemptions and special depreciation allowances will have been reduced. German statutory corpo-

rate tax rates will be placed in a middle position among OECD countries, down from the top position it held until now (Table 10, and Box 4). As Table 11 shows,

Table 10. **Statutory tax rates within the OECD**

Per cent

	1986	1991	1998
United States	46.0	34.0	35.0
Japan	43.0	38.0	37.5
Germany [1]	**56.0**	**50.0-36.0**	**45.0-30.0**
France	45.0	34.0-42.0	33.3
Italy	36.0	36.0	37.0
United Kingdom	35.0	34.0	31.0
Canada	36.0	28.0	28.0
Australia	49.0	39.0	36.0
Austria [2]	50.0	30.0	34.0
Belgium	45.0	39.0	39.0
Denmark	50.0	38.0	34.0
Finland	33.0	23.0	28.0
Greece [3]	49.0	46.0	35.0-40.0
Iceland	51.0	45.0	30.0
Ireland	50.0	43.0	32.0
Luxembourg	40.0	33.0	30.0
Mexico [4]	39.2	35.0	34.0
Netherlands	42.0	35.0	35.0
New Zealand	45.0	33.0	33.0
Norway	28.0	27.0	28.0
Portugal	42.0-47.0	36.0	34.0
Spain	35.0	35.0	35.0
Sweden	52.0	30.0	28.0
Switzerland [5]	4.0-10.0	4.0-10.0	7.8
Turkey	46.0	49.0	25.0
Average basic rate [6]	43.5	36.3	33.2
Range (maximum to minimum) [6]	28.0	27.0	20.0
Standard deviation [6]	7.0	6.8	4.5

Note: This table represents the rates of central government corporate income tax. Austria, Canada, Finland, Germany, Italy, Japan, Norway, Portugal, Switzerland and the United States also have sub-central corporate taxes. Rates rounded to nearest percentage point. Many countries also have special rates for firms with lower profits and for particular sectors.

1. First figure = retained; second figure = distributed earnings. In 1995, a "solidarity" surcharge was introduced on the corporate income tax (5.5 per cent on the tax liability in 1998). It is not included in the figure shown in the table.
2. 1986 data are national data.
3. The last column applies to 1997 instead of 1998.
4. Mexican corporate tax data for the first year considered are from national sources and concern 1988 instead of 1986.
5. The figure for 1998 is a net rate. The federal tax is deductible from its own base.
6. Excludes Switzerland, takes into account the rates on retained earnings for Germany (respectively 50 per cent in 1991 and 45 per cent in 1998), 34 per cent for France in 1991 (rate applicable from the second part of the year onwards), 40 per cent for Greece in 1997 (rate applying to non-listed corporate profits), and 42 per cent for Portugal in 1986.

Source: Owens and Whitehouse (1996). The OECD Tax Data Base (1998).

Table 11. **Marginal tax on income implied by the planned business tax reform**[1]

Per cent

Income from:	Income distributed				Income retained	
	Recipient has the lowest marginal income tax rate[2]		Recipient has the highest marginal income tax rate[3]			
	December 1998	Post-reform	December 1998	Post-reform	December 1998	Post-reform
Incorporated business	25.9	32.5	53.0	43.2	45.0	25
Unincorporated business	25.9		47.0		25.9-47.0	25
Version *a)*		32.5		43.2		
Version *b)*		19.9		43.0		
Wages	25.9	19.9	53.0	48.5		

1. The derived marginal tax rates are net of the business trade tax (*Gewerbesteuer*) and the "solidarity" surcharge on tax liability. Marginal tax rates for wages are exclusive of social security contributions. For incorporated companies it is assumed that full profits are taxed at the company level with a rate of 25 per cent and 50 per cent of distributed profits are taxed at the personal income tax rate. For unincorporated companies, two reform options are considered that have been proposed by the Tax Reform Commission: in Version *a)* it is assumed that profits are taxed in the same manner as corporated companies; in Version *b)* it is assumed that distributed profits are fully taxed at the personal income-tax rate.
2. From 2002, 19.9 per cent, according to the Tax Relief Law (*Steuerentlastungsgesetz*).
3. From 2002, 48.5 per cent for all cases except version b) for unincorporated enterprises where 43 per cent is assumed; based on the Tax Relief Law (*Steuerentlastungsgesetz*).

Source: OECD

marginal tax rates on dividends and profits would be substantially lower, with the notable exception of dividends from corporations accruing to low-income shareholders. The decline in statutory rates will be particularly pronounced for retained profits which will be taxed much more favourably than distributed profits, given the high personal income tax rates in combination with the "half-base" taxation principle for distributed profits. A possible exception, depending on the reform option chosen, are the owners of unincorporated business with a low personal income tax rate (the lowest income tax rate being 19.9 per cent compared with 25 per cent on retained income).

In some respects, the reform would also increase the neutrality of the tax system, notably with respect to treatment of retained earnings across different forms of enterprises, which will be taxed at a standard rate of 25 per cent. Also, the decrease in the depreciation rate for investment in machinery and equipment should reduce the present bias of the tax system in favour of capital-intensive companies. However, other proposals introduce important non-neutralities. The tax preference in favour of retained profits is granted for the purpose of fostering investment and employment creation, but this feature runs the risk of being counter-productive. German enterprises finance their investments both internally via retained earnings and reserve creation and externally via funds provided by

banks and the capital market in general. Tax preferences for retained profits would hamper the re-allocation of funds via the capital market. Such a policy would run the risk of producing a misallocation of resources.[16]

Furthermore, the split rate approach to distributed and retained profits will provoke reactions on the side of the shareholders to reclassify their income. In particular, the system provides incentives to avoid higher taxation of distributed profits by cumulating income within the company and realising capital gains by selling the firm at a later stage. To counteract this possibility, the government plans to extend the full taxation of gains from selling stakes in incorporated companies (income from selling a share of a partnership is already taxed) to include all holdings of 1 per cent or more of the company capital (compared with 10 per cent or more at present) (see Box 2). While such a measure would place incorporated and non-incorporated businesses on a more equal footing, it would appear to be at variance with the goal of making the provision of equity and venture capital more attractive. To the extent policy makers wish to increase taxation of consumption relative to the taxation of savings, increasing consumption taxes would be more appropriate.

The preference in the present system by which income from personal entrepreneurship is taxed at lower marginal rates than labour income, including certain types of self employment (*freiberufliche Tätigkeiten*), will become more pronounced, given the envisaged substantial cut in the statutory tax rate for retained earnings for both partnerships and corporations (Table 12).[17] Apart from the fact that this approach might raise constitutional issues,[18] it is questionable on economic grounds. As has been argued in the OECD *Jobs Study*, high taxation of labour income tends to increase unemployment and foster black market activities. In

Table 12. **Marginal taxes for resident and non-resident shareholders**[1]

Per cent

	Imputation system		Half-base system	
	Marginal tax rate of shareholder[2]		Marginal tax rate of shareholder[2]	
	19.9	48.5	19.9	48.5
Domestic shareholder	19.9	48.5	32.5	43.2
Foreign shareholder	43.9	63.9	39.9	61.3

1. The derived tax rates are net of the business trade tax and the "solidarity" surcharge on tax liability. In the half-base system, profits are taxed at a rate of 25 per cent at the company level. For residents, 50 per cent of distributed profits are subject to the personal income tax (after company taxation). In the imputation system profits are taxed at a rate of 30 per cent at the company level, and residents receive a tax credit for the corporate tax paid on the dividends.
2. As legislated in the Tax Relief Law for 2002. It is assumed that residents and non-residents have the same personal income tax rate.

Source: OECD.

addition, the differentiation between income considered to originate from personal entrepreneurial activity and other types of (self-employment) income is ambiguous and can give rise to a redefinition of market activities as "entrepreneurial" for the purpose of avoiding higher taxation.[19] Hence, cuts in tax rates should not be focused on "entrepreneurial" and corporate income, but should include other types of income, such that top marginal tax rates are more or less equalised. An income tax system with marginal tax rates increasing in a few steps to a relatively low top level, which has recently been suggested in the German policy debate, could meet these requirements and is relatively transparent.

The present imputation system avoids double taxation of distributed profits for residents, but involves giving tax credits to residents which are not available to non-residents. Although the system has clear advantages when viewed from a domestic perspective, there are disadvantages from the viewpoint of attracting foreign capital. The envisaged half-base system avoids double taxation of residents only partially by reducing personal income taxation of distributed profits, but has the advantage that the tax inequality between foreign and domestic shareholders can be reduced. Table 12 provides an illustration for marginal tax rates that would be faced by domestic and foreign shareholders under the present imputation system – with legislated tax rates for 2002 – and the envisaged half-base system if domestic and foreign shareholders faced the same personal income tax rates. As the table shows, the gap in marginal tax rates on domestic profits between resident and non-resident shareholders would remain high for those paying the top income tax rate (some 18 percentage points in comparison to 15½ percentage points in the present system). At the lowest marginal tax rate, the gap would decline to 7½ percentage points but the bulk of this convergence effect would be due to higher marginal tax rates of residents. Indeed, the new system would only slightly lower combined marginal taxes for foreign shareholders. Also, according to one estimate some 85 per cent of all German shareholders would pay higher taxes on dividend income than under the present system. In assessing the policy response to the issues of double taxation and international discrimination it is important to note that the lower personal income tax rates are, the less need there would be for dividend relief schemes to avoid double taxation of residents. Hence, the need to reduce personal income tax rates is reinforced from this perspective.

Pension reform

Previous *Surveys* have demonstrated that pension reform is necessary to put the system on a sound fiscal basis against the background of population ageing. While the shift in the age distribution of the population will only acceler-

ate in a decade, fiscal action is required now if the burden of fiscal adjustment is to be minimised. From the perspective of inter-generational equity, the fiscal burden shifted to the younger generation will increase the later pension reform will be implemented. Demographic developments, in combination with incentives for early retirement, are already exerting a negative impact on the financial situation of the pension system: in 1998 the 3 per cent increase in pension outlays substantially exceeded the annual per capita adjustment in pensions (implemented in June 1998) of 0.44 per cent in the west and 0.89 per cent in the east.

The pension reform strategy

The pension policy of the new government has proceeded on two tracks. Firstly, savings measures put in place by the old government were withdrawn in January 1999 as socially unacceptable. Key to the pension reform of the old government was the introduction of a demographic life-expectancy adjustment into the pension adjustment formula. This measure, which was legislated to become effective in 1999, was projected to cause a gradual decline in the reference pension replacement rate by 6 percentage points until 2030. Also, pensions for reduced capability to work (R*ente wegen geminderter* E*rwerbsfähigkeit*) and for invalidity (B*erufsunfähigkeitsrente*) were scheduled to be combined in 1999 into a single disability pension with eligibility to be assessed solely on health grounds without reference to labour market conditions which had hitherto been the case. Both measures have been withdrawn by the new government with the proviso that a new reform package should be implemented not later than in 2001.

Second, new measures target both the revenue and the spending side of the system. From a longer-term perspective, the Minister of Labour aims at stabilising the pension contribution rate at around 19 per cent until 2010. The government puts a high priority on lowering pension contribution rates, mainly for the purpose of lowering non-wage labour costs. To this end, the proceeds from the new energy taxes, introduced in April 1999, have been utilised to engineer a proportionate cut in pension contributions by 0.8 percentage points (equivalent to DM 8½ billion revenues from the energy tax) to 19.5 per cent of the wage base. Further cuts in contribution rates are planned for succeeding years, as higher energy taxes are phased in. Also, the revenue base has been widened by extending the liability to pay social security contributions to casual employment and certain types of self employment (“apparent self-employed”, S*cheinselbständige*). Employers are obliged to pay pension contributions even if the casually employed exercise their right to abstain from coverage by the pension system. But the employees can obtain such coverage if they also contribute to the system.

On the spending side, the government decided to suspend the pension adjustment formula based on net wages for 2000 and 2001, indexing pensions to inflation instead. This move was motivated by the fact that net wages were

boosted by the 1999 income tax reform, which under the present adjustment formula – in combination with higher wage settlements – would imply a substantial fiscal burden for the pension system and the federal budget. Also, pension adjustments after 2002 will be affected by the fact that the child benefits will be netted out from the relevant net wage base. Without this change the increases in child benefits scheduled for 2000 and 2002 would boost pension payments in later years.

In addition, the Minister of Labour has suggested two major elements of structural reform. Firstly, employees would be obliged to contribute to a funded pension scheme of their choice. Under this proposal required contributions would gradually rise from 0.5 per cent to 2.5 per cent of gross wages. The proposal would also have spending implications for the pay-as-you-go scheme: since pension adjustments are indexed to net wages the replacement rate in the pay-as-you-go scheme would be lowered to the extent net wages are reduced by the contributions to the funded scheme. While the proposal was not supported by the cabinet, it continues to be an important element in the current policy debate. Furthermore, the Minister of Labour has suggested to introduce into the pension system a minimum pension which is means tested.

Evaluation of progress

Focusing attention too narrowly on the policy goal of stabilising pension contribution rates does not necessarily improve the fiscal conditions of the pension system. Indeed, the present partial change from contribution-based finance towards greater budgetary transfers, which has brought down contribution rates, does not improve the fiscal balance of the system. On the contrary, the task of putting the system on a sustainable fiscal base might have been complicated by linking pension finances to environmental taxes, since the latter are subject to a different set of economic policy considerations, implying a higher degree of uncertainty about the mode of revenue generation. Similarly, temporary *ad hoc* revisions of the indexation formula may make a significant contribution to the budget consolidation effort, as in the present case, but such interference with benefit-defined pension systems can have negative confidence effects and should not replace rules-based pension consolidation measures. Also, it is not yet clear to what degree the inclusion of casual employment and certain types of self-employment will increase revenues. But to the extent a wider circle of contributors will generate additional claims for pension benefits, budgetary pressures are only shifted into the future.

Curbing early retirement should continue to have high priority on the policy agenda. As has been argued in previous *Surveys*, experience in Germany and elsewhere indicates that, unless labour market conditions are excluded as qualifying criteria, invalidity pensions tend to function as substitutes for other

early retirement programmes. It is therefore appropriate to set the eligibility for invalidity pensions exclusively on health grounds without reference to labour market conditions. The revised legislation which relaxed this requirement does not appear to be appropriate in this respect. It has also been suggested that the possibility for early retirement be extended by establishing an additional system which pays full pensions financed by employers' and employees' contributions based on agreements by the tariff partners. This is currently being debated in the "Alliance for Jobs" consultations between the tariff partners and the government (see Chapter III). While it is not clear yet that this approach would produce lower labour costs than the present one, under the fiscal perspective a minimum requirement should be that the supplementary system be actuarially neutral.

The proposal to reduce the pay-as-you-go part of pensions and to supplement it with a funded component appears promising. To the extent funded systems induce higher national savings they can reduce the negative impact of demographic changes. Also, a mixed system can be considered as "diversifying" the specific risks associated with its components: while the pay-as-you go part is sensitive with respect to demographic developments it is largely invariant with respect to swings in capital market conditions. For funded pensions the opposite tends to be true.

Similarly, the proposal to introduce a means-tested minimum pension into the system might help to put the pension finances on a sustainable base. Much depends on the design. Targeting public pensions on those in need would only make sense fiscally if the overall volume of benefits financed out of obligatory contributions or taxes were cut back. Otherwise a means-tested component in the pension system would only increase the amount of redistribution within the system without contributing to fiscal consolidation. The introduction of a means-tested minimum pension would be fiscally advantageous if obligatory contributions and benefits in the overall system were reduced and more scope given to top-up statutory pensions via voluntary contributions. Combining means-tested minimum pensions with a flexible funded layer of the pension system would be a suitable reform option. In view of the redistributive nature of minimum pensions they might be best financed out of general tax revenues rather than wage based contributions. This does not preclude keeping a contribution-based pay-as-you-go segment within the overall pension system.

Developments in the health sector

The health care system: a new round of reforms

The health-care system was discussed at depth in the 1997 *Survey* and a detailed set of reform recommendations was set out. The new government has

moved immediately in the area of health sector reforms and, in line with its overall strategy to reduce social security contributions, has emphasised the need to maintain stable health insurance contribution rates. This will be achieved by a legislated budget ceiling for the total expenditures of the statutory health funds, which will grow at the same rate as the contribution base. Rising demand for health services and increasing costs are to be offset by increased efficiency. At the same time, it has sought to re-emphasise the "solidarity" approach to funding over the "insurance" approach to health care, while aiming to improve the health care quality and to strengthen the rights of patients in seeking redress against faulty practice. Rationing of health care through, for example, waiting lists is not foreseen in the government's plans.

The emphasis given to expenditure control comes at a time of rapid expansion: expenditures by the statutory health insurance funds grew markedly in the first half of 1999 and some projections indicate that the system could finish the year in deficit once again. Particularly worrying has been the exceptionally rapid growth in pharmaceutical expenditures, which seems to be only partly related to a flu epidemic in the first quarter, and in hospital costs. The latter increase is partly related to a high public sector wage increase at the beginning of the year which has not been compensated by efficiency improvements. Overall, the pattern has been as observed in other countries: unless fully implemented, health sector reforms are often successful in stabilising finances for only a year or two before the old growth path reasserts itself. The 1998 *Survey* concluded that reforms were incomplete, with health funds lacking the power to control the costs of health supply, while the system of health suppliers was inadequately integrated and inefficient with excess capacity evident.

On coming to office, the government moved to reverse several reforms introduced by the previous administration. Co-payments for medicines and for other services were removed or reduced, and the possibility for health funds to provide no-claim rebates and to reimburse patients, rather than paying the health provider directly, were abolished. At the same time, payments to the new Länder health funds out of the risk equalisation fund, which were intended as a temporary measure, were extended indefinitely, on the grounds of re-establishing "solidarity". While removing no-claim bonuses might act to restore consistency in a solidarity-based system, rolling back the co-payment measures (which were flanked by provisions to help the chronically sick) removes an incentive to the more responsible use of medicines which did not seem to violate the solidarity principle. Indeed, the rapid rise in pharmaceutical costs in the first half of 1999 may be partly related to the reduction of this incentive. Extending the risk equalisation fund to the new Länder acts as an additional fiscal equalisation.[20] It may have the effect of raising incentives for western-based health funds to take on clients in the eastern part of the country thereby potentially raising competition.

At the heart of the new system, which should come into force at the start of 2000 if both houses of parliament approve,[21] is a global budget for each statutory health fund, as well as for the system as a whole, with the growth of total health expenditures limited by law to the growth rate of the contribution base.[22] The actual budget will be built up in several stages, with an important role for the health funds acting co-operatively at the Länder level: total budgets for hospitals and for doctors will be agreed at the Länder level, rather than at the level of the individual health fund. General practitioners will now receive a fixed budget and their numbers will be restricted to 60 per cent of all registered doctors (GPs and specialists) in that area.[23] The health funds will have greater powers to negotiate directly with groups of doctors and clinics to supply a package of health services without the need to seek permission of the doctors association. However, the funds must act collectively. If an individual fund exceeds its budget it will have two years to correct it and progress in doing so will be overseen by the authorities and by the respective Länder association of the health fund. In the event of advanced warning of an overrun, the authorities and the associations may take pre-emptive action, including appointing consultants to examine the fund.

Pharmaceutical budgets will be benchmarked to the insurance areas with the lowest volumes (corrected for age and some other factors). Pharmaceutical expenditures have long been a major problem in Germany, as elsewhere, and indeed this year the system threatens to exceed the budget limit. The government now proposes to introduce a positive list and to more firmly control prices. If in the future the pharmaceuticals budget is exceeded by doctors, there will be some reparation in succeeding budgets, although this has not proved an effective instrument in the past.

With respect to health suppliers, there are important changes in hospital financing and in the integration of the ambulatory and stationary sectors. After a number of delays, performance-related pricing will be finally introduced for hospitals from 2003, at which time the health funds (acting collectively) will have the right to terminate contracts with expensive clinics as long as a minimal service in the region is guaranteed.[24] The health funds will gradually take over investment and maintenance financing of the hospitals from the states, the process to be completed by 2008.[25] In return they will be able to participate in hospital planning, although ultimate planning authority will stay with the states. With respect to improved integration between ambulatory and stationary sectors, long a problem in the health sector, hospitals will be restricted in their ability to undertake operations normally performed by the ambulatory sector but will be able to provide services in some areas where there is a difficult sickness profile. Doctors in the ambulatory sector will have some rights to follow the treatment of their patients in the hospitals. It is intended that general practitioners will act to some extent as gate-keepers and a bonus will be given to patients who first visit their doctor before turning to a specialist.

Competition between funds and with the private sector will be restricted. In a move to support the regionally based public funds (AOK), health funds which are based on companies (*Betriebskrankenkassen*) will only be able to accept members from the broader public after ten years of operation and all administrative costs must be charged to clients. For the private insurance companies, lower premiums must be charged for older clients (thereby introducing non-insurance elements into this sector), to be financed by a 10 per cent surcharge for all those between 21 and 60 years of age.

Adequacy of the reforms from a medium-term perspective

A key question is whether the proposed health reform places the system on a sustainable path in the medium term or whether, as in the past, it will result in only short run savings with the need for another round of reforms at a later stage. At this stage it appears that the underlying objectives might not be achieved:

- With the well-established tendency for health expenditure to rise faster than incomes, it is doubtful whether a fixed contribution rate represents an appropriate target, although it is a key labour market objective of the government. As argued in the 1997 *Survey*, an option would be to freeze employer's contributions and let consumers decide what they are prepared to pay, but this presupposes channels for choice and for competition which have not been developed. To achieve the stated objective might well ultimately require some restriction through either direct rationing of health services or restriction of demand through incentives such as co-payments. The problem is best illustrated this year by the threatened overrun of the pharmaceutical budget where, if the budget is to be adhered to, some form of rationing will be necessary.
- The health strategy relies on the effectiveness of tight global budgets but it is not clear how effective these might be since sanctions and instruments do not appear to be strong. If an individual fund is in breach of the budget it is subject to peer group pressure in the form of review from other health funds. But unless the fund in question has suffered from quite specific problems with respect to administrative costs, an investigation by other funds is likely to find that general factors affecting all of them are the cause. Such a finding may allow increases in contribution rates. The fact that all health funds must deal collectively with health suppliers makes such an event likely since there is little room for manoeuvre for individual health funds.
- A key element to support fixed budgets is the introduction of a positive list for medicines but the hoped for savings might well prove illusory as

has been the experience in other countries. There is a danger that non-listed medicines will be substituted by more expensive ones on the list and the bureaucratic problems should not be underestimated: a previous proposal for a positive list went over 1 000 pages. A negative list would be more appropriate and simpler. There is a more general problem with binding pharmaceutical budgets: the objective of integrated health care is to save overall costs for a given standard of health care; pharmaceuticals are just one input to health care and if tighter restrictions are placed on them than on other forms of medical treatment, the result may be less effective and more costly health care.

– The reform gives health funds collectively a great deal of formal monopsonistic bargaining power in dealing with health suppliers. But the powers of the health funds to reduce excess capacity among hospitals is limited because the Länder maintain final responsibility for any restructuring. Moves to more closely integrate the ambulatory and stationary sectors appear to be formal rather than substantial.
– Even though individuals are still free to change their health fund, the reform effectively moves the health system towards a universal health fund since the ability of individual health funds to compete is highly restricted. Health funds have greater possibilities to negotiate with groups of health suppliers without the need for approval by the doctors association; but the fact that all health funds need to negotiate collectively could reduce the flexibility of this mechanism, which still relies on local adaptation and experimentation.

Overall, the reforms introduced into the system may be characterised as dirigiste, and it is not evident that a consistent system is evolving, nor that sustainability over the medium term will be achieved.

III. Structural policy: a review

Recent *Surveys* have recommended a broad range of specific structural measures for Germany, aimed at increasing the flexibility of labour markets and enhancing product-market competition. In assessing medium term prospects, the 1998 *Survey* noted that the series of policy measures already introduced, though incomplete, was beginning to show through in higher productivity and employment growth. The assessment was that structural unemployment could be falling, though it remained too high and further wide-ranging reforms were needed to ensure that labour costs and employment flexibility were adequate to make the best use of Germany's substantial human resources. The new coalition government has since reversed a number of recent labour market reforms on which the assessment was based. The reduction of unemployment remains a key priority, but the wide agenda for reform signals a substantial change of approach in certain areas of labour-market policy advocated in the OECD *Jobs Strategy*. The first section of this Chapter reviews the broad shift in policy emphasis that has occurred with respect to labour-market policies. The second section examines the accumulating experience in deregulating the public utility sector and in promoting goods market competition more generally, while the final section presents an agenda for further action.

Tackling high structural unemployment: recent developments in labour-market reform

A shift in policies: seeking a new consensus

The new government is following a different approach to labour market issues from that of its predecessor. The aim of fostering an employment-oriented development of wages was initially interpreted as utilising productivity gains entirely for wage increases in 1999; and while there is agreement on the need to cut non-wage labour costs, this has been seen to entail shifting the tax burden from labour to corporations and consumers, rather than reducing the overall burden of taxation and public spending. Reform of the wage-bargaining system is

not high on the agenda, even though previous *Surveys* have noted that this is an area where there was a possibility to exploit existing spontaneous trends towards more market-based outcomes. Moreover, to an important degree, the approach to unemployment reduction relies on restricting labour supply, via a desire to promote early retirement and to encourage part-time work by older workers. The same aim is behind the desire to achieve a redistribution of labour input between workers by cutting overtime work. There is some continuity in the continued emphasis on active labour market policies, the modernisation of the dual vocational training system and the acquisition of further vocational qualifications; but even here there are differences as to policy approach.

In general, the new policy agenda underlying Germany's *National Action Plan for Employment* is one that preserves existing market institutions, bargaining structures and regulations, combining a cautious approach to innovation with a desire to achieve consensus. The government has not felt constrained to have policy initiatives approved by the social partners before initiating them. However, in an effort to generate consensus, as part of its *National Action Plan for Employment*, it has instituted round table talks between the social partners and the government aimed at developing a joint strategy for employment and growth (the *Alliance for Jobs*). Within the *Alliance* framework, working groups have been set up to address the various topics on which the new strategy will be based. Major issues on the agenda of the Alliance are:

- Fostering part-time employment and flexible working time, reducing overtime work and improving the compatibility of employment and family life;
- creating more flexible opportunities for early retirement;
- fostering an employment-friendly wage policy and lowering non-wage labour costs;
- creating employment opportunities for low-qualified persons;
- modernising the dual vocational training system and strengthening the acquisition of further vocational qualification;
- improving the competitiveness of enterprises via a comprehensive business tax reform.

In July 1999, the Federation of German Unions (*Deutscher Gewerkschaftsbund*) and the German Employer's Association (*Bundesvereinigung der Deutschen Arbeitgeberverbände*) released a joint statement noting, *inter alia*, that a medium-term oriented wage policy of the tariff partners is needed to reduce unemployment, and that productivity gains should be utilised for the purpose of fostering employment. However, no consensus has yet begun to emerge as to the modalities by which this may be achieved. In general, most policy measures remain in the planning stage or are undecided. Whether the current policy process will evolve into a coherent and effective policy programme is thus at this stage unclear.

Increasing wage and labour cost flexibility

While it was too early to say whether improved labour-market flexibility had permanently changed wage behaviour, the 1998 *Survey* was cautiously optimistic that falling unit labour costs suggested that the natural rate of unemployment could indeed be declining. However, with unemployment at 9 per cent in the west and 18 per cent in the east (and the overall level of labour underutilisation in 1998 at 15 per cent if active labour market and related schemes were taken into account), further reforms were needed to ensure a better alignment of wage costs with the level of productivity compatible with full employment. Particular problems attached to the new Länder, where the excessive level of wages relative to productivity biased growth towards substituting capital for labour. While labour-market institutions in the east have been adapting, the pace of adjustment has been insufficient relative to the scale of the problems involved.

The above assessment was made against the background of an informal trend towards greater decentralisation. The German wage formation system is responding *de facto*, if not *de jure*, to the economic pressures confronting not only the new Länder but also the old states. Although the branch-wide tariff agreement remains dominant, a growing number of firms are seeking to negotiate their own wage contracts with the unions or with the work councils. Since 1991 the number of companies with tariff contracts bargained directly between a union and the firm has increased by 70.5 per cent, and this trend continued in 1998 with 9.5 per cent more enterprise contracts in the west and 4.7 per cent in the east.[26] But branch-wide agreements still dominate: in 1998 some 21.5 million employees were covered by branch-wide tariff contracts with firm-specific contracts accounting for 3 million employees.[27] In addition, there is an increasing tendency to unify the branch-wide wage rounds for the new and the old Länder, albeit with some allowances for special conditions. Collectively-agreed tariff contracts are sometimes associated with opening clauses for plant-level agreements, and companies with a small number of employees – notably in the new Länder – may be allowed to pay lower wages. The most important instruments to incorporate company-specific aspects into collective bargaining agreements are given by employment protection and emergency clauses, which allow companies to pay lower wages, provided the level of employment is guaranteed. While these instruments implicitly recognise to some extent a trade-off between wages and employment, they are often confined to situations where the economic conditions of the enterprise have already deteriorated to a critical point.

The deficiency of existing instruments to accommodate the needs of individual companies and their workforce has led an increasing number of enterprises to exit collective branch wage agreements. As a rule, this requires leaving the contracting employer's association. The associations have experienced a secular decline in their membership which has accelerated in recent years.[28] This

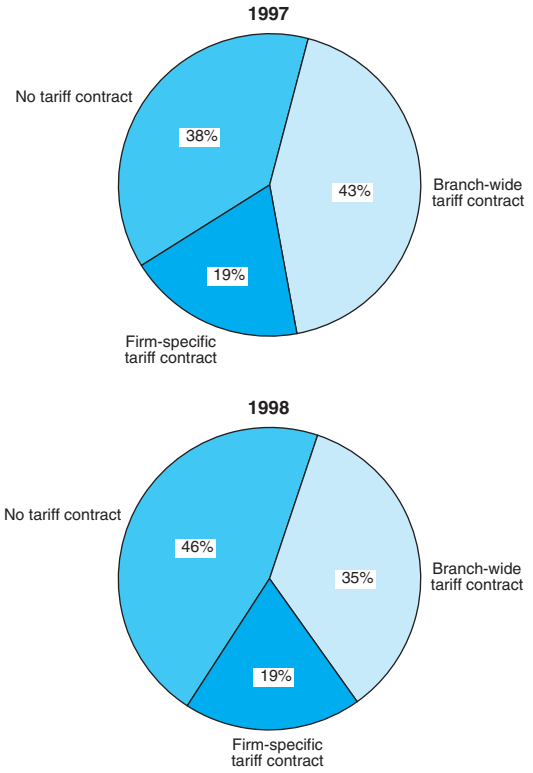
trend appears particularly pronounced in the new Länder which have been subjected to high increases in tariff wages in the past to achieve wage convergence with the west. According to a recent study, the number of employees subject to a branch-wide wage agreement in the private sector declined between 1995 and 1998 by 4.4 percentage points to 67.8 per cent in the west and between 1996 and 1998 by 5.7 percentage points to 50.5 per cent in the east.[29] According to another two-year panel survey of industrial companies in eastern Germany, the number of enterprises which are not subject to a branch-wide tariff contract increased from 38 per cent in 1997 to 46 per cent in 1998 (Figure 12). Two-thirds of companies with a firm-specific or no tariff agreement pay below tariff [30] and correspondingly, in most cases where branch-wide tariff contracts were abandoned wages declined below the tariff norm. The study found that companies which did not apply a tariff agreement break even or generate profits, while most enterprises with losses apply a tariff contract.

Non-observance is also important: one study estimates that 18 per cent of companies in the west and 30 per cent in the east apply plant level agreements which are strictly illegal in that they deviate from collective tariff agreements which the companies are bound to observe. In the absence of opening clauses, enterprises which are bound by tariff contracts are only allowed to deviate from collective agreements in favour of employees, and this has been interpreted by the courts to mean higher pay or shorter working hours. By contrast, lower pay or more flexible working time arrangements are usually accepted by a company's work force because they are expected to lead to greater job security or avoid lay-offs, and often job guarantees are an explicit part of the deal. However, the high level of non-compliance with tariff agreements has led to demands from the unions to give them the legal right to pursue class actions (*Verbandsklagerecht*), and the government is considering changing the labour legislation accordingly.

A ruling by the Federal Labour Court (*Bundesarbeitsgericht*) in April 1999 has been widely interpreted as implying that the unions already possess such a right for a large range of company agreements. In addition, the Court has confirmed the existing legal principle that employment guarantees cannot compensate for lower wages or longer working hours. The court ruling is only with respect to the existing law, leaving the responsibility for any modernisation with the legislators. The high rate of non-compliance with tariff agreements, especially in the new Länder, should be taken as an indication that collective agreements can fail to meet the requirements of individual companies and their work force. Results from the panel of eastern German industrial companies supports this conclusion. Of the enterprises with below-tariff wage payments, three-quarters report that they would either slide into losses or increase their losses if they had to increase their wages to the tariff level. A tenth of the companies with below-tariff wages would consider their viability to be threatened, while four-fifths reported they would respond by shedding labour and one third by revising downwards their invest-

Figure 12. **Tariff contracts in industry in eastern Germany**

Percentage share of firms[1]



1. Number of sampled enterprises: 171.
Source: Institut für Wirschaftsforschung Halle, *Industrieumfrage 1998,* Halle, 1999.

ment plans. Hence, if unions successfully exercised class actions (*Verbandsklagerecht*), a large share of enterprises, in particular small and medium-sized enterprises in the east, would be at risk of becoming entrapped in arrangements with adverse implications on employment and competitiveness. In view of the evolving legal situation, institutional reform which widens the scope for arrangements at the company level has become even more urgent.

Administrative extensions of wage agreements to non-contracting parties (*Allgemeinverbindlichkeitserklärung*) in the new Länder have risen nearer to levels in

the old states, with the number of tariff agreements subject to such extension increasing from seven in 1991 to 163 in 1998, when there was a marked increase. Over the same period the number declined in the west (from 615 to 425), although there was also a pick-up in 1998.[31] While the share of extended tariff arrangements in overall tariff contracts in Germany is small (1.2 per cent) the procedure conflicts with the requirements of more flexible contractual arrangements, especially in sectors such as construction where contracts have been bound but in which the pressures for restructuring are great.

Since 1999, the Minister of Labour can, on his own authority, extend wage agreements to cover employment of foreign workers in the construction sector on assignment from a company in another country. Previously, the approval of the representatives of the employers' associations and the unions was needed for the compulsory extension of tariff agreements by the Minister, and this is still the case in all other areas. In addition, the regulation can now be extended to all wage groups considered in tariff agreements rather than being confined to the minimum wage. As a first application of the new powers, the minister has set an ordinance which declares higher minimum wages in the construction industry compulsory for the purpose of protecting German construction workers against low wage competition from foreign workers, including those from EU countries.

Previous *Surveys* have argued that a greater wage spread at low wage levels is important to create employment opportunities for the poorly-qualified. This question has gained increasing importance in the policy debate and is being considered in the *Alliance for Jobs*. The discussion so far has been largely concentrated on "subsidising" wages at the low end of the wage distribution. The "benchmarking group" of the Alliance for Jobs suggested a subsidy on social security contributions of the lower paid at a regressive rate.[32] While the proposal is still under consideration, a number of studies indicate that a general wage subsidisation for the low paid would imply substantial budgetary burdens if significant employment effects were to be generated.[33] The effectiveness would also depend on whether the benefits were paid to the workers or the firms. Reduced contributions for employees would raise incentives for the unemployed to take up a job, but without a corresponding downwards adjustment of wages this is likely to have little influence on labour costs and demand, and unemployment may be little affected. Similarly, while reducing social security contributions for employers (or granting some other form of wage subsidy) would directly lower labour costs, a larger wage spread would be necessary to limit the implied fiscal burden. It would therefore be inappropriate to create a "low-wage sector" by relying on fiscal measures alone. What is necessary is a wage bargaining system – supported by appropriate changes in the system of social assistance – which improves the chances for the unemployed (or those not in the labour market) with low qualifications to price themselves back into the labour market. Without this precondition, wage subsidies could even turn out to be counter-productive in the

medium term in that they provide incentives for moral hazard on the part of the social partners: bargaining to raise wages in this part of the labour market.

Non-wage labour costs in Germany are high and constitute an important impediment for higher employment by creating a large wedge between labour costs and take-home pay. In addition, marginal effective tax rates for the unemployed can rise to very high levels due to benefit withdrawal. As was discussed in the 1996 *Survey of Germany*, recipients of social assistance face marginal effective tax rates of 100 per cent over a large income bracket as benefits are withdrawn. For certain types of occupations and family status the difference between market wages and social assistance payments might be either low or even non-existent. In accordance with these disincentives for labour supply and demand, studies indicate that the number of social assistance recipients being employed totals only some 10 per cent of recipients who are employable. More generally, these features encourage market-inactivity or black market activity on the side of the unemployed and discourage labour demand on the side of the firms. Reforming the tax and transfer system to increase work incentives and lower labour costs is therefore important.

Increasing working time flexibility and easing employment security provisions

Substantial progress has been made during recent years with respect to allowing more flexible working time arrangements at company level. Such arrangements have also served to support reductions in working time; average hours worked per week, as determined by tariff agreements, continued to decline marginally in 1998, to 37.38 hours in the old Länder and 39.25 hours in the new Länder. Further working time reductions are scheduled for this year and 2000. As a rule, periods of overtime work can be compensated by reduced hours over a time period of one year. Some tariff agreements even allow for longer equalisation periods of up to two years or more, and in the chemical industry the working time for project-related employment can be equalised over a time span of three years. Some tariff agreements also allow for flexible working time arrangements across the work force. In the chemical industry the normal working time of 37.5 hours can vary between 35 and 40 hours according to plant level agreements. Some agreements in the metal and electrical industry foresee working time reductions to 30 hours for the purpose of avoiding lay-offs. The shorter working time can be compensated by permanently longer working times of up to 40 hours for a part of the workforce. Despite this flexibility, the desire on the part of both the unions and the government is for a reduction of overtime in favour of full-time employment.

Arrangements providing for part-time employment of older employees (*Altersteilzeit*) have gained importance in tariff agreements: in the past year the number of such agreements has almost tripled from 70 at the end of 1997 to 200 at

the end of 1998. Since 1996 the Federal Labour Office financially supports the part-time employment of elderly employees by guaranteeing both an earnings level of 70 per cent of the employee's former full-time net income and social security contributions of up to 90 per cent of the wage base, provided a company compensates for reduced working hours by hiring unemployed or newly graduated apprentices. Most tariff agreements on part-time employment for the elderly foresee a further topping up of the wage premium by the employer, and often they determine the distribution of part-time employment over the admissible time span of up to ten years. The number of workers applying for such schemes has been quite small although recent developments are not known.

Job protection was re-tightened in 1999 by lowering the threshold for the application of the dismissal protection law from ten employees per establishment to five. This reverses legislation introduced in 1996, which was gradually phasing in the higher threshold for existing workers. Despite the rather restricted easing, there is some evidence that small firms were encouraged to expand beyond the five employees threshold.[34] Social criteria for selecting those to be dismissed in the event of mass layoffs have also been largely re-introduced.

Reducing the distortions arising from unemployment insurance and related benefits

The new government has also reversed a number of measures, introduced in recent years, which tightened the provision of social benefits.[35]

- Recipients of unemployment insurance benefits (*Arbeitslosengeld*) are freed from the obligation to report to the labour office every three months in order to reduce the associated administrative burden. The maximum commuting time for admissible job offers has also been reduced from three to 2½ hours (full-time employment).
- Redundancy payments (*Abfindungen*) will no longer be credited against unemployment insurance benefits as had been the case since 1997. Redundancy payments are used by enterprises to increase the acceptability of lay-offs and avoid legal disputes relating to the fulfilment of the provisions of the dismissal law. Under the 1997 legislation the indirect subsidisation of dismissals would have been phased out, which would also have increased incentives for job search. The government expects that talks between the social partners will lead to recommendations to wage negotiators to incorporate a substitute scheme involving greater support for training in tariff agreements.
- The legal minimum level for sick pay by employers, paid for the first six weeks of sickness, has been raised to 100 per cent of normal pay. This reverses a provision, introduced in October 1996, which reduced the legal minimum to 80 per cent. However, overtime premia remain

excluded from the wage base (in contrast to the situation prior to 1996). The policy reversal may not have made a big difference in practice since sick pay of 100 per cent has remained a feature of many wage contracts.

Another potentially important change for the unemployed concerns benefits in cases of repeat unemployment spells. As in the past, the unemployed who accept work at wages below those earned in their previous occupation obtain unemployment insurance benefits which are based on the wage in the first job if they re-enter unemployment during a three-year period. However, these benefits could not exceed the net earnings in the last job. This ceiling has been dropped, with the intention of increasing the incentive to accept lower-paid jobs.

The obligation to pay social security contributions has now been extended to casual employment (jobs entailing a small number of hours per week and with pay not exceeding DM 630 in the west or DM 590 in the east) and to the "apparently" self employed. Plans to restrict the range of employment which is exempt from paying social security contributions has been an issue in the policy debate for some years (see OECD *Survey of Germany*, 1998). The government's objective was to shift revenues from the income tax system to the social security funds , extend the coverage of social insurance, reduce the effective subsidisation of small jobs, and abolish the perceived abuse of small jobs as a vehicle by employers to avoid paying social security contributions.[36] For both types of employment, the new regulations have turned out to be rather complex and opaque[37] involving substantial administrative costs for enterprises employing workers on casual jobs and for the tax offices. Several million claims for entitlements to social security-free jobs need to be approved by the tax offices (out of the first 1 million applications, 200 000 were rejected) and employers now need more personal details than before. For those workers who have an approval, little changes for either the employer or the employee. For others, the main effect is that, if the social charges are not compensated by higher wages, the take-home pay would decline by some 22 per cent (net of taxes), forcing workers either to quit such jobs or eventually to demand higher wages.

First reports indicate that employment in small jobs declined markedly after the new regulation had been introduced, mainly as a result of employees quitting. A survey conducted by the Chamber of Commerce Association (*Deutscher Industrie- und Handelstag*) in May 1999, projected that minor employment would decline by 800 000 jobs,[38] and preliminary results from another survey indicate a drop in the second quarter of 1999 by 500 000[39] – although it also recorded a similar increase in the first quarter. There is evidence that this mainly affects persons whose minor employment is a sideline job to their main occupation. It is too early for an assessment to what extent small jobs will be transformed into regular employment.[40] The government now plans to revise the new criteria for

the "apparently" self-employed to improve their applicability. Indeed, clearer and more transparent criteria relating to the conditions under which social security contributions have to be paid are necessary for labour supply not to be discouraged.

Enhancing active labour market provisions and re-entry to employment

Following a period of gradual decline, active labour-market measures (retraining and job-creation schemes) were increased significantly during the course of 1998 by the previous government. Responding to the steep increase in unemployment in the new Länder immediately following reunification, active labour market measures in Germany were largely concentrated on the east. But the role attached to ALMPs has changed over time. Their importance first peaked in 1992 with an average number of 388 000 people in the east participating in job-creation schemes and another 489 000 participants in training measures. With employment stabilising, and based on less favourable assessments of the effectiveness of ALMPs, their provision was gradually reduced to cut government spending.[41] By January 1998 participation in work provision and training schemes in the new Länder had fallen to 177 000 and 125 000 respectively. But this phase was apparently followed by a re-assessment of the role of ALMPs and the introduction of new programmes to cushion high unemployment in the short term. By November 1998 participation in work provision and training schemes in the east had reached new highs of 442 000 and 181 000 respectively. In the old Länder active labour market schemes were stepped up as well[42] (Figure 4).

A major goal of the new government's labour market policy is to prevent a build-up in long-term unemployment, and to reach this goal several criteria for the provision of ALMPs have been eased and outlays have been held at a high level. For 1999 as a whole more people are planned to be placed in active schemes than in the year before. Total outlays for active labour market measures are budgeted to increase by 16.1 per cent to DM 45.3 billion (1.2 per cent of GDP)[43] accounting for 56 per cent of the amount spent for unemployment insurance benefits and unemployment assistance. The measures are also seen as contributions to comply with the employment policy guidelines of the EU. Major changes are:

- Job creation schemes (*Arbeitsbeschaffungsmassnahmen*) have been opened for persons who have been unemployed for at least six months rather than twelve months as has been the case hitherto.
- Persons aged 55 and over, who have been identified as another problem group, can be supported for up to five years. Conditions for obtaining job familiarisation support for older workers (*Eingliederungszuschüsse*) have been relaxed.

- Greater emphasis is given to job creation measures carried out within the public sector – communes, Employment Associations – as opposed to those carried out within firms. "Structural Adjustment Measures" (SAMs, *Strukturanpassungsmassnahmen*).[44] carried out within companies are being concentrated on particularly disadvantaged persons.
- Structural Adjustment Measures that up to now were confined to the new Länder have been made available for the old Länder as well, and new fields have been introduced for the whole of Germany (fostering business- and tourist-oriented infrastructure).
- Provision of support funds (*Überbrückungsgeld*) to unemployed people founding their own enterprise has been extended in scope.
- A new programme to combat youth unemployment has been created (see below).

The experience with ALMPs in eastern Germany has not been favourable, probably due in part to their natural lack of focus in the circumstances: experience in other OECD countries indicates that ALMPs should be tightly targeted at problem groups to be effective instruments for re-integration of unemployed into the regular labour market.[45] The potential for job-creation measures to provide the unemployed or those at risk with immediate occupational experience can be seen as an advantage on its own. But a full assessment has to consider other benefits and costs as well. For Germany, where ALMPs have been applied on a large scale in the new Länder,[46] recent econometric studies based on panel micro data sets indicate that job creation schemes have hardly been effective in bringing people back into regular employment if compared with unemployment benefits. Training measures, on the other hand, appear to have a somewhat better record: although micro-econometric investigations for eastern Germany yield mixed results, there appear to be modestly positive employment effects from participation in training or re-training schemes[47] (Annex I). In addition, the deadweight effects associated with ALMPs can be large, and there is a risk that private sector entrepreneurial activity is crowded out by public work programmes.[48] Moreover, some calculations suggest that the cost of job creation schemes exceed the savings in terms of replaced benefits and additional revenues from higher employment.[49]

Improving skills, entrepreneurial dynamism and competition

Improving skills and technological know-how

As noted in the 1998 *Survey*, adapting the apprenticeship curricula to new professional demands is a crucial element in modernising Germany's hitherto successful dual vocational training system, but progress appears to have slowed.

Thirty new training curricula have become available for the first time at the start of the apprenticeship training year 1999/2000. A further new twenty curricula are to be introduced in 2000/2001. The development of the apprenticeship system is also a topic in the *Alliance for Jobs* talks. In the past, the discussion has often been concentrated on whether binding commitments should be made on the number of training places offered by employers each year. So far the employers' associations have rejected this as unrealistic.

To reduce youth unemployment, a new public support scheme came into operation in January 1999, which aims at making available an additional 100 000 places in training programmes or work provision schemes with training content, to young people who are unemployed or threatened by unemployment. The main goal of the initiative is to increase the chances for youths to obtain vocational training within enterprises. For this purpose, short courses of one to three months duration are offered to those who have not yet obtained an offer for a placement within the dual training system. In addition, apprenticeship places outside of enterprises are being created to compensate for a lack of training opportunities within companies, especially in the new Länder. Supplementary occupational training, extending over one year, is given to young people who turn out not to be suitable for participation in regular training within the dual system. As a final pillar of the programme, financial support is given to regional projects which seek to generate additional training places within companies.

First experience with these measures show high take-up rates among youths, and a considerable number of participants have already completed the short-term schemes offered. However, with 60 per cent of participants still in training, it is too early to assess the success of the programme.

Increasing product-market competition

Subsidies and state aid

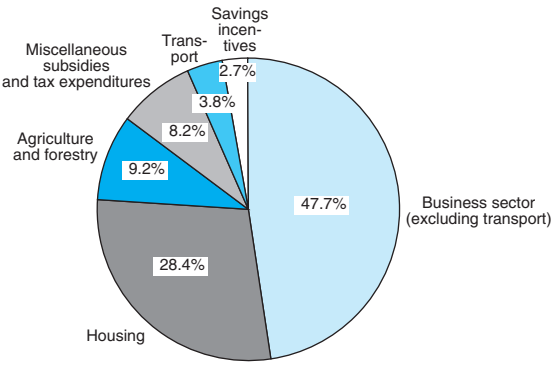
Progress in reducing subsidies has been slow but steady, although the improvement from 1999 onwards will be slowed by higher tax expenditures associated with the preferential energy tax rate for business and agriculture. Total subsidies (defined in the government's subsidy report as direct payments by government and tax expenditures) by all levels of government and the EU are set to decline from DM 120.8 billion in 1997 to DM 115.6 billion this year, of which roughly three-quarters accrue to the old Länder. But only DM 1 billion of this reduction is due to federal and Länder governments: DM 2 billion is due to lower payments by the EU following reforms of the agricultural price support system. The primary reason for such modest progress in domestic subsidy programmes has been an increase in tax expenditures of some DM 3.5 billion, due to the preferential energy tax rate (20 per cent of the normal rate) granted to business and agriculture (see Chapter IV). Subsidies (including tax expenditures) granted

by the federal government are set to rise from DM 43 billion this year to DM 45 billion in 2000, reductions due to the announced tax reform measures being more than offset by higher expenditures associated with the preferential rate for the energy tax. As before, around half the subsidies will accrue to the business sector (Figure 13), the most important single programme being aid to the coal industry (Chapter IV). Federal subsidies for the new states are expected to decline only marginally to some DM 12 billion.

A number of policy issues have arisen in the areas of subsidies and state aid. Germany has proposed that rules governing state aid for industry be retained within the framework of regional development for the old as well as the new Länder. This would mean that 40 per cent of the population would be classed as living in areas needing development assistance, divided equally between new and old Länder. However, although the area and population coverage would appear generous, the expenditures in the west are only approximately one tenth of those devoted to the new Länder. Specific programmes in the new Länder and the old have also come under criticism and review by the EU. These include repeat assistance for some firms in the new states and financial support and guarantees to the state banking sector (*Landesbanken*). The EU has decided that a *Landesbank* received subsidies in the form of excessively cheap capital injections by a state government and has demanded repayment. Despite its stated stance

Figure 13. **Subsidies granted by the federal government**

Percentage share of total[1]



1. In 1998. Including tax expenditures.
Source: Bundesministerium der Finanzen, *Subventionsbericht,* 1999.

to cut subsidies, the government has strongly defended the present system. Another policy issue discussed in previous *Surveys* concerns the government guarantee extended to these banks which allows them to borrow with a triple "A" rating, thereby reducing their cost of capital. The guarantee distorts competitive conditions in the banking industry and could be dealt with in a manner similar to Austria by simply charging for the service.

A number of privatised enterprises in the new states have had difficulty in restructuring and this has resulted in re-negotiations to change their privatisation contracts (*Nachverhandlungen*): in 1998, 850 new cases were opened while 1 250 were completed.[50] In some cases the re-negotiation is required to save the company from closure and since 1997 the BVS (*Bundesamt für Vereinigungsbedingte Sonderaufgaben*), which took over responsibilities from the Treuhandanstalt, has been entitled to contribute financial aid in these cases (*Konzertierte Aktionen*). The BVS has participated in 156 such actions to date in co-operation with the Länder and other parties. Twenty-seven new cases were undertaken in 1998 (less than in previous years), but the cost per case has doubled to around DM 10 million. The mandate of the BVS has now been extended indefinitely and there is pressure from the new Länder for them to assume more control of the organisation. However, it appears unlikely that eastern German companies which cannot stand the test of the market after nearly ten years of support can be turned round further by subsidisation. Rescue operations for enterprises in the new Länder should not therefore be financed via subsidies, but left to banks or other suitable agents in the market. With respect to the general support programme for the eastern Länder, which will remain in operation until 2004, investment subsidies have become more focused on the manufacturing sector and inner city areas (see OECD *Economic Survey of Germany* 1998) but co-ordination between the numerous programmes remains under-developed. Looking ahead, a strategy for a re-orientation of financial aid in eastern Germanys needs to be developed soon. In view of the fact that deficiencies in eastern Germany as a business location have been eased substantially, and some regions in the east are even better equipped with infrastructure than some regions in the west, subsidisation of the new Länder as an entity should not be continued after 2004. Rather, subsidies need to be administered on the basis of rules applicable to the country as a whole.

Liberalisation of the electricity sector

The new law liberalising power generation and distribution has only been in force since mid-1998, but prices for industrial clients have already fallen by some 20-30 per cent. These reductions have resulted from the threat of third-party access to the network, since actual access remains limited. Network owners (who are also generators) have often refused third party access on the grounds of technical problems or on the basis that existing contracts to supply clients were binding. In response, the competition authorities have argued that denying

access would represent an abuse of a dominant market position and have implemented a number of proceedings to test the claims that technical problems prevented access and that contracts agreed under previous monopoly conditions are inviolable. Transmission fees are still widely judged to be too high and will restrict the development of competition.[51] Under pressure from the government to introduce new regulation, the Association of German Industry (BDI) and the Association of Electrical Producers are now negotiating a new agreement which foresees, *inter alia*, markedly lower transmission fees for long distances.

A number of issues are becoming apparent relating to the role of city-based generators/distributors (*Stadtwerke*) and the suitability of the competition framework within which they operate. (Issues arising from the emphasis given to environmental objectives in the liberalisation of the electricity sector are discussed in Chapter IV). The *Stadtwerke* (of which there are some 550, 500 of which are only involved in distribution) are owned by cities or local government, but large generators/network owners often have significant stakes. They are thus for many purposes in a dominant market position. Under the existing law, up to 2005 they can opt to become single buyers for a period and 150 have done so. Despite this seemingly strong position the companies are in financial difficulty leading to demands on the federal government to introduce protection for them. This needs to be avoided. The problem with the *Stadtwerke* is that they are usually small and cross-subsidise other local government activity such as public transport.[52] Their ability to adjust is also limited. They are not permitted to export electricity to other cities and it is often not possible for them to merge because of local and Länder laws. Within the context of a liberalised electricity market, consideration needs to be given to changing the framework conditions under which they operate and putting them on an equal basis with private firms. It may even be questioned whether this is an area in which local government activity is at all necessary.

With respect to competition issues in the electricity sector at large, it seems clear that an aggressive enforcement of competition law is achieving results. However, the competition authorities have gone about as far as they can go and much will depend on how the courts will eventually review the decisions, especially with respect to the validity of contracts. Enforcing competition policy in a market still characterised by dominant companies and by the potential to raise barriers to entry is difficult and the problems have been compounded by the lack of power on the part of the competition authorities to demand immediate implementation. Such powers have been granted to the regulator in the telecommunications sector. Under the existing powers operating in the electricity sector, any order to cease or desist can be immediately challenged and could take some five years to be decided by the courts. In the meantime, competition would remain hindered and competitors suffering from the abuse of dominant market position might even be forced to quit the market. Under immediate implementation, the

dominant player would have to comply immediately pending any appeal to the courts.

Telecommunications

As with electricity, the immediate effect of liberalisation of the telecommunications market since the beginning of 1998 has been a marked reduction (by about 85 per cent) in the costs of long-distance calls using the fixed net. In this area new competitors are active and some are rapidly developing their own nets. Competition has been fostered by liberal regulation with respect to inter-connection fees, by the absence of a requirement to invest in own infrastructure and by a user-friendly access to competing suppliers through the same call number. By contrast, local call charges have not changed significantly, the old monopolist remaining dominant.

There are several grounds why competition has not developed more rapidly in the area of local calls, both structural and regulatory. At the local level the greatest potential for competition to develop is via the broad band cable net. However, at the time Telekom was privatised the government decided to leave the old monopolist in control of the net rather than to establish a potential competitor. Access to the cable net is also under its control, and the regulator has accepted the high prices charged for this on the grounds that, despite low levels of utilisation, investment costs (including over-investment) should be fully passed to clients.[53] No allowance appears to have been made for inefficient use of the net and for the possibility of stimulating competition.[54] There is also a pricing problem with respect to the interconnection of local networks: the "last mile" problem. While the regulator reduced the price from that demanded by the monopolist, it still seems to be a barrier to competition. The rental price for local lines is actually higher than the monthly fee paid by consumers. The decision was also based on cost calculations which accepted a short depreciation period for the network – and therefore high prices. However, the granting of frequencies for local wireless networks should help to establish competition in the local net.

Operating restrictions in the retail sector

Shop-opening hours in Germany remain excessively restrictive in comparison with other OECD countries (Table 13). Several Länder have taken unilateral action to extend shop-opening hours, thereby bringing the issue to a head. To avoid a conflict with the relevant federal legislation, several Länder have restricted longer shop-opening hours to tourist centres and spas or have associated it with cultural events, an approach which they consider to be legally admissible. It has been argued that this initiative leads to competitive distortions within the retail sector. Some Länder have responded to this criticism and have demanded further deregulation by tabling a draft law in the *Bundesrat* allowing a general extension of opening hours.

Table 13. **Shop-opening hours in European countries**[1]

	Weekdays	Sunday
Austria	6 h-19.30 h; Saturday – 6 h-17 h	Exceptions for family business
Belgium	5 h-20 h; Friday – 5 h-21 h	One rest day per week
Czech Republic	No regulation	
Denmark	0 h-24 h; Monday from 6 h, Saturday – 0 h-17 h	Closed
Finland	7 h-21 h; Saturday – 7 h-18 h	12-21 h
France		Closed
Germany	6 h-20 h; Saturday – 6 h-16 h	Closed
Greece	6 h-20 h; summer – 6 h-21 h	Closed
Hungary	No regulation	
Ireland	0 h-24 h	As on weekdays
Italy	7 h-22, maximum – 13 hours	Eight Sundays per year open
Luxembourg	6-20 h; Once per week to 21 h; Saturday – 6 h-18 h	6 h-13 h
Netherlands	6 h-22 h	Twelve Sundays per year open
Portugal	6 h-24 h	8 h-13 h
Spain	0 h-24 h	
Sweden	0 h-24 h	As on weekdays
Switzerland	Canton regulation	
United Kingdom	0 h-24 h	10 h-16 h

1. In some countries such as France, Germany and Austria different rules apply to tourist centres.
Source: OECD.

Assessment and scope for further action

An overall assessment of structural policy reform in Germany has to start with the recognition that much of the previous moderate progress in reform, which was judged by the OECD as an appropriate national adaptation of the *Jobs Strategy*, has been reversed. In combination with enhanced unemployment related benefits and measures which effectively reduce labour-force participation, the employment generating capacity of the economy is not likely to have been improved. In the important area of wage bargaining, policy is based on resisting spontaneous developments towards greater differentiation via proposals to grant wider powers to the social partners to challenge local wage bargaining and to extend the powers of the government to compulsorily extend wage deals to non-contracting parties. As yet, the aim of achieving an effective commitment to aggregate wage flexibility on the part of the social partners (which it could be argued might be at least some (imperfect) substitute for microeconomic structural reforms) remains only a vague statement of principle. It is also unclear what will emerge from the round table discussions taking place in the *Alliance for Jobs*. The OECD view remains that achieving an adequate rate of job creation in Germany requires both greater flexibility for industry to adapt to changing economic conditions and greater

incentives and scope for individual initiative. Box 5 gives a summary of OECD policy recommendations for the creation of modern and dynamic labour and product markets, based on past *Jobs Strategy* reviews and the national policy initiatives which have followed.

In the area of wage setting, there is a need to redefine the legal status and powers of some institutions such as works councils. The scope for plant level agreements on wages and work conditions is insufficient and there is a need to resolve the inconsistency which exists between the official resistance to local wage adaptation and the already-accepted expedient that concessions on wages and working conditions can be bargained against greater job security. Decisions about the precise nature of such trade-offs are best handled at the local level, and in this respect proposals to strengthen the powers of the social partners to challenge local wage agreements need to be restricted to clarifying legal questions. Widening of the scope for compulsory extension of tariff agreements to non-contracting parties is also unhelpful and goes against the principle of preserving tariff autonomy among contracting parties. Unifying the wage rounds for eastern Germany with those for the west, which can be increasingly observed, also does not seem to be appropriate for improving the competitiveness and employment generating capacity of east German enterprises. While this is a decision for the social partners, the government should discourage such a development.

At the same time, attention should be given to reducing benefit withdrawal rates in the social assistance system and to lowering replacement rates where the difference to market wages is low. A great deal of labour market flexibility has been lost with the new rules covering the payment of social security charges for casual jobs and for the "apparently" self employed. It is thus somewhat paradoxical that the policy debate has moved toward stimulating a low wage sector through reducing social security contributions. With respect to small jobs and self-employment the rules for paying social security contributions should be redefined, such that the flexibility in this segment of the labour market is enhanced and no new barrier for self-employment is erected. As for the low wage sector more generally, the tax/transfer system needs to be reviewed but policy here cannot substitute for a wage distribution which increases employment opportunities for the less qualified. A comprehensive approach is required rather than resorting to subsidising social security contributions or wages within an otherwise unchanged policy setting.

The substantial increase in resources devoted to ALMPs and the relaxation of eligibility criteria for participation does not appear to be justified in view of the evidence, in particular from eastern Germany, that large-scale spending on such programmes is ineffective in creating jobs. They are associated with large dead-weight effects and crowd out economic activity and unsubsidised employment in the primary labour market. Active labour market measures should be

Box 5. A review of progress on structural reform

Since the last review a number of policy measures have been implemented and there is a change in approach in certain areas. This box summarises actions taken over the entire period since the original review in the OECD *Economic Survey*, 1996.

Previous proposal	Action taken	OECD assessment/ recommendations
A. Progress in labour market reform		
I. Increase wage and labour cost flexibility		
– Encourage wage differentiation, greater plant-level bargaining and opening clauses.	Minimum wage in construction was declared binding in mid-1997. The scope for the government to extend wage agreements to non-contracting parties was widened in 1999	Wage bargains should incorporate greater flexibility; widen the legal framework covering wage negotiations; eliminate administrative extension of wage agreements.
– Facilitate the employment of older workers and reduce incentives for early retirement.	Early retirement age to be increased; part-time work of older workers is subsidised under some conditions; eligibility conditions for invalidity pensions tightened. Actions reversed in 1999.	Consider further means to raise the retirement age (stricter eligibility criteria, actuarial discounts for pension benefits) and keep transitional arrangements to a minimum.
II. Increase working-time flexibility and ease employment security provisions		
– Reform regulations underpinning inflexible working practices.		Review effects of regulations.
– Liberalise terms for renewing fixed-term contracts.	Renewal period and admissible frequency of renewals increased; scope for secondments extended. Period since extended and approval requirement reduced.	Monitor employment effects and continue with reforms if necessary.
– Facilitate part-time work.	More flexible work arrangements in the public sector; subsidies to part-time work for the elderly.	Avoid further subsidisation of part-time work for the elderly.
– Reform dismissal protection.	Exemption from general dismissal law widened; social criteria for dismissals relaxed. Both reversed in 1999.	Liberalise dismissal protection.

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(continued)

Previous proposal	Action taken	OECD assessment/ recommendations
III. Reduce the distortions arising from unemployment insurance and related benefits		
– Improve the incentive structure of benefits.	Minimum ages for receiving unemployment benefits increased by three years; redundancy payments credited against unemployment benefits, reversed in 1999.	Abolish the extended period of receiving unemployment benefits for the elderly; increase the gap between social assistance payments and the earnings of low-income families.
– Restrict the criteria for not accepting job offers, strengthen penalties, tighten controls.	Qualification no longer reason for rejecting job offers; definition of acceptable wage narrowed; internal auditing introduced of criteria; controls tightened, but reversed in 1999.	Monitor the revised criteria for refusing job offers or taking up employment and narrow if necessary.
– Lower social security contribution rates.	Temporary curbing of certain social transfers; increase in VAT to avoid further rises in rates; pension reform legislation passed (reversed in 1999); pension contribution rates lowered by introducing new eco-taxes; obligation to pay contributions extended to casual employment and "apparently" self-employed.	Curb social spending; consider further measures to secure financing of the pension system (retirement age, basic pension, funding). Keep the flexibility of casual employment and self-employment by redefining the pertinent rules for paying contributions
– Simplify the personal tax system and lower marginal taxes.	Income tax reform laws implemented in 1999. Further tax reform being elaborated.	Proceed with tax reform and lower tax rates.
– Reduce the generosity of sickness benefits and strengthen incentives to avoid abuse.	Statutory sick pay lowered from 100 per cent to 80 per cent, reversed in 1999.	Reduce the generosity of minimum sickness benefits.

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(*continued*)

Previous proposal	Action taken	OECD assessment/ recommendations
IV. Enhance active labour market provisions		
– Exclude measures from the requalifying period for unemployment benefits.	Training periods will not re-establish eligibility for another entitlement period.	Continue reform.
– Target ALMPs at at-risk groups.	More flexible work contracts for formerly unemployed. Communes given greater powers to introduce job-creation programmes and new loan facility given for associated non-wage costs; ALMPs extended (both training and work provision) and eligibility conditions relaxed.	Monitor the efficiency of work contracts for the unemployed, in particular in relation to employment provision measures. Tighten new powers of communes to organise job-creation programmes and seek to avoid simply transferring benefit recipients from one level of government to another. Target ALMPs on at-risk groups and reallocate funds in favour of training. Evaluate all programmes.
– Base employment subsidies on wages below market rates.	Admissible wage base lowered; special conditions for the new Länder phased out.	
B. Improving skills, entrepreneurial dynamism and competition		
V. Improve labour force skills		
– Preserve and restore the attractiveness of the dual vocational training system, clarify its relationship to higher education.	Curricula for apprenticeships revised and new occupations introduced. Public support scheme to create additional training introduced.	Continue to revise vocational training schemes and preserve benefits from training apprentices. Do not introduce a training levy; monitor the efficiency of public support schemes; avoid attaching subsidies to training.
– Shorten higher education and focus it on more occupational-oriented studies.	Agreement in principle on university reform between federal government and states.	Shorten higher education and make it more occupationally oriented. Make universities more performance-oriented. Greater use of fees or student loans.

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(continued)

Previous proposal	Action taken	OECD assessment/ recommendations
VI. Enhance creation and diffusion of technological know-how		
– Deepen financial markets, foster venture capital and reduce regulatory barriers.	Laws implemented to improve flexibility in financial markets, ease access of companies to equity capital and change corporate governance.	Monitor development of risk capital; introduce pension funds. Avoid introducing tax disincentives for venture capital in the tax reform.
VII. Support an entrepreneurial climate		
– Facilitate the establishment of new enterprises.	Accelerated depreciation allowance for newly-founded enterprises; wage subsidies for small new-founded firms.	Continue to reduce regulatory impediments.
– Increase the knowledge of the new owners of enterprises in the new Länder.	No major changes.	Support projects to diffuse knowledge.
– Reduce the overall tax burden.	Abolition of wealth tax and the business capital tax; reduction of the inheritance tax; reduction of household tax but temporary increase in business tax; business tax reform in preparation.	Implement a tax reform with sharply reduced tax expenditures and lower marginal tax rates.
VIII. Increasing product market competition		
– Simplify planning approvals.	Approval procedures simplified and entry into trades made somewhat easier.	Monitor the effectiveness of the new regulations; continue reform and modernisation of law on trades.
– Liberalise shop-closing hours.	Statutory shop opening hours extended in 1997.	Liberalise further statutory shop opening hours.

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(*continued*)

Previous proposal	Action taken	OECD assessment/ recommendations
– Encourage competition in the network sectors.	Energy and telecommunications sectors liberalised and competition is developing rapidly. Competition law widened to cover network sectors and transport. Telecoms regulation established.	Monitor whether regulations to enforce the opening up of networks are effective and promote contestable markets. Support restructuring of community generators and resist demands to protect them. Resist demands to protect the interests of the old telecommunications monopolist.
– Market test services by communities need to be more closely market-tested.	No major changes.	Abolish preferential tax treatment of public utilities.
– Pursue privatisation.	Telekom partly privatised.	Step up privatisation.
– Reduce subsidies.	Subsidies benefiting construction reduced overall but reduction of coal subsidies slowed. Subsidies to east placed on new basis.	Avoid new industrial subsidies; set up schedule for reducing existing subsidies; close tax expenditures. Reform regional support.

confined to high risk groups and not used to reduce effective labour supply, as in the new Länder. Saved financial resources could, *inter alia*, finance a reduction of labour taxation. To raise the efficiency of ALMP instruments, all programmes should be subjected to evaluation studies and funds reallocated in favour of training measures rather than job creation projects. From the perspective of training in general, it is important to secure and further develop the high performance of the German dual apprenticeship system, which requires a further adaptation of skill profiles and a widening of the apprenticeship curricula. A "tool kit approach" which allows for a combination of several shorter training blocks associated with particular skill profiles may be worth considering. Measures designed to stimulate short-term demand for apprentices run the risk that trainees may not move into occupations with the most promising longer-run prospects. Such schemes need to be carefully monitored to minimise the risks and should not give rise to diminishing the role of enterprise-related training.

With respect to product markets and economic efficiency more generally, extensive subsidisation is a major remaining problem, and it is essential that the government moves ahead in its stated policy to reduce subsidies. Regional support for development in the new Länder needs to be based on a common set of criteria applying to the whole country. The other major problems in the competition area are the restrictions on shop-opening hours, which need to be liberalised further, and the remaining barriers to competition in the energy and telecommunications sectors. The federal competition authorities have been active in opening up these markets, but there is a need for action in more specific areas:

- Competition in the electricity sector needs to be promoted through a regulatory framework which, *inter alia*, ensures non-discriminatory access to the grid and no special protection should be extended to the community generators (*Stadtwerke*). Consideration needs to be given to giving them greater powers to restructure, perhaps via privatisation. Environmental objectives in this area should be reached through more efficient means (Chapter IV).
- In the field of telecommunications the government needs to promote competition, especially in the area of local calls. Demands to protect the interests of the old monopolist should be rejected. In order to reduce the potential for conflict of interest the government should dispose of its remaining shares in Telekom as quickly as possible.

The effect of market liberalisation has been marked price reductions in the areas most affected, particularly for business clients, but significant scope for further efficiency gains exists.

IV. Making growth more environmentally sustainable

Introduction

Sustainable development issues have increasingly started to influence policy making in the OECD area, especially since the Rio Conference in 1992, and they have been made a top priority for the work of the OECD. Germany has also been strongly influenced by this trend and indeed since 1994 the concept of sustainable development has been incorporated into the federal German constitution, which recognises the preservation of "the means of life for future generations" as a fundamental aim of the state. The country's future environmental goals are ambitious and far-reaching, based on an impressive performance over the past fifteen years, which have seen marked improvements in air and water quality and a proven degree of flexibility in achieving this. Environmental objectives include reducing carbon dioxide emissions by a quarter by the middle of the coming decade, gradually phasing out nuclear power and further improving environmental conditions in both cities and the countryside. However, the easiest and least costly improvements have probably already been exploited, and herein lies the key economic issue; how to achieve environmental targets at the least cost to existing generations in terms of employment and growth. Care will be needed to ensure that the tradeoffs – or indeed complementarities – between environmental and economic objectives are properly evaluated. This involves a complex set of issues, and especially so since policy choices with respect to climate policy will also depend crucially on the ability to persuade other countries to co-operate and this will influence the type of instruments to be considered.

The chapter examines the economic issues which are emerging as environmental policy shifts from dealing with the effects of local environmental excesses to defining and achieving an environmental strategy for achieving sustainable growth. In the first section it describes Germany's environmental objectives and provides a short review of past and likely future environmental performance. It then describes the political and institutional background, which has resulted *de facto* in a complex regulatory structure in which the links between

environmental and economic efficiency objectives are not only unclear but have often not been fully assessed; particular controversy exists as to the employment effects of environmental policies. Against this background, the third section outlines the economic policy issues and choices that arise in attempting to meet current and future environmental objectives, given that the evidence suggests that some targets might not be reached unless further policy measures are taken and that the associated costs are likely to be greater than in the past. In the same vein, the fourth section analyses important selected sectoral issues in more detail, including policy in the electricity, coal, transport and agriculture sectors. A final section draws together the broader issues and presents conclusions.

Environmental issues: concerns, objectives and performance

Environmental concerns...

Germany is characterised by a high level of political and societal sensitivity to environmental issues and there are a number of reasons for this concern. Population density and industrialisation are high and agriculture is intensive. The transport network is dense with high rates of traffic growth. Germany has become a major transit route for traffic in all directions as European integration has strengthened. There is also a long history of having to cope with trans-boundary problems: the coastline is long; there are borders with nine other countries, and most of the rivers have their sources in other states. Imported air pollution from the planned economies of eastern Europe was a major concern in the past and Germany is presently both an "importer" and "exporter" of pollutants.

Economic growth was exceptionally rapid following the war but by the 1960s, such development was seen as being accompanied, in the popular view, by deteriorating water quality and increasing air pollution, which were considered as a potential threat to human health. Pollution of both the air and the aquatic environment by household and industrial effluents became more evident to the public through dying and damaged forests and by major outbreaks of algae in both the North Sea and the Baltic. At the same time, several accidents in the chemicals industry heightened public awareness – as did Chernobyl. Sensitivity to environmental matters continued to increase through the 1980s and 1990s – reinforced by the environmental situation in the new Länder – with the focus broadening from local environmental problems toward global issues – and global commitments – related not only to greenhouse gases but also to natural resource use more generally. By the mid-1990s fully half the population perceived the environment to be bad or very bad (*Umweltbundesamt* 1997*b*).

... have led to wide-ranging environmental objectives

In response to rising public sensitivity environmental policy has been both broadened and deepened in the 1990s, leading to the adoption of a broadly based notion of "sustainable growth" as an overall policy aim[55] and it is in this context that the current debate about natural resource use is taking place. German environmental strategy is based on the attainment of six goals which are considered as underpinning "sustainable development". They are:[56]

- protection of the climate and the ozone layer;
- preserving an ecological balance;
- reducing the use of natural resources;
- safeguarding human health;
- developing environmentally compatible forms of mobility;
- propagating environmental ethics.

These goals have been widely discussed with a view to setting more specific environmental objectives and targets for achieving sustainable development. The Environmental Action Programme of the EU also lays down specific objectives.[57] A non-exhaustive list of goals and proposed targets is presented in Table 14. So far, the specific commitments of the German government are relatively few, key exceptions being international commitments to the reduction of greenhouse gas emissions, the elimination of ozone depleting substances and the phasing out of nuclear power.[58]

Central to achieving environment goals are both natural resource management, such as the preservation of biodiversity by better land use management and the preservation of forests, and environmental management. Three longer-term rules have been adopted which should form the basis for the complex task of resource and environmental management:[59]

- R*egeneration*: the extraction rate of renewable resources should not exceed the regeneration rate.
- S*ubstitution*: the extraction rate of non-renewable resources should not exceed the rate of substitution with other resources or capital.
- A*daptability*: emission levels should not exceed the rate of assimilation of the environment.

Environmental management covers urban pollution problems such as smog and acidification, which has been responsible for reducing water quality (which in turn threatens biodiversity) and the destruction of forests, avoiding algae plagues in coastal and inland waters (eutrophication) and the avoidance of global warming. To achieve these goals involves both the reduction of greenhouse gas emissions, the most important of which is carbon dioxide, and reductions in the emissions of sulphur and nitrogen oxides into the air and nitrates into the water. For the other greenhouse gases (methane (CH_4); nitrous oxide (N_2O); fluoro carbons (PFCs,

Table 14. **Environmental goals and targets**

Goals	Environmental targets		Targets proposed by other institutions
	German Government	EU	
Preserve the climate	CO_2 reduction of 25 per cent by 2005 (basis 1990)	Kyoto-Protocol: reduction of 21 per cent by 2008/12 within the EU envelope of an 8 per cent reduction.	Diverse proposals, far beyond the political targets.
	UNFCCC: stabilisation of greenhouse gas concentrations.	EAP: stabilisation of greenhouse gas concentration.	
Stop depletion of the ozone layer	Montreal Protocol	Elimination of ozone depleting substances. Exceptions only up to 5 per cent of 1990 consumption.	Reduction of concentration in stratosphere to 1,3ppb chloride (UBA, 1995).
			Prohibition of HCFCs (SRU, 1996).
Reduce eutrophication in rivers and lakes	Third North Sea Conference Agreement to cut nitrates by 50 per cent	EAP (1992): Significant reduction of nitrates in rivers and lakes with more than 50mg/l. Stabilisation and reduction of N-concentration in groundwater by 2000.	Wuppertal Institute (1995): Elimination of chemical fertiliser by 2010; N-emission reduction by 60 per cent for 2010 and by 80 to 90 per cent by 2020.
Lower acidification of rain	SO_x – lower the 1990 emission levels by 80 per cent by 2005 NO_x – expect decline by 19 per cent but no further target	EAP (1992): SO_x reduction 35 per cent in 2000 (basis 1985). NO_x stabilisation on 1990 levels	Wuppertal Institute (1995): SO_2 reduction 80-90 per cent by 2010. NO_x reduction 80 per cent by 2005. NH_3 reduction 80 to 90 per cent by 2010. SRU (1994): NO_x reduction by 80 per cent in 2010.
Improve the urban environment		EAP (1992): SO_x and NO_x as above. Noise levels below 65dB(A) in nightly equivalent levels.	Wuppertal Institute (1995): NO_x and SO_2 as above. SRU (1994): VOC reduction 80 per cent by 2005 (basis 1987). Noise: 55dB(A) daytime, 35-45dB(A) night. UBA (1995): Noise: below 65dB(A) daytime, later below 50dB(A).

Table 14. **Environmental goals and targets** *(cont.)*

Goals	Environmental targets		Targets proposed by other institutions
	German Government	EU	
Preserve biodiversity	10 per cent of area of Germany as an interconnected net of protected areas.	EAP (1992): Interconnected net of protected natural areas.	SRU (1994): 10 per cent of the area of Germany should be protected; 5 per cent without any human interference. Wuppertal Institute (1995): Stabilisation of urban land use.
Decrease material waste	BMU (1997): Product stewardship Management priorities: Avoidance, recycling, disposal. Closed substance cycles.	EAP (1992): Efficient and sustainable resource use. Prevention of waste; Recycling: waste treatment with energy recovery, incineration and deposition. Recycling ratios of more than 50 per cent for paper, glass, plastics. Waste production per capita not beyond 300kg/cap in each member state.	SRU (1996): Preference for material recycling.
Improve water resources, water quality	Quality Class II in rivers.[1]	EAP (1992): Sustainable use of water resources.	SRU (1987): At least a water quality of Class II should be achieved in all water resources.
Preserve forests and fish resources	BMU (1997): Preservation of ecological functions of forests. Sustainable fisheries.	EAP (1992): Preservation of ecological functions of forests. Increased afforestation.	UBA (1995): Preservation of local, diversified forests.
Preserve and restore soil quality	BMU (1997): Minimisation of problematic inputs. Use of land with low impact on ecology.	EAP (1992): Establishment of land use planning for sensitive land areas.	UBA (1995): Reduction of soil erosion to 5 to 10t/ha/a; in the long run stabilisation of soils. Wuppertal Institute (1995): Reduction of soil erosion by 80 to 90 per cent in 2010.

Note: for definitions and sources see glossary and bibliography.

1. Quality class II refers to water which is minimally polluted but suitable for human consumption and a wide range of aquatic life.

Source: OECD.

HFCs); sulphur hexafluoride (SF_6); carbon monoxide (CO) and nitrogen oxides (NO_x), no comprehensive reduction policy with specific targets has been formulated so far, although the "Carbon Dioxide Reduction Inter-ministerial Working Group"[60] has elaborated targets, as has the Enquete Kommission of the German parliament (1995). Nevertheless, despite the lack of clear quantitative targets, a number of measures have been introduced by the government. Great emphasis has also been placed on reducing refuse not only to avoid local environmental problems such as the destruction of the soil and pollution of water supplies, but also to preserve world natural resources more generally.

While the reduction of specific emissions might impact on several environmental objectives, there are also in some instances policy conflicts. For example, reducing NO_x emissions from cars through the use of catalytic converters so as to reduce acidification and improve the urban environment involves greater emissions of carbon dioxide for any given level of traffic; diesel cars may reduce green house gas emissions but can raise the emission of carcinogenic substances.

Progress has been impressive although problems remain

The improvement in environmental performance has been impressive over the last two decades. Remarkable progress has been made in reducing pollutants emitted from stationary sources (*e.g.* industrial and thermal power plants), such as sulphur dioxide (SO_2), particulates and some heavy metals, and large investments in municipal and in industrial waste water treatment plants have led to marked improvements in water quality. Thus by 1993, the OECD's *Environmental Performance Review* was able to conclude that Germany had been able to "decouple" economic growth from the flow of several major pollutants:

- Greenhouse gas emissions, carbon dioxide in particular, fell rapidly between 1990 and 1995 but this development was mainly due to the economic restructuring in the new Länder where emissions have been cut by almost one half (Tables 15 and 16). This special development can be mostly attributed to the process of de-industrialisation although new investment has also played a role by raising energy efficiency. Only traffic related emissions have been increasing strongly due to the catch up process in private car ownership. The most recent estimates indicate that carbon dioxide emissions for Germany have declined at a slower rate since 1995, with increases in 1996 attributed to an abnormally cold winter (Figure 14).[61] However, these figures are preliminary and need also to be interpreted in the light of a slower GDP growth since 1995.
- German water management policies have improved the quality of surface waters, especially in the old Länder. With a few exceptions, Quality Class II (broadly speaking, water suitable for human consumption) has

Table 15. **Carbon dioxide emissions**

Million tonnes

	Old Länder		Per cent change	New Länder		Per cent change	Germany		Per cent change
	1990	1995	1990/1995	1990	1995	1990/1995	1990	1995	1990/1995
From incineration in									
Energy production	275.6	276.7	0.4	163.9	96.5	–41.1	439.4	373.2	–15.1
Industry	120.7	111.5	–7.6	49.0	15.3	–68.7	169.7	126.8	–25.3
Traffic	138.8	144.6	4.1	19.8	26.2	32.1	158.6	170.7	7.6
Households	138.8	157.2	13.3	59.4	28.8	–51.5	198.2	186.1	–6.1
Small enterprises	45.5	41.4	–9.0	24.3	9.5	–60.8	69.8	50.9	–27.1
Others (military etc.)	12.9	10.2	–21.2	7.7	2.3	–70.7	20.6	12.5	–39.6
Industrial processes	22.0	22.0	0.2	5.6	3.2	–42.5	27.5	25.2	–8.4
International air traffic	11.1	13.5	22.2	0.5	0.3	–32.8	11.6	13.9	19.8
Total	708.8	722.1	1.9	305.4	172.3	–43.6	1 014.2	894.5	–11.8

Source: Stein and Strobel (1997).

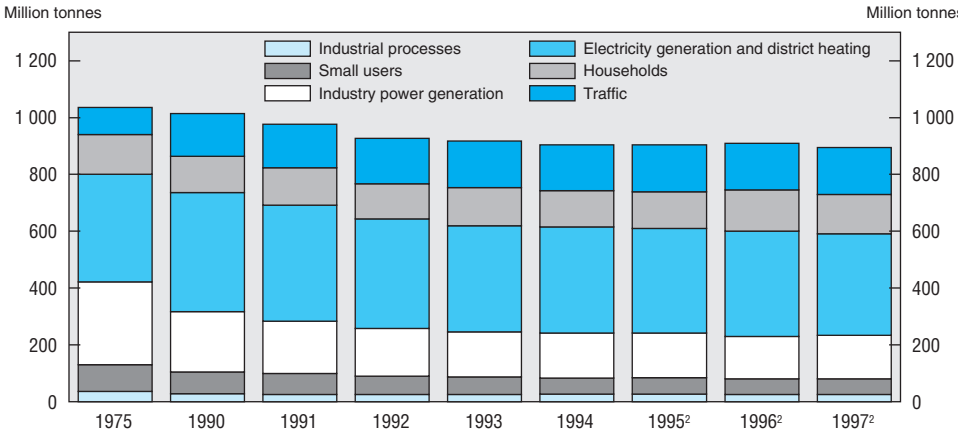
been achieved in most rivers. The most important pollutants such as nitrates, heavy metals and hazardous substances (see Figure 15) etc. have been drastically reduced and ecological functions in many rivers have been restored.[62] Remaining pollutant loads such as cadmium are very small and can hardly be controlled because they come from non-point sources and often occur as erosions from historic deposits.[63]

Table 16. **Emissions of greenhouse gases**

	1990	1995[1]	Forecast for 2005[1]	Change 1990/1995	Change 1990/2005
	Megagrams			Per cent	
CO_2	1 025 000 × 10^3	908 300 × 10^3	883 200 × 10^3	–11	–14
CH_4	5 682 000	4 788 000	3 004 000	–16	–47
N_2O	226 000	210 000	159000	–7	–30
CF_4	355	218	105	–16	–71
C_2F_6	42	27	11	–35.7	–74
SF_6	163	251	186	+54	+14
NMVOC	3 155 000	n.a.	2 700 000	n.a.	–14
NO_x	2 640 000	n.a.	2 130 000	n.a.	–19
CO	10 742 000	n.a.	5 400 000	n.a.	–50
H-CFC	200	2 214	7 991	+107	+3 896

Note: For definitions, see glossary.
1. Including climate policy decided by 1996.
Source: BMU (1997).

Figure 14. **Carbon dioxide emissions**[1]

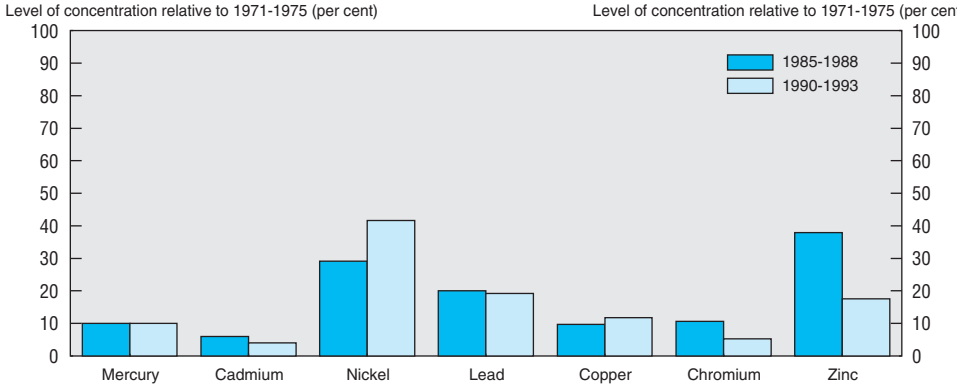


1. Total Germany from 1991.
2. Provisional data (December 1998).

Source: Bundesministerium für Wirtschaft und Technologie (BMWi).

Figure 15. **Heavy metal concentrations in the Rhine at the Dutch border**[1]

Level of concentration 1971-1975 = 100



1. Expressed in mg/kg. Concentrations in the Elbe have been reduced by similar percentages but the initial pollution was much higher so that improvements are still necessary to reach the absolute levels comparable to the Rhine.

Source: Bundesumweltministerium (BMU).

– Emissions of oxides of nitrogen and volatile organic compounds have fallen in the 1980s and 1990s.

Despite these advances, achieving environmental goals is proving to be difficult in a number of areas:

– Estimates of carbon dioxide emissions up to the year 2005 indicate that, taking account of all policy decisions made up to the year 1996, Germany is likely to fall short of its target emission level by a large margin (Annex II). Subsequent policy initiatives, including the energy tax, which became effective in April 1999, have not been sufficient to alter this conclusion significantly.[64]
– Emissions of nitrogen oxides (NO_x) and volatile organic compounds, having fallen in the 1980s and 1990s, are expected to increase again (Annex II). Emissions of both are considered too high contributing to forest damage and to summer smog in urban areas (Box 6).
– Point sources of pollution in surface waters have been effectively brought under control, but non-specific sources remain a problem: about 60 per cent of the nitrogen load in rivers originates from such sources even though there is some tendency for improvement. A large proportion is due to run-offs from agriculture, from drainage and from erosion, as well as, most importantly, groundwater contamination (Table 17). This pollution load from non-point sources apparently requires further and different policy initiatives.
– Although nitrogen emissions were reduced by about a quarter between 1985 and 1995, this fell well short of the 50 per cent reduction requirement demanded by the Third North Sea Conference in 1990. The coastal areas in the North and on the Baltic Sea receive pollutants from rivers, as well as through the air. The nutrient load from rivers and from direct discharges into the sea continue to add to eutrophication problems especially in the Baltic Sea. The main nutrient loads come from agriculture and sewage plants, while for other pollutants, the main sources include air traffic and power stations.

In sum, with respect to environmental quality, significant progress has been achieved in many areas, and projections show important improvements in comparison with the *status quo ante*. However, significant slippage from some policy targets is apparent, for example, carbon dioxide emissions and nitrate pollution in rivers. In other areas the targets, even if achieved, could be insufficient to meet environmental goals with respect to urban air pollution and forest preservation. Moreover, further challenges need to be faced: the intention to close nuclear reactors eventually must make, *inter alia*, carbon dioxide reduction targets more difficult to achieve, while national targets will need to be further tightened in the decades ahead if the global concentration of carbon dioxide is to be stabilised

Box 6. Despite lower levels of pollution, damage to forests has not been reduced significantly

The damage to forests (*Waldschäden*) has been monitored since 1984, when the problem was already widespread with half of all trees showing symptoms of environmental stress. Until the early 1990s, the proportion of trees without any damage (category 0) was falling and that of trees with significant and severe damages (categories 2 and 3), and dead trees (category 4) was increasing. Since 1991, data for unified Germany show a decrease in damage. However, this is mainly due to a significant improvement in the new Länder, where apparently the immediate emission impact on trees has been reduced due to the closing down of industrial emission sources, a strong reduction in the use of brown coal in domestic heating, and modernisation of the capital stock. In the old Länder damage is still rising in some states, mostly in the north-western part of the country.

Complex chemical reactions between different pollutants subject to sunlight contributes to forest damage. The most important pollutants are sulphur dioxide (SO_2), nitrogen oxides (NO_x), ammonia (NH_3), and ozone (O_3). In combination, they damage trees directly by reducing the functions of leaves, or by accumulating substances in the soil which then lead to long-run damage of the roots. These substances are removed from soils only slowly, so that a reduction in emissions will only result in a very gradual diminution of damage from contaminated soils.[1] Despite the emission reductions achieved so far, the deposit levels in forests remain too high. Acidification and eutrophication, through sulphur and nitrogen emissions, remain the major contributors to damage. According to the German government, in 1993 only 16 per cent of the forest area was subject to acidifying depositions below the critical load; for nitrogen eutrophication, only 5 per cent of the forest area is below the critical level.[2] Although emission reductions from national and foreign sources have, been significant, harmful effects of pollutants accumulated in the soil are long lasting.[3]

1. BML (1996*b*), (1997).
2. BML (1997).
3. For the year 2005 a further reduction to 0.26 million tonnes of dust, to about 0.55 million tonnes of SO_2, to 1.74 million tonnes of NMVOC (see glossary) are expected. The reduction of nitrogen oxides (NO_x) is only expected to reach 2.14 million tonnes in 2005 (BML, 1997).

Table 17. **Nitrogen emissions into rivers**

1995

	Thousand tonnes	Per cent		Thousand tonnes	Per cent
Non-point sources	460	60	**Point sources**	315	40
of which:			*of which:*		
Precipitation	20	3	Industry	60	7
Agricultural run-off	20	3	Treated rain water	20	3
Drainage	45	6	Waste water from households	235	30
Erosion	45	6			
Ground water[1]	330	42			

1. 90 per cent of which from agricultural sources.
Source: BMWI (1996).

(OECD 1999*a*). These environmental policy issues raise questions which go beyond simply tightening existing policies: how should policies and targets be formulated in the most cost effective manner? The question involves not only issues of economic principle but also ones of institutional, legal and political arrangements since at the end of the day, judgements about the benefits of environmental targets – and often about the costs involved – are highly subjective.

The framework of policy formation

Regulations versus economic instruments

The cost-effectiveness of environmental policies

A consistent theme of the discussion below is whether the benefits of environmental policies, in terms of improved environmental performance, have been achieved in a cost-effective manner. On the one hand, the rationale for government intervention is clear: the social costs of economic activity are not taken fully into account in current relative pricing – transport is one example, coal-fired energy consumption is another. In principle, the economic answer would be to incorporate social externalities into prices, and attempt to reach the "right" level of environmental quality (or pollution) at lowest cost by equalising the marginal (aggregate) damage to marginal abatement costs. This would ensure both that polluters pay in proportion to the damage they cause and that resource allocation was based on decisions where environmental consequences were fully taken into account. But, on the other hand, such externalities may be difficult to measure – indeed, the concept of marginal damage is far too static for a situation where "sustainable development" could be threatened by cumulative environmental damage. And incorporating social externalities into prices might have important short-run costs in terms of economic dislocation. In practice, therefore, as the discussion below indicates, the setting of specific targets and objectives for environmental policy has taken place mainly through the political process, involving most interest groups in the economy. The process is not an explicit one since even at the project level – with the exception of some transport projects – formal cost-benefit analysis has played very little role compared with legal and regulatory considerations, of the administrative courts in particular.[65] The policy issue which needs to be considered is whether the process has effectively related the benefits of environmental policy to the costs.

Nevertheless, even where policies addressing specific environmental problems are not optimal in the sense of being allocationally efficient, they can be made individually cost-effective. Cost minimisation requires solely that the marginal abatement cost from all sources of pollutants or emissions is equalised.

If it is not, resources could be diverted from sectors and activities where it is relatively costly to comply, offering potential efficiency gains. This is particularly important in Germany where environmental policies cover a wide range of sectors and activities (Table 18), each of which might be associated with differing marginal costs of abatement, including monitoring costs. Indeed, even within broadly defined sectors, marginal abatement costs could differ widely between plants and industrial processes. The instruments being used to achieve environmental targets being mainly regulatory, they have been highly specific to each sector and even to individual processes. The issue is whether there is an integrating system which takes into account the time-varying abatement costs in other sectors or sources of emissions. Without this process, advances in environmental performance might be largely borne by particular sectors and activities which are easy to regulate or control but not necessarily those associated with the lowest marginal cost.

In theory, regulations could be designed so as to achieve an efficient outcome, but the necessary information to achieve this is rarely available. In the German case, as will be seen below, regulations have been implemented in close consultation with polluters and have often been so differentiated that marked variations in marginal abatement costs across sectors and sources might have been avoided. Such an outcome can be facilitated where many of the actions implemented belong either to "no-regret" strategies or to policies implemented in an environment where marginal abatement costs are in any case low. Several German studies[66] have found that resources may not be used efficiently, due to a lack of information, education, or institutional/market failures. In this case, environmental policy measures can be introduced without imposing welfare costs on the economy resulting in a "win-win" situation. Such "no-regret" options can be realised where the actors concerned receive information about the profitability of

Table 18. **Key environmental targets and economic activities**

Economic sectors or activity involved	CO_2 and climate change gases	Air pollution including SO_x and NO_x emissions	Water	Soil	Biodiversity
Agriculture	**	*	***	***	****
Consumption/household	***	*	*		**
Industry	***	***	**		*
Energy generation	***	***	*		**
Transport	***	***			**
Waste disposal	***	**	***	***	*
Imported		**	**	*	

Note: Illustrative level of importance *** = highly important; * = less important.
Source: OECD.

such actions or where, for example, best technology has already been discovered by others and is now enforced by regulation. The implications are that information policies may make an important contribution to reducing the marginal costs of pollution reduction: carbon-dioxide emissions are a case in point where, until 1997, cuts were relatively easy to make (see below). Voluntary agreements are usually predicated on the assumption that certain anti-pollution gains are easy to make and that the sharing of information is important.

The U*mweltbundesamt* has conducted several *ex post* cost-benefit studies of environmental policies and concluded that the returns from environmental protection (the ratio of costs to overall benefits) ranged from 1:1.5 for noise abatement to 1:20 in the case of desulphurisation of power stations, depending on the instruments chosen.[67] High rates of return might indicate a period where "no regret" measures were possible or where marginal abatement costs were low. The above studies do not, however, shed light on whether the most cost-effective instruments were in fact adopted (*i.e.* whether the same goals could have been achieved at even lower cost) and whether costs for the economy as a whole have been minimised. Moreover, even where regulations have been cost effective, the prospects both of higher marginal abatement costs in the future and of a tightening of environmental targets have focused attention on the correct balance between regulation and economic instruments, such as ecological taxes and/or tradable emissions permits to achieve future environmental goals. Under such circumstances, regulations could need to be adapted and co-ordinated and greater emphasis given to policies which make greater use of economic instruments to achieve an equalisation of abatement costs. Economic instruments are not a complete substitute for regulation; indeed, there is often strong complementarities. However, the regulatory system does need to be compatible with a greater use of economic instruments where these are appropriate.

Links to economic objectives

In allocating to economic instruments a larger role in meeting environmental targets, the issue of efficient markets comes to the fore. Thus far, subsidies and tax exemptions (for car-transport expenditures, subsidisation of agriculture and coal-mining etc.) and competition-inhibiting regulations have been used to try to meet employment and competitiveness objectives while often ignoring any environmentally harmful side-effects. There have thus been inconsistencies between macroeconomic policy objectives and ecological ones. In moving ahead towards a more cost-efficient environmental policy, clearer and more consistent objectives will be required across all public programmes. Instruments which are targeted toward the source of emission or environmental damage can only be effective in markets where there are proper economic incentives and adequate price signals. In this sense, economic and environmental efficiency are mutually reinforcing. Indeed, where competition is constrained and individual markets are

distorted by regulation and government intervention, economic instruments such as ecological taxes cannot be clearly focused and there is a danger that policy may risk falling short with respect to both economic and ecological objectives.

In the process of shaping ecological instruments to meet economic as well as environmental goals, attention in Germany has turned to the specific proposition that the revenues from ecological taxes can be used to achieve higher employment by reducing the tax burden on labour (and by raising the relative price of the environment). Fundamental to the strategy of the government is the judgement that there is a potential for a "double dividend". Two studies ("DIW/Greenpeace Study" and the IEW study) find strong positive employment effects from such a policy. Other studies are less sanguine about positive employment effects and the possibility of a "double dividend". A broader study, which looks at German environmental policy in a European context and examines the role of policy co-ordination, finds some potential for a double dividend in the EU; but there are numerous uncertain influences which might alter the sign of the welfare effect. Employment could rise but not to any great extent and not by enough to solve the unemployment problem. By contrast, two other studies come to the opposite conclusion: that an ecological tax increases the unemployment rate. The analyses undertaken relate to different types and levels of efficiently designed environmental taxes and it is worth noting that the implied carbon dioxide tax is often high (as are the revenues), a result confirmed with the OECD Green model.

The "double dividend" controversy, which revolves, in essence, around whether a move to ecological taxes (which are borne partly by inactive members of the population) can shift the incidence of taxation away from labour, illustrates the extent to which environmental policy effectiveness depends upon a policy setting which is not distorted by non-environmental sectoral considerations. Environmental effects and economic benefits will be the more significant where marginal-abatement costs are equalised, as discussed above, which requires a non-discriminatory approach. In this respect, attempting to graft a sectorally-focused ecological tax regime (*i.e.* one that differentiates rates among polluters and emissions) onto a selective regulatory system would tend to reduce the possibility of a double dividend occurring. The following discussion, which tracks German environmental policies through from the creation of a complex regulatory system to the introduction of the energy tax system in April 1999, is based on the proposition that achieving a significant "double dividend", in terms of both employment and environmental performance, depends on the evolution towards a broad-based and mutually consistent approach to the use of economic instruments.

Implications of a federal institutional structure

The institutional framework has exercised a powerful influence in the setting of policy objectives and targets, as well as in the choice of instruments,

and proposals for future policy directions need to take the factor into account. Germany is a federal state, with the power to determine environmental policies divided among the federal, state (Länder), and local authorities. Following an amendment to the German constitution, the federal government has obtained "concurrent" legislative power in the areas of waste management, air pollution control, noise abatement, protection from radiation, and criminal law related to environmental protection matters. In these policy areas federal regulations supersede state and local decisions. In the case of water management, regional planning, nature conservation and landscape preservation, the federal government can only issue so-called framework laws, which oblige state governments to pass supporting legislation. Federal legislation and ordinances may, in turn, be constrained by minimum standards defined at the European level if such requirements exist. In those cases in which the federal government has only "framework competence", co-ordination and subsequent legislation at the Länder level is necessary. The Länder legislation often defines the framework in which local authorities implement environmental policies. Especially in those cases where direct regulatory instruments are used, substantial discretion remains at the local level.

Within the federal government, resource and environmental policies are not exclusively assigned to the Federal Ministry for the Environment. This is, in part, due to the fact that resource or environmental policies have historically been allocated to ministries within whose domain such policies impacted. For example, in the case of water management, drinking water quality and supply policies are dealt with in the Ministry of Health, while water management in the rural sector, as well as flood protection and coastal zones management, is in the responsibility of the Ministry of Food, Agriculture and Forestry. Waterways are managed by the Ministry of Transport and energy policy is under the auspices of the Economics Ministry. Such an allocation of responsibilities may enhance the co-ordination between environmental and other policy objectives, but it requires intense efforts to achieve consistency between environmental policy instruments in different areas. Each Ministry has its own environmental group and there is a dense network of inter-Ministerial Committees although how co-ordination works in practice is unclear. Co-ordination might be improved by underpinning decisions through formal cost/benefit or cost effectiveness analysis.

Influence of EU regulations

Community-wide environmental regulations in the EU increasingly both stimulate and constrain the ability of the federal and the Länder legislatures in formulating their environmental policies. In the case of air pollution, minimum ambient air quality standards for several substances (SO_2, VOC, Lead, NO_x) have been the subject of several directives.[68] Although the management of water

resources and water quality contains local as well as trans-frontier aspects, the European Commission issued several directives on the preservation of water resources as early as 1972. Ambient quality standards have been set for many types of water resources, including monitoring and information requirements for national governments, which effectively limits the principle of subsidiarity. One of the major concerns of the European Commission is the possibility that national environmental regulations may limit the free movement of goods in the internal market. For example, in the case of waste management policies, Germany has been criticised for choosing waste policies which could create barriers to entry for

Box 7. The role of the EU in environmental policy

The European Community now has an important influence in defining the options for and the implementation of environmental policy in Germany. At the end of 1997, 212 directives, 43 regulations and 77 decisions of the EU were in force.* Until 1987, EU environmental policy had no explicit legal foundation, so that measures were based on internal free market requirements. However, since the Amsterdam Treaty entered into force in 1997, the EU has at its disposal several means for policy intervention: the common environmental policy, single market policy and common policies with side effects on environmental issues. Each mechanism involves different decision making processes, including different powers for the EU parliament.

The *common environmental policy* recognises the importance of sustainability and is subject to the subsidiarity principle. The parliament has co-decision making powers. Decisions covering taxes such as a carbon dioxide tax and national energy supply are exceptions and can only be decided unanimously. Member states are allowed to adopt stricter measures but these have to be notified to the Commission. The Commission has moved to specify national minimum standards (such as for water quality and energy taxes), which serve to restrict the principle of subsidiarity.

Single market based environmental policy is made on a qualified majority voting basis and in co-decision with the European parliament. Individual countries are allowed to retain or adopt stricter standards if these do not restrict trade between EU members and if they are based on new scientific information or on specific country problems. However, the Commission decides on the legitimacy of the standards, so that the possibility for countries to react to national problems is in practice restricted. Community policy competencies are not limited by explicitly specified supra-national environmental problems.

Common policies in areas such as *transport, agriculture, energy and regional development* have to take into account the requirement of environmental protection, especially the aim of sustainability. However, majority voting in some cases, and unanimity in others can weaken this mechanism, especially if environmental objectives would call for fundamental changes in sectoral policy approaches such as in agriculture.

* Welfens and Hillebrand, 1998.

foreign suppliers of consumer goods.[69] Similarly, the scope for structuring energy taxes is constrained by the need not to discriminate, directly or indirectly, against energy imports from other member countries.[70]

The EU framework tends to favour regulatory instruments and minimum standards (Box 7) and forecloses a number of policy options. Regulations and directives are often preferred because the competence of the Commission is more easily exercised in this area than that of taxation, and decisions require a simple majority. But majority voting also makes it unlikely that very restrictive standards are adopted. On the other hand, EU-wide economic instruments such as environmental taxes require unanimous voting which in a community characterised by widely different interests makes it less likely that they will be adopted. The failure to reach agreement on the Commission's carbon dioxide/energy tax proposal is a case in point.

A complex regulatory structure...

The effects of the above institutional structures have been to create a highly complex regulatory system with minimal use of economic instruments. There is no single environmental code in Germany (U*mweltgesetzbuch*), but rather a tangle of laws and regulations which are sometimes inconsistent. The initial emphasis on public health, which coincided with a strong belief in state control and in social planning, led to legal and regulatory measures which increased in complexity over time as new issues were identified and handled individually. This also tended to lead to an emphasis on sector specific policies, such as those focused on power plants, rather than on policies aimed at emissions from all sources, or at the cause of environmental problems more generally. Many environmental policies (air pollution control, many water quality policies) rely on a direct regulation of emissions. The laws and directives involved often require best available technologies to be used in a particular plant. The local regulatory agency now has only limited discretion in deciding which abatement activities are appropriate. To be economically efficient, direct regulation requires considerable co-ordination vertically and horizontally among government agencies, which in practice is scarcely possible. Co-ordination issues are reduced by economic instruments.

... based on precautionary action, co-operation and the polluter pays principle

In formulating specific policies and targets, three basic principles have been utilised: precautionary action; co-operation, and the "polluter-pays". The precautionary principle has been a key element since 1971, when the federal environment programme was updated. It is important since it has been interpreted as requiring the *ex ante* regulation of potential emission sources. This has,

in turn, led to regulation which specifies (requires) best available technologies in terms of emission prevention, at the end-of-pipe stage as well as in the integrated production processes. Emission standards are also based on this principle. The second principle, of co-operation, requires a policy approach that emphasises consensus between the affected parties, especially the emitter and

Box 8. Voluntary agreements/arrangements in environmental policy

Voluntary agreements have been used extensively in Germany to implement environmental policy especially with respect to climate change and waste disposal: there are some 100 such agreements and only the Netherlands has anywhere near this number of agreements in OECD Europe.[1] The agreements have been negotiated between industry associations and the government and usually concern the specification of pollution abatement objectives (i.e. they are target-based). However, in the waste sector there are also implementation-based agreements, which specify measures to achieve targets set by legislation. In Germany the agreements are not legally binding since the government has no legal powers to sign agreements with associations. However, there has often been a threat of additional regulatory action in the event of no agreement or if its aims are not achieved.

The motivation for the government in seeking voluntary agreements include the desire to hold down implementation costs and to preserve international competitiveness. They are also viewed as useful in new areas of environmental policy where laws and regulations are still untried. Moreover, co-operation between government and industry has a long history. For industry associations such agreements offer the prospect of raising flexibility in comparison with regulation. There may also be significant "soft effects" in terms of dissemination of information and awareness raising.

There are also significant negative features associated with such agreements which are noted in this *Survey*. Agreements which set environmental targets are not likely to be ambitious and exclude the interests of other members of society. Indeed the OECD has concluded that their environmental effectiveness is generally low. This conclusion is also applicable for Germany. For example, the agreement to reduce fuel consumption of cars roughly corresponds to what in any case appears to be the trend while the agreement with industry to lower CO_2 emissions appears to set objectives below the business as usual path. The agreement to take back old cars free is also of little ambition since the cars involved still have a market value (see below).[2] In other cases discussed in this chapter, voluntary agreements can be used as a barrier to entry against new rivals thereby reducing competition and welfare. Agreements often attribute identical targets to individual firms rather than concentrating abatement where costs are lowest. This reduces their cost effectiveness relative to the use of economic instruments.

1. See *Voluntary Approaches for Environmental Policy In OECD Countries: An Assessment*, OECD, Paris, 1999.
2. K. Rennings *et al.*, "Voluntary Agreements in Environmental Protection: Experience in Germany and Future Perspectives", *Business Strategy and the Environment*, 6, 1997.

the regulating agency. The third principle, that the "polluter pays", has been interpreted along the lines of the OECD as meaning that the cost of avoiding or remedying environmental pollution must be borne by the "party responsible for its occurrence".[71] The principle has been applied in so far that emitters are obliged to pay for the cost of the technology or other actions demanded by the regulator but the coverage is not complete. For instance, Germany has adopted a number of direct and indirect subsidies to promote environmentally-sound practices in the private sector, especially in agriculture (see below). However, the polluter-pays principle is rather limited in that there is no requirement to compensate society as a whole for the damage imposed by an emitter from ongoing pollution.

A resulting key feature of the institutional setting is the role assigned to voluntary agreements and the generally close relations between polluters and the regulatory authorities, both of which stem from the emphasis on co-operation in determining policy. Voluntary agreements (Box 8) cover areas as diverse as carbon dioxide reduction by industry, the phasing out of fluorocarbons, taking back of batteries and the disposal of old cars.[72] Indeed, the law covering waste disposal foresees voluntary agreements and the EU has also recommended their adoption. In addition, voluntary agreements are also being used to promote clean technologies such as agreements on the reduction of car fuel consumption and the promotion of the three litre/100 km car. The availability of this close negotiating mechanism has limited policy options (see below) and may account for the hostility shown by industry to environmental taxes which need to be approved by the parliament and might thus be vulnerable to more general fiscal pressures.

Public outlays

Environmental policy has involved significant outlays, although individual indicators should be treated with care. Pollution abatement and control expenditure in the old Länder as a share of GDP is around 1½ per cent and is rather high among OECD countries (Table 19) while the capital stock devoted to environmental protection is now also significant (Table 20). However these expenditures refer only to end-of-pipe costs ("additional" costs) and not to equipment integrated into production processes to reduce emissions or to lower consumption of resources. More importantly, they exclude the significant expenditures which have been undertaken to improve environmental performance in the new Länder. The volume of expenditures in the new states is not known with certainty and many projects have been undertaken in conjunction with work-creation schemes which has served to raise costs. In addition to the re-equipment of power plants and the installation of water purification facilities – both of which have led to significantly higher user charges in the east than in the west – significant expenditures have been devoted to the rehabilitation of polluted industrial and military

Table 19. **Environmental expenditure on pollution abatement and control**

Per cent of GDP, mid-1990s or latest available year[1, 2]

	Public and private sectors	Public sector only
United States	1.6	0.7
Japan[3]	1.1	0.9
Germany (western)[4]	**1.4**	**0.8**
France	1.4	1.0
Italy	0.9	0.5
United Kingdom	0.3	0.4
Canada	1.1	0.7
Australia	0.9	0.6
Austria[4]	1.7	1.0
Denmark	0.9	0.6
Finland[5]	1.1	0.6
Greece	0.8	0.5
Hungary	0.7	0.3
Iceland	..	0.3
Netherlands	1.8	1.3
Norway[6]	1.2	..
Poland	1.2	0.3
Portugal	0.7	0.6
Spain	0.5	0.5
Sweden	1.2	0.8
Switzerland	1.6	1.0

1. Figures are not necessarily comparable, since methodologies and coverage differ across countries.
2. Data coverage across countries differs from year to year. In general, changes have been rather modest in the time period covered (1985-97); with some exceptions (which could be the result of changed data coverage, etc., and not necessarily changes in PAC expenditures).
3. Data on business sector current expenditure not available.
4. These data are not comparable with other data because the definitions and methodology used are different. The OECD estimate for public and private sector PAC expenditure is 1.7 per cent of GDP.
5. Includes an estimate for public sector PAC expenditure.
6. OECD estimates for 1990.

Source: OECD, *Environmental Indicators,* 1998.

Table 20. **Expenditure on environmental protection in the old Länder**

In DM billion (in nominal prices)

	1975	1980	1985	1990	1994	1995
Environmental protection assets, government-owned[1]	72	124	165	220	305	316
Environmental protection assets belonging to the manufacturing industry[2]	22	35	51	83	105	107
Environmental protection investments by the government	4.7	8.1	6.8	10.4	10.8	10.6
Environmental protection investments by the manufacturing industry	2.5	2.7	5.6	7.2	4.8	4.2
Current environmental expenditure by the Government	3.0	4.7	6.4	9.3	13.5	13.5
Current environmental expenditure by the Manufacturing industry	3.2	5.2	7.9	11.2	14.0	14.8
Environmental expenditure as a proportion of GDP	1.3	1.4	1.5	1.6	1.5	1.4

1. Publicly-owned environmental assets, 1995: 91 per cent water protection, 7 per cent waste disposal.
2. Privately-owned environmental assets, 1995: 55 per cent water protection, 29 per cent waste disposal.

Source: *1998 Report on the Environment,* BMU (1998).

sites.[73] Whether these projects made sense in terms of meeting environmental objectives in a cost effective manner or whether they were primarily driven by local labour market conditions is unclear. An indication of the total outlays involved in both the new and the old Länder can be gained from employment statistics: a joint survey published in 1996 by the four leading economic institutes concluded that some 956 000 people were employed in environmental protection in 1994, about 2.7 per cent of the workforce or nearly the same share as vehicle manufacturing.[74]

Policies in force and directions for change

With the economic consequences of environmental policy in mind, it is useful to separate the instruments currently applied into five categories:

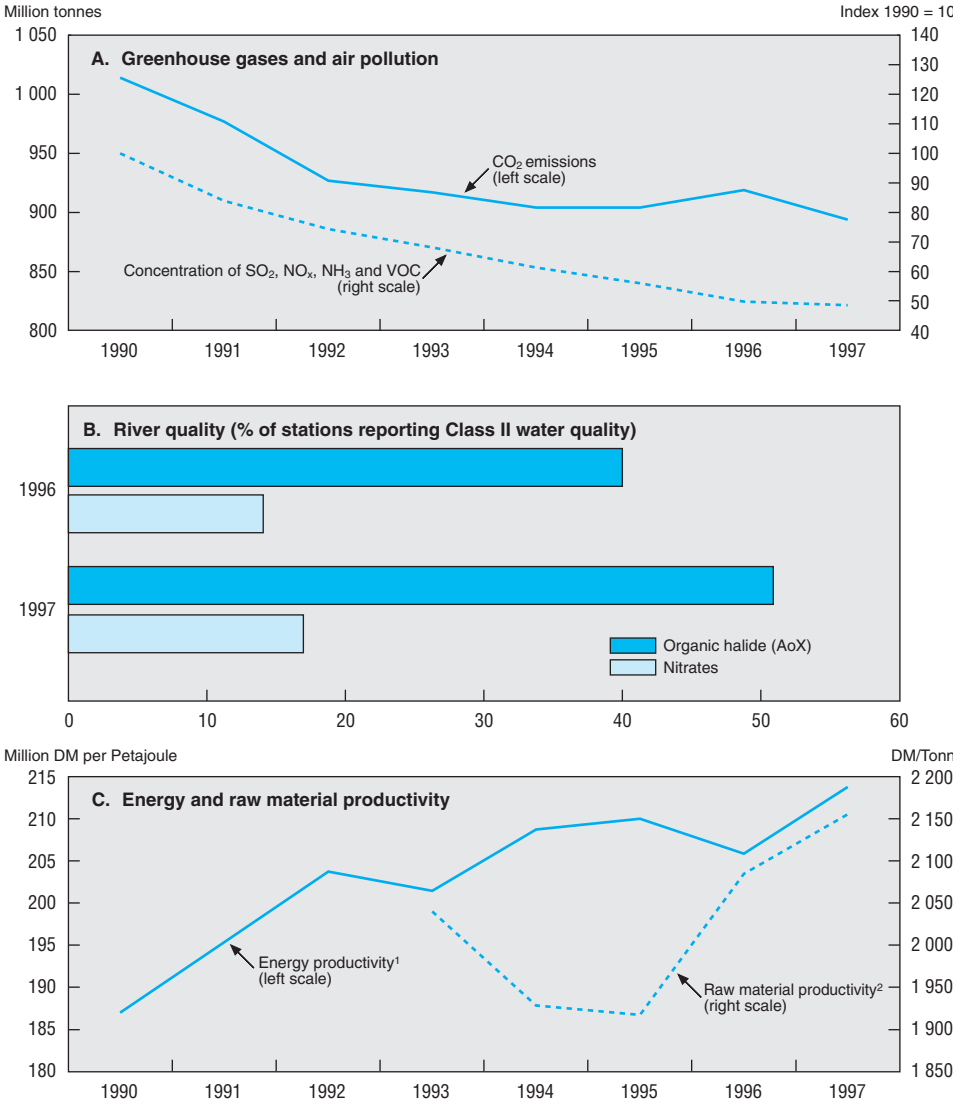
- *Regulations, laws, guidelines*, such as emission related laws and ordinances.
- *Economic instruments*, including energy taxes and subsidies for alternative energy sources, tax exemptions, subsidised loans, public infrastructure investments, and funding of research and development.
- *Voluntary agreements* by which the business sector, such as automobile producers, agree to meet specific environmental objectives.
- *Information, education and training*, to raise the awareness of the German public about environmental policy.
- *Research and development*, including interdisciplinary approaches to climate impact research, energy saving technologies, etc.

Of these instruments, direct regulations and voluntary agreements rely on non-market mechanisms; hence, the possibility of using relative price signals is largely non-existent. However, as regulatory intervention becomes more costly, especially as the scope for "no-regret" policies or low abatement cost opportunities are used up, the current policy configuration may have to give way to a different policy mix.

The management of natural resources

With respect to the use of natural resources, both renewable and non-renewable, the debate in Germany has often focused on the use of total resources, one measure of progress towards sustainability being aggregate material usage (defined as tonnes) per unit of GDP (Figure 16).[75] The justification for recycling and for material audits is often given as the need to preserve world resources rather than reduce pollutants, despite the fact that for many resources the elasticity of supply appears to be high in the long term. Moreover, with respect to renewable resources such as tropical forestry and fishing, over-exploitation may well be a global problem, but domestic policy in a single

Figure 16. **National indicators of progress toward sustainable development**



1. Defined as the ratio of GDP at 1991 constant prices to primary energy supply (expressed in Petajoule).
2. Defined as the ratio of GDP at 1991 constant prices to national extraction of raw materials (mineral, clays, etc.) and imports of goods (raw materials, half-finished goods, finished goods). All goods are expressed in tonnes.

Source: BMU; BMWi.

Box 9. Taking into account the external benefits of forests

Forests not only produce wood for commercial use but also supply a large number of non-market services, such as protection of biodiversity and recreation. The question therefore arises as to how important such externalities might be and how best they can be incorporated into the decisions of forest owners.

In a densely populated country like Germany, forests are used intensively for recreational purposes. Several studies have attempted to estimate the willingness-to-pay for such services. Results of contingent valuation and travel cost studies vary around DM 100 per visitor per year. These yearly numbers are extrapolated from responses from visitors with different lengths of stay in forest areas.[1] Since the frequency of visits is higher in urban, densely populated areas, the social value of forests in such areas would be correspondingly higher.

Fewer studies have estimated the willingness-to-pay for biodiversity, protection from noise and air pollution and water quality. Those which have been done do not estimate the contribution of forest areas separately but include other natural areas as well. A nation-wide survey on biodiversity preservation found a monthly willingness-to-pay of DM 20 to DM 33.[2] A survey of all ecological functions resulted in monthly values of DM 45 to DM 67, of which recreation was only DM 2.50 to DM 5 whereas species preservation contributed DM 16.[3] Since separate willingness-to-pay estimates are usually sub-additive, these estimates are at least not contradictory.

The contingent valuation studies show a comparatively large non-market value of recreational and ecological services provided by forests. A willingness-to-pay of more than DM 100 DM – possibly around DM 200 – per person per year and a population of just over 80 million leads to an overall willingness-to-pay of DM 8 to 16 billion. This should be compared with timber sales of the German forest sector in 1996 of DM 3.2 billion.

Considering the large non-market value of forest services relative to sales of lumber, the question arises as to whether the forest area in Germany is optimal and whether the owners of forests should be given incentives to take into account the ecological and recreational functions of forests. The results of the contingent valuation studies cannot be taken as a guide to defining the optimal forest area, but the type of forest to be aimed for seems to be clear: mature stands with natural habitat. The maximisation of profit from timber production is unlikely to meet the broader objective of maximising the social value added of forests if only price incentives from wood prices are used.

Private as well as publicly-owned enterprises managing forests already receive subsidies, *e.g.* for the repercussions following the 1999 storms, for the costs of preventing further damage to forests from air pollution, or simply direct income transfers. Most of these programmes are not explicitly related to the non-market services of forests. A redirection of these subsidies according to the services provided by forest owners as measured through a system of eco-points – an indicator for the achievement of a socially desirable forest quality – has been proposed.[4] Such a system could replace some of the direct regulations concerning ecological objectives in forest management.

(*continued on next page*)

(*continued*)

The social efficiency of forest management may be better achieved with such price incentives for the provision of ecological and recreational services.

1. For a review of the results see Elsasser (1996).
2. 1991.
3. Holm-Müller *et al.* 1991.
4. Linckh 1997.

country can do little to alleviate the problem directly. International agreements are necessary.

From the viewpoint of domestic resources, issues arise with respect to forests, bio-diversity and, to a lesser extent, management of water resources.[76] Forest management has been operated since the nineteenth century on the basis of tree-felling rates close to the rate of regeneration.[77] To correct pollutant damage to forests will require policies oriented to reducing a wide range of emissions (Box 6), while to secure the external benefits of forests arising from leisure activities and protection of bio-diversity might require restructuring financial supports for forest owners, so as to more closely reflect social preferences (Box 9). With respect to biodiversity, around one half of all vertebrates and about one third of ferns and flowering plants are regarded as under threat while two thirds of biotopes are endangered.[78] The cause for this situation is the level of pollutants and nutrients as well as more intensive land use in the past on the part of agriculture. A policy framework which takes account of effects across sectors is clearly required with respect to air pollution, as is closer attention to agricultural policy – to some extent in the domain of EU policy. In this sector the polluter pays principle does not seem to be applied. Rather, current agricultural practices are considered *status quo* and improvements are interpreted as land conservation and the maintenance of biodiversity which are to be paid for through subsidies.[79]

The climate protection programme

The Federal government's climate protection programme[80] contains over 130 measures for reducing emissions of carbon dioxide and other greenhouse gases, including actions aimed at the sequestration of carbon in forests. In this list of activities all types of policy instruments are represented:

- Most of the programmes contain *subsidies*, such as those for alternative energy sources, tax exemptions, subsidised loans, public infrastructure investments, and funding of research and development. A correction for the difference between market prices for the use of fossil fuels and

their social cost has only been implemented partially, since April 1999 when a *new energy tax* has been applied. The absence of a tax which fully reflects the environmental externalities has led to the need to subsidise or otherwise protect alternative energy sources. The government considers an EU-wide carbon dioxide/energy tax to be necessary. Traffic-related carbon dioxide emissions are expected to fall due to increased gasoline taxes and measures which increase the attractiveness of local public transport systems.

- A considerable number of *emission related laws and ordinances* are in force in Germany and these have been tightened. A significant contribution to carbon dioxide-reduction is expected from the Amendment to the Thermal Insulation Ordinance (W*ärmeschutzverordnung*, WS*ch*V) with a reduction of 7 million tonnes of carbon dioxide in 2005,[81] the Amendment of the Heating Installation Ordinance (a reduction of 9.7 million tonnes of carbon dioxide in 2005), and the amendment of the Ordinance on Small Firing Installations (a reduction of 1.4 million tonnes of carbon dioxide in 2005[82]).
- Two important *voluntary agreements* are currently in force: the German Motor Industry Association (VDA) promised in 1995 to reduce the specific fuel consumption of cars manufactured and sold in Germany by 25 per cent by the year 2005, relative to the year 1990; industry, represented by the BDI, promised in 1996 to reduce carbon dioxide emissions in 2005 by 20 per cent compared with 1990 levels. The reduction targets are, however, often specified in terms of specific ratios such as carbon dioxide per unit of output rather than as absolute emission levels. In return, the federal government has refrained from introducing additional regulatory instruments covering fuel efficiency for reducing carbon dioxide emissions. Moreover, industry is only subject to 20 per cent of the standard rate under the new energy tax. The commitment by industry is accompanied by a list of reduction targets and a system for verification. The 20 per cent overall reduction is composed of specific sectoral targets agreed upon by different associations organised in the BDI.
- The climate policy initiatives of the German Government include substantial financial resources directed towards *information and education* about climate change as well as the training of experts.[83] The awareness of the German public about climate change and climate policy seems to be quite high compared with other European countries or the rest of the world.
- R*esearch on climate change*, especially in the natural sciences, has long been supported. Recent efforts go towards interdisciplinary approaches of climate impact research as well as climate policy.

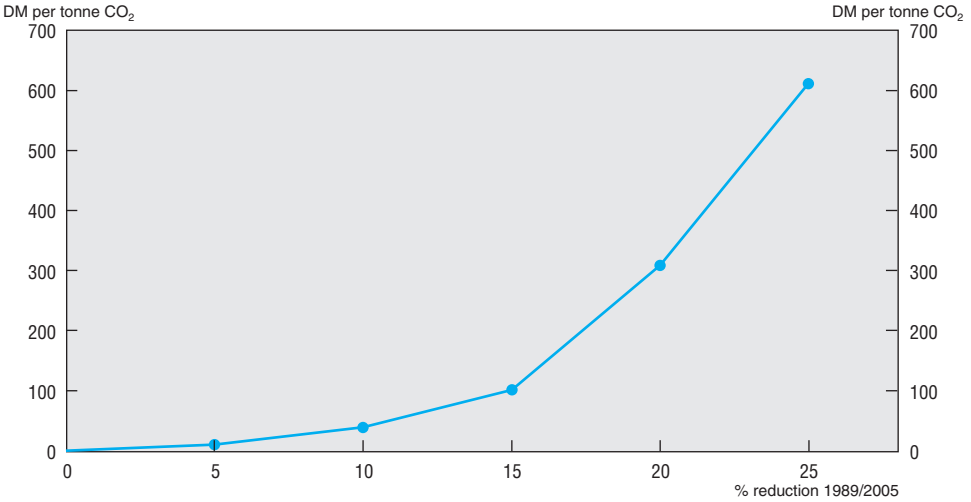
- The development of non-fossil energy sources and energy-saving technologies is also supported by R&D *spending*. The historically heavy bias of funding towards nuclear energy research has often been criticised. The IEA's Review of Energy Policies went further, noting that "energy R&D funding has thus become an instrument for political forces to show that they care about the environment" which involves support for pet projects rather than for promising technologies.[84]
- On the other hand, the government is committed to phasing out nuclear power which is responsible for some 30 per cent of electricity production at present. What such a phase out will imply for green house gas emissions is unclear depending on the pace of closure and the alternatives which might be available to replace the capacity.[85] However, it is evident that the phase-out will make the achievement of the governments environmental objectives even more difficult to achieve and strengthens the need for economically efficient policies.

Moving on from the period of low abatement costs

After a period when improved environmental performance was probably relatively costless to achieve, further improvements are likely to be costly. Figure 17 indicates that the marginal cost of reducing carbon dioxide emission by up to 10 per cent could have been quite low, and similar conclusions can be drawn from the voluntary agreement of German industry, where the reductions achieved to date have been realised quickly in the first years and apparently relatively easily. Three-quarters of the proposed 20 per cent specific reductions (*i.e.* emissions per unit of output) had been already achieved by 1996. However, in 1997 carbon dioxide emissions increased to 170.3 million tonnes in the concerned industries from 162.3 million tonnes in 1996 (Table 21). The RWI institute, which monitors the voluntary agreement, attributes this relapse to the problem of moving from low cost ("no-regret") to more costly reduction strategies[86] and if this is so, it seems unlikely that the speed of reduction can be maintained. Similarly, the patterns for other emissions indicate that further improvements are likely to be more difficult to achieve.

The policies implemented so far concentrate on direct regulations, subsidies, and voluntary agreements, so that relative price incentives for an economic use of fossil fuels have been largely non-existent. This may well have had already negative implications for efficiency. Table 22 shows the projected sectoral emission reductions from the policies implemented up to 1996. A reduction strategy which better exploited the available technological possibilities would require a somewhat dif ferent sectoral composition (column 4). Additional reductions would be needed in private transport, household energy use and energy production. For these reductions to be realised would, however, require changes in relative prices through for instance taxation to take account of the negative externalities arising

Figure 17. **Marginal abatement cost of CO_2 reduction in the old Länder**



Source: IKARUS model results from Stein and Strobel (1997), Table 139.

from the emissions of greenhouse gases and other substances. When even more ambitious environmental targets are considered, the efficiency losses from inappropriate instruments would usually increase more than proportionately.

Such price incentives as already exist may be impeded by existing regulatory and legal barriers. Thus, investment in energy-saving insulation of apartments is often not undertaken because owners are not permitted to recoup their costs via higher rents and tenants cannot risk having to write-off there own investment if they move.[87] For similar reasons, the Bundesbahn has had little incentive to keep costs down through greater energy efficiency, while electricity producers have not been under competitive pressure to raise fuel efficiency until the recent liberalisation. Incentives for individuals and companies to invest in energy saving requires that there be a rate of return to the investor, as well as to society at large.

Ecological tax reform

One of the new government's major priorities is an ecological tax reform, which has widened the scope for economic instruments. However, implementing the reform has proved difficult for institutional and political reasons. The reform, which came into force in April 1999, raised the gasoline and diesel tax slightly by 6 pfennigs per litre. Light oil and gas taxes were raised considerably and a

Table 21. **Carbon dioxide reduction achieved by industries signing the voluntary agreement**

Association	Base year (1990)	1996	1997	Reduction		Target achievement[3]
	Million tonnes of CO_2				Per cent	
Potash industry	4.8	1.1	1.1	3.7	77.7	102.5
Cement industry[1]	13.0	10.2	10.0	3.0	23.0	75.3
Lime industry[1, 2]	2.7	2.4	2.5	0.3	9.2	78.6
Ceramic tiles and slabs	0.7	0.5	0.4	0.2	35.2	87.0
Brick industry	2.4	2.3	2.2	0.2	6.6	68.4
Refractory industry[1, 2]	0.4	0.3	0.3	0.1	28.2	112.4
Iron and steel Industry	69.9	57.9	63.4	6.5	9.3	67.2
Non-ferrous metals industry	14.6	12.7	13.4	1.2	8.4	79.2
Chemicals industry	65.5	49.2	51.1	14.4	22.1	..
Wood pulp, paper and cardboard industry	14.4	13.2	13.5	0.9	6.2	117.2
Glass industry[1]	6.4	6.0	6.2	0.2	2.5	70.6
Textile industry[1]	5.8	3.9	3.9	1.9	33.4	163.2
Sugar industry	4.5	2.6	2.3	2.2	48.0	70.0
Total	205.1	162.3	170.3	34.8	17.0	85.0
Public electricity supply	289.0	264.0	261.0	28.0	9.7	80.8

1. Base year 1987.
2. West Germany.
3. Targets are differentiated between sectors.

Source: Buttermann and Hillebrand (1998).

Table 22. **Optimal composition of carbon dioxide emissions from energy use, old Länder**

	Actual 1989	Expected[1] 2005	Percentage change 1989/2005	Optimal 8 per cent reduction[2]
	Million tonnes of CO_2			
Energy production	227	209	–8	–13
Industry	160	131	–18	–19
Traffic	134	159	+19	+12
Households	115	119	+3	0
Small enterprises	54	46	–15	–19
Total	690	664	–4	–8

1. Expected emissions given climate policies in force in 1996; according to Stein and Strobel (1997).
2. Percentage change in emissions relative to 1989 according to IKARUS-Model simulations. To the baseline cut from existing policies of –4 per cent have been added results from the IKARUS-Model optimisation for lowest-cost cuts of up to 10 per cent.

Source: Calculations by the OECD from Stein and Strobel (1997), Tables 113 and 138.

taxation of electricity was introduced. Nuclear fuels and coal are not subject to a tax and for the latter, domestic production remains subsidised. Because of public concerns about competitiveness and employment, industry and agriculture are subject to only 20 per cent of the standard tax rate (Box 10). In addition, the net tax burden on companies has been capped: companies are eligible for a refund if the cost of the new energy taxes exceeds the notional savings from reduced social security contributions (the employers' contribution rate has been reduced by 0.40 percentage points) by more than 20 per cent. Originally it was intended to exempt the energy intensive sector but this proved impractical and might have contravened EU laws on state aid. Revenues from the tax have been used to reduce social security charges, on the grounds that this will produce the 'double dividend' described above, in the form of lower unemployment and improved environmental performance. The tax reduction for industry and the main focus on end-users as in the taxes on electricity has distributional effects, shifting the tax burden from companies to consumers and towards low income groups.[88]

The tax rates on energy are probably inefficient, since they are not based on emissions of carbon dioxide from each energy source. Tax rates in terms of carbon dioxide vary considerably (Table 23).[89] Gas and oil used in the generation of electricity are not subject to the new tax, although for most plants they remain subject to pre-existing taxes; only electricity itself is subject to the tax. The most carbon dioxide-intensive fuel, coal, is not taxed at all. Gasoline taxes are about ten times as high as the taxes on oil and gas, and indirectly on the fossil fuels used in electricity production. Moreover, gasoline appears to be taxed heavily in comparison with diesel which has been kept relatively inexpensive for transport and competitiveness considerations.[90] While diesel taxes are low, the fixed costs of operating either trucks or diesel cars are high. Local employment considerations have clearly been important in the decision not to tax coal, although another factor was the need not to confer a relative advantage to nuclear power, which would have been contrary to the key objective of phasing out nuclear power generation.[91] A coal tax would also have favoured imported over domestic energy production, particularly of nuclear power: under EU regulations, border taxation of energy imports according to the fuel used in generation is not permitted and would in any case be impractical.

As part of its overall budget strategy for 2000 and beyond, the government announced in June that the specific tax rates on petrol and diesel would be raised between 2000 and 2003 by 6 pfennigs a litre each year, while the tax on electricity would be raised each year by 0.5 pfennigs per Kwh. Apart from DM 200 million (the sum will rise in line with the revenues accruing from the electricity tax), which would be used to promote renewable energy sources, revenues will be used to lower the pension contribution rate from 19.5 per cent at present to 18.5 per cent in 2003. The reimbursement scheme for enterprises is to be maintained for the present although it may become increasingly untenable

Box 10. Competitiveness considerations in the environmental tax debate

Competitiveness issues have always been a stumbling block to the implementation of either energy taxes or a pure carbon dioxide tax. Indeed, in countries applying a carbon dioxide tax (*e.g.* Sweden, Denmark, Norway) energy-intensive industries have always been either tax-exempt or subject to special rates. The arguments traditionally used against the unilateral implementation of a carbon dioxide/energy tax applied to all sectors may appear even stronger in the case of Germany. Energy-intensive industries represent a significant proportion of manufacturing output and industry is strongly export oriented, with markets being characterised by strong price competition. In addition, compared with France, Germany's main competitor on the electricity market, the energy supply is much more carbon-intensive. A high tax on coal-produced electricity would substantially reduce the competitiveness of Germany's electricity production.

Given competitiveness concerns, the government coalition agreement stresses the need for harmonisation of energy taxes at the European level, favouring the introduction of energy taxes at a sufficiently high level to steer behaviour. However, it is unlikely that such a harmonisation will take place quickly. Three proposals have been made for an EU-wide energy taxation system, without agreement. The first two envisaged broad-based additional minimum tax rates on all fuels (except renewables) with a carbon dioxide component and an energy component. It was to be phased in progressively up to a level of US$10 per barrel-equivalent. Exemptions were proposed for energy-intensive industries. The proposals met strong opposition from most of the member countries. A new proposal by the Commission in 1997 included a minimum rate taking into account all existing energy taxes, instead of minimum additional tax rates, and the taxation of electricity as a final product. Member states could choose the way the tax would be calculated and energy-intensive industries could be exempted.[1] This proposal was not accepted by the member countries at the 1999 Cologne summit. Tax measures require unanimity to be accepted.

Competitiveness arguments should be treated with caution. First, the carbon dioxide tax revenues could be used to reduce other distorting taxes (such as labour or corporation taxes), which would favour the competitiveness of less energy-intensive industries.[2] Second, the effects of a carbon dioxide tax on the competitiveness of energy -intensive sectors are small in comparison with other factors, such as the exchange rate, the wage rate and the overall regulatory burden. In general, the approach to competitiveness is too narrow. What matters in the end is the dynamic comparative advantage – which in the long run may not be in energy intensive sectors – and the capability and flexibility of the economy to adjust to changes in relative prices.

In addition, the evidence for significant leakage effects (*i.e.* the displacement of emissions from countries applying environmental regulation to countries with no or less stringent regulation) is quite weak.[3] It is true that the uncoordinated introduction of a carbon dioxide tax could lead to an undesirable switching of productive capacity, which would be costly to reverse, if other countries were to introduce a carbon dioxide tax later on. However, a gradual phasing-in of the tax (including, for instance, a temporary compensation scheme not related to emissions) could largely avoid such

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unnecessary costs by giving economic actors due time to adjust and provide sufficient time for the various national authorities to co -ordinate their climate change strategy.

As an alternative to an EU-wide energy taxation, it would be in Germany's interest to support the creation of an international market for greenhouse gas emission permits. It would reduce the overall costs of meeting the Kyoto target by allowing the emission reductions to take place where abatement costs are lowest. To the extent that emission quotas can be partly "grandfathered" for a while (*i.e.* allocated in proportion to past emissions) a trading system would reduce the cost for the industries compared with a pure carbon dioxide tax, therefore making it more acceptable, while at the same time providing the necessary abatement incentives at the margin. The same could be achieved with an EU-wide carbon dioxide tax combined with a compensation scheme or tax credit which does not modify the incentives to reduce emissions.

1. Bill, 1999.
2. Barker, 1999.
3. Adams, 1997.

as ways are implemented to circumvent the provision. To guard against such a development as well as to preserve competitiveness, the Ministry of Finance has proposed that the energy tax should be levied on end use only. Proposals to introduce energy audits, which might offset a tax liability for firms, are still under discussion. By and large, it would appear that while the overall level of energy taxation will move in the right direction, the inefficient structure in terms of different implied rates of CO_2 taxation remains essentially untouched. Moreover, from the environmental perspective moving to an end-use tax would be a step in the wrong direction since the incentives for an optimal combination of energy inputs would be reduced.

Air pollution

Air pollution control has been quite effective in reducing emissions from stationary sources and in accelerating the introduction of best available technologies.[92] Detailed direct regulations with tight guidelines for granting permission to run a stationary source of air-borne emissions have led to a lower level of pollution. By prescribing maximum permissible concentrations of emissions from facilities falling under the regulations, the end-of-pipe technology that can be used is effectively determined. Traffic related regulations also require specific technologies such as catalytic converters. These are, however, backed by tax subsidies. Improvements from these levels of emissions are difficult to imagine given the current state of technologies.

Table 23. **Effective taxation of energy sources**

Product	Tax burden[1]		Energy intensity	Emission factor[2]	Tax burden[1]		
	1998	Increase since 1 April 1999			1998	Increase since 1 April 1999	Total since 1 April 1999
	DM/litre	DM/litre	M. Joules/litre	Kg CO_2/MJ	DM/tonne CO_2		
	(1)	(2)	(3)	(4)	(5) = [(1)/(3)/(4) × 1 000]	(6) = [(2)/(3)/(4) × 1 000]	(5) + (6)
Petrol[3]	0.98	0.06	34.20	0.072	397.99	24.37	422.36
Diesel	0.62	0.06	38.60	0.074	217.06	21.01	238.07
Fuel oil	0.08	0.04 (0.008)	40.80	0.074	26.50	13.25	39.74
	DM/kWh	DM/kWh		Kg CO_2/kWh	DM/tonne CO_2		
	(1)	(2)	(3)	(4)	(5) = [(1)/(4) × 1 000]	(6) = [(2)/(4) × 1 000]	(5) + (6)
Natural gas	0.0036	0.0032 (0.000640)	..	0.202	17.86	15.87	33.73
Electricity	..	0.02 (0.0040)	..	0.560[4]		35.71	35.71

Note: In brackets special price for industry and agriculture.
1. Without VAT.
2. Source: *Umweltbundesamt*.
3. Tax rates are differentiated between leaded and unleaded.
4. For electricity a CO_2 emission of approximately 0.560 kg per produced kilowatt hour results from the underlying energy mix (in the year as well as 1994) and the estimated efficiency factor of power plants. CO_2 neutrality is assumed for nuclear power, water power, wind power and other regenerative substances.

Source: Council of Economic Advisors.

Although abatement costs vary significantly across different types of mobile emission sources, and are likely also to vary for stationary facilities in the future as emissions are further reduced, the scope for economic instruments to improve efficiency is rather circumscribed. Emissions need to be measurable close to the source if taxes, charges or pollution permits are to be introduced. This is possible for many gases in stationary and large facilities, but expensive in small facilities and often impossible in mobile emission sources such as cars and trucks. A study of nitrogen oxides (NO_x) emission controls through tradeable permits found that only stationary sources, which are relatively homogenous, could be controlled with such an instrument.[93] Nevertheless international experience does show that gains can be expected from closely-focused emission taxes even in homogenous sectors, particularly with respect to nitrogen oxides emissions (Box 11). For mobile sources a mixture of instruments, including economic ones based on taxing inputs rather than emissions *per se*, may prove necessary

Box 11. NO_x emissions charge in Sweden: positive experience

In contrast to carbon dioxide, nitrogen oxides emissions (NO_x) are strongly influenced by the combustion conditions so that any environmental tax has to take the form of a direct emissions tax rather than as an energy excise. However, the cost of emissions measurement is high. In order to avoid these problems when Sweden introduced a nitrogen oxides charge in 1992, it was confined to a small group of large industrial plants and power stations where monitoring costs were likely to be low relative to potential abatement cost saving.* The tax is based on measured emissions or on presumptive emissions with firms opting for one system or the other. To avoid distorting the pattern of competition between large sources subject to the charge and smaller competitors (and to avoid any incentive to substitute small boilers) almost all the revenues from the tax are returned to the participating sources in proportion to their final energy output. Thus, sources with high emissions relative to their energy output are net payers.

After 20 months in operation, a commission examined the initial experience. Emissions had been reduced by 55 per cent at a cost of between SKr 5 to SKr 20 per kilo with an average cost of about SKr 10 per kilo. On the assumption that the tax rate reflects the social value attached to reducing NO_x emissions, the commission concluded that the increase in welfare amounted to some SKr 250 million. It was also found that changes in operating procedures for a given plant could have a significant impact on the level of emissions without any change in the technologies employed. In order to encourage emissions conscious behaviour by employees (and hence savings in tax payments) some plants introduced wage bonuses relating to emissions reductions which the plant had achieved.

* For details see S. Smith, "Environmental and public finance aspects of the taxation of energy", *Oxford Review of Economic Policy*, 14, No. 4 , 1998.

although the underlying principle that policy should seek to equalise marginal abatement costs by all sources remains valid.

An important condition for the effective use of economic instruments is that emissions should not be concentrated in local areas termed "hot spots". Otherwise, economic instruments need to take account of the local emission situation in terms of timing and locality. In the extreme, where instruments have to be shaped to local circumstances, they are little different from direct regulation of emission sources.[94] A further inhibiting factor is that a facility such as a fossil fuel combustion plant usually emits a large number of different hazardous substances. Tax rates would need to be set for each of these and it is not clear how this should be done. For example, because many of these emitted substances are jointly produced, this would require tax rates which are conditional on the complementarity of the processes creating emissions. Since this is unlikely to be achievable in many circumstances, the regulating authorities may need to resort again to the direct regulation of plants or processes. Large combustion facilities may therefore be easier and more efficient to control through the existing system than through a complex set of tax rates. What is needed is a careful assessment of the costs and benefits of differing policies for abatement control. In this respect, international experience shows that in those cases where only a single or few substances are emitted, or where the substances are emitted in fixed proportions, economic instruments could play a role. In particular, the establishment of a trading programme for sulphur dioxide from electricity generation for the United States as a whole, to address the acidification problem, has allowed cost savings from emission reduction compared with command and control instruments (Box 12).[95] This system has also proved compatible with control of particularly high local levels of pollution (hot spots).

In Germany, existing regulations sometimes treat sources differently, leading to an inefficient level of emissions. One example is the incineration of waste. The new Waste Law allows competition between waste incinerators and industrial incinerators for burning waste. The required standards for emission concentrations are identical for both facilities, but the required standard of best available technology is defined differently for the two processes, resulting in different costs of abatement equipment.[96] The apparent over-capacity of waste incinerators may, among other things, be attributable to the cost burden imposed on the disadvantaged sector by the regulation. Another example concerns the different treatment of stationary and mobile sources. Emissions from non-stationary sources – mostly coming from traffic – have not been reduced nearly as much as those from stationary sources in the case of nitrogen oxides. As a result, the share of traffic-related emissions has been increasing, to almost 70 per cent (see sectoral discussion below). It is clear that further emission reduction measures will need to concentrate on such mobile sources, and for this traffic-related policies will be indispensable. A reduction in traffic will probably be needed as

Box 12. Emission trading in the United States

The United States has two large-scale "cap-and-trade" schemes currently in operation. Permit trading in a nation-wide SO_x scheme covering emissions by electricity generators began in 1995 and that in a NO_x scheme covering twelve north-eastern states began in May 1999. The SO_x scheme is operated under federal legislation in an 1990 amendment to the Clean Air Act, whereas the NO_x scheme is operated under a voluntary agreement between the states concerned. Although emission trading had been an option under a number of regulatory programmes from the 1970s onwards, the SO_x scheme is the first to be thought successful; the failure of earlier schemes – in the sense that no true markets ever developed – is attributed to the fact they were adjuncts to command and control schemes, with restrictions on trading that rarely made it worthwhile. The SO_x scheme was the first in which the cap and trade system is relied on almost entirely for emission control.

The SO_x scheme

This scheme has a number of notable characteristics: permits are geographically undifferentiated; allocation is almost entirely based on the product of a performance standard and representative utilisation levels, though a small proportion are auctioned each year; in the first few years of the scheme only large emitters were covered, smaller emitters being included from 2000; trading is very flexible – "banking," in which unused allowances can be carried forward, is allowed, as is forward trading; trading is not restricted to emitters and reporting requirements are not burdensome; non-compliance penalties are severe (a fine of $2 000 per ton, some ten times current permit prices, and at least three times most earlier estimates of the permit price). The aim is to reduce overall SO_x emissions from power generation to 50 per cent of their 1980 value in a progressive tightening from 1995 onwards.

Evaluation of the scheme is complicated by the fact that caution over whether trading would be feasible led many emitters to cut their emissions more than necessary, and that expanding rail capacity and deregulation meant that a low-cost option of switching to low-sulphur fuels became much more widely-available. Actual emissions have thus been substantially under the target, with some banking in anticipation of a tighter future market, and permit prices were much lower than initial projections for several years, though they have risen since mid-1998. After a period of about a year during which trading was thin, and it was not clear what "the" market price was, trading grew rapidly and independently of the official auctions, and a going price has been well established since 1996.

The absence of geographical differentiation has not led to emission "hotspots" as some had feared, with some of the largest emission cuts coming in areas with high initial emissions and sources are still required to limit emissions to prevent violation of health-based air quality standards. Reasonable estimates of the savings due to trading suggest that abatement costs were reduced by between a quarter and one third, though these depend on what alternative scheme is considered (Schmalensee *et al.*, 1998).

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The NO_x scheme

This is similar to the SO_x scheme, with twelve of the states that comprise the Ozone Transport Commission having delegated the running of the scheme to the federal Environmental Protection Agency. The aim is to reduce summertime (May-September) NO_x emissions to less than half their 1990 level, with the reduction taking place in two steps – in 1999 and 2003. An interesting difference is that the penalty for non-compliance is specified as confiscation of permits equal to three times the size of the violation; individual states retain the power to fine non-compliers up to $25 000 per ton in their own state. Another important difference is that emitters remain subject to various technology standards set by the EPA. Although permits were allocated by each state over a period of a year beginning in September 1998, conditional trading began in late 1997; since then 1999 vintage prices rose to around $3 000 per ton by late 1998, spiked at over $6 000 in March 1999 before declining to under $2 000 in July.

well as refinements to fuels and engines which will require a balance of economic and regulatory policies.

In sum, with respect to the control of air pollution, Germany has chosen an approach setting allowable emission concentrations, specifying best available technologies, and licensing emitting facilities. This policy has resulted in remarkable reductions in the emission of most hazardous substances from stationary, mainly large, facilities although whether this has been at least cost is not clear. Smaller facilities have been subjected to similar regulations, whereas mobile sources such as traffic-related emissions account for an increasing share of pollutants. Further reductions will need to come from mobile sources, since additional contributions from stationary sources might only be possible with large and increasing abatement costs. In moving forward greater reliance may need to be placed on economic instruments so as to achieve ambitious environmental targets at least cost. Experience in other countries shows that this is a viable complement to regulation.

Water pollution

The regulatory system requires every firm or institution which discharges waste water into surface waters to obtain a permit from the local authority.[97] This is only granted if waste water is treated according to the generally-available technology. In addition, a charge is levied on waste waters, this being practically the only emission which has been directly targeted by an economic instrument in Germany.[98] However, the charge is applied in a discriminatory way. Charge rates are reduced for emitters who use best available technology and they can also be

reduced for local reasons. Having been designed to create incentives for additional reductions in those cases where marginal abatement costs are comparatively low, recent amendments have been strongly criticised as undermining its original incentive function.[99] Charge rates differentiated according to the technology used and the emission concentrations distort the incentive of the waste water charge.[100] And a major inefficiency arises from provisions which permit the charge to be offset against investments in new sewage systems or their repair and with investments in the treatment of a particular substance which are made in a treatment plant. This is believed to lead either to symbolic investments in order to avoid the waste water charge[101] or to a subsidy on investment which a company is legally bound to make anyway, thus increasing the deadweight loss of the charge.

The control of non-point water-pollution emissions has not kept pace with the control of point sources, the former now contributing 60 per cent of the nutrient load, mostly from agricultural sources. So far the nutrient input in agriculture is partially regulated through regulations of agricultural practices, *e.g.* for the use of natural manure or maximum levels of fertilisers to be used annually. However, price signals such as a nitrogen tax on fertiliser or tradeable permits, are not used, although they might contribute to reaching a more efficient level of fertiliser use. In their absence, and although a great deal has been achieved in reducing water pollution, the instruments in place have not been effective enough to meet the environmental targets in a cost-efficient manner.

Waste disposal

Germany has a number of waste problems including hazardous and nuclear waste but packaging and other consumer related waste issues have received disproportionate public attention.[102] During the 1980s there was increasing concern about the future volume of waste which was expected to be generated, most projections indicating that continuing rapid growth would bring increasing pressure on scarce landfill capacity, together with dangers from soil and water contamination. In response, the waste management Law of 1986 specified a hierarchy of priorities: waste avoidance, waste recycling and waste removal, in declining importance.[103] This policy logic was extended in 1996 with the concept of "product responsibility" by which producers are responsible for their products from "cradle to grave". Most importantly, the law determines that producers and distributors of goods can be obliged by ordinances to take back their products after use and to "reprocess" a predetermined share of the waste (*Verwertung*). Similarly, the last owners of a product can be held responsible for returning the goods after use. The government can also assign the costs associated with the various stages of waste management to the different parties involved. In implementing the policy, considerable emphasis has been given to establishing volun-

tary agreements with industry (Box 13). Measured in terms of waste reduction and the proportion of waste that is being recycled, the German waste policy has been quite effective. At present, about one-third of all waste is being recycled, with the recycling quota of industrial waste, which is relatively homogenous, amounting to some 60 per cent (Figure 18). With respect to packaging, which only accounts for about 5 per cent of total waste, recycling ratios ranging from 79 per cent for composite materials to 94 per cent for paper and board have been achieved and per capita consumption of packaging material has declined from 95 kg in 1991 to 82 kg in 1995.

Quotas for re-use, recycling and energy and non-energy recovery have conflicted with underlying market developments and have led to a number of economic and environmental issues:

- Quotas for recycling etc. are second best to a policy for landfilling based on differentiated fees targeted directly at the externalities (volume of and emissions from landfilling).[104] This would require, however, that fees for waste processing be corrected for the costs of emissions into air and soil. Public waste management facilities are currently not even allowed to charge the full cost of waste collection, recycling and removal due to accounting regulations for public institutions.[105] They are also not allowed to charge shadow prices for incineration and for landfill. Such a system would improve efficiency in the disposal of waste.
- Waste intended to be reprocessed is often transported over long distances, and the associated pollution would have to be taken into consideration for a comprehensive environmental evaluation of the waste processing strategy.[106] The restrictiveness of waste-processing quotas for packaging material has been evidenced in increased inventories of waste material. (In response, the range of admissible modes of reprocessing packaging material has been widened in 1998, including now certain types of combustion.)
- Identifying the material to be subjected to quota opens another source of distortion. The packaging ordinance requires that packaging materials be separated from other recyclable waste of the same material in the collection and sorting stages, whereas efficient recycling should treat the same material equally to avoid unnecessary costs.[107]
- Quotas restrain the utilisation of other forms of waste processing which might turn out to be more environmentally friendly and economically efficient. For example, following the marked reduction of polluting gas emissions due to technological progress, incineration and energy recovery facilities could offer a more efficient and environmentally friendly means of recycling.

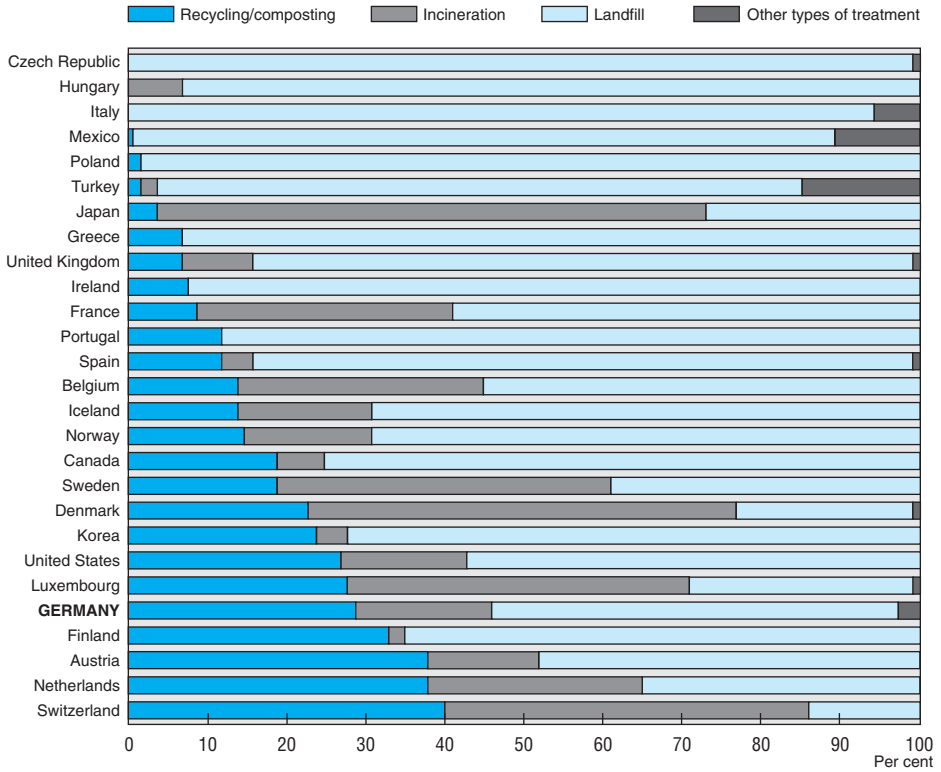
Box 13. Voluntary agreements covering waste disposal

The goals of the Waste Management Act have been implemented through voluntary agreements (*Selbstverpflichtungen*) which in some cases have been underpinned by ordinances.

- *The packaging ordinance*, which came into force already in 1991, is the most far-reaching ordinance on waste processing. Producers and distributors of packaging material have been made liable to take back their packaging material. In addition, the ordinance determines what fraction of the waste – broken down by material – must be "reprocessed" (*verwertet*) rather than being dumped. The definition of "reprocessing" has been widened with the most recent amendment of the ordinance and now also includes forms of combustion for the purpose of energy recovery, in addition to pure recycling which was the focal point of the first version. Firms have the choice of meeting their obligations individually or to participate in a collective scheme which organises the relevant aspects of waste management. The majority of firms choose the collective solution. This led to the establishment of a single private waste management company, the "Dual System Germany" (*Duales System Deutschland*), which commissions collection of waste packaging and grants contracts with firms to guarantee its processing. Producers of packaging material participating in the scheme label their wrappings with a green dot (*grüner Punkt*) for which they have to pay licence fees that cover the costs of the system. This has been extended by an ordinance in 1998 specifying recycling standards and collection quotas.
- In 1988, the *battery industry* committed itself to establish a take-back system for batteries and improve the labelling with respect to hazardous substances in the batteries. The scheme has been extended by an ordinance in 1998 which specifies recycling standards and collection quotas.
- The *paper industry* entered into a voluntarily agreement in 1994 to raise the recycling quota for printed material to 60 per cent by the year 2000. In 1995 as much as 72 per cent of paper was already recycled.
- The *construction industry* – apart from the large construction companies which develop their own recycling strategy – has announced that it will voluntarily reduce the quantity of construction waste. Between 1995 and 2005, the amount of waste relative to the value of new buildings and constructions is to be reduced by 50 per cent. It is not clear whether this goal will be reached (SRU 1998).
- The *Automobile Industry Association* – in a reaction to a planned ordinance on the recycling of used cars – signed a voluntary agreement on a modified take-back guarantee of used cars and car parts and on reducing the share of non-recyclable parts in cars. In response, the government did not further pursue the planned ordinance in the original form. Instead, the voluntary agreement was specified in the new ordinance in 1997. In the self-binding arrangement, the car producers have bound themselves to accept used cars without charge which are up to twelve years old. An EU proposal for a European wide ordinance was vetoed by several governments, including Germany, on the grounds that it would put their producers at a competitive disadvantage.
- In 1991 the government presented a draft ordinance on the waste management of *electrical appliances* and *electronic components*, specifying take back and reprocessing obligations. The ordinance is still being debated, and proposals for voluntary agreements have been made, but no final agreement has been reached.

Figure 18. **Treatment of municipal waste**[1]

As percentage of amounts disposed of:



1. Mid-1990s. Municipal waste is waste collected by or on the order of municipalities. It excludes waste from municipal sewage networks and treatment, as well as municipal construction and demolition waste. Recycling data not available for Italy: incineration data not available for Ireland.

Source: OECD, *OECD Environmental Indicators.*

The German system needs to be developed further by widening the admissible channels of waste processing so as to allow price signals to play a larger role in the waste allocation process.

Greater use of market mechanisms presuppose the creation of efficient markets but in the waste area a great deal of progress is necessary. The establishment of a single waste management company, which works on behalf of all packaging-producing firms participating in the Dual System, has been repeatedly criticised for restraining competition between processing firms at different stages of waste disposal. Moreover, the waste disposal industry is highly concentrated, due in great measure to the strong position of local utilities. Without some form of

central waste management each firm using packaging would be individually responsible for securing waste collection, sorting and processing, which would clearly be less efficient than a co-operative solution. Nevertheless, the system has revealed deficiencies with respect to competition which point to shortcomings in the regulatory framework. Competition between waste collectors is virtually absent. Contracts for waste collection are granted to just one firm per region, often without the need to re-tender for a period of ten years. While the revised ordinance of 1998 now requires open tendering and shorter contracts, market contestability will remain restricted by the long duration of existing contracts. More importantly, the market structure has been adversely influenced by the provision in the Packaging Ordinance requiring that waste collection and processing within the Dual System has to be co-ordinated with the collection and processing systems of the local communities. Not surprisingly, some 25 per cent of waste collectors under contract to the Dual System are communal firms or firms with communal participation.

As noted in Box 13, "self-binding arrangements" (*Selbstverpflichtungen*) by industry have been used extensively, but some have been criticised for achieving only what would have materialised without intervention, as well as for giving industry a disproportionate say in environmental policy.[108] For example, it has been suggested that the free take-back guarantee for cars up to a maximum age of twelve years does not essentially contribute to the environmental objective of reducing the volume of shredder waste (which is classified as hazardous).[109] Most cars for scrap are not under twelve years of age, the mean age at disposal being 13.2 years. The pledge to take back cars free is also redundant since such cars still fetch a positive market price. More generally, to the extent voluntary arrangements are costly for participating firms, they may be unstable if free-riding cannot be excluded.[110] Moreover, there might be a danger that self-binding arrangements allow insiders to the agreement to set standards in their own interest and at the expense of potential competitors.

In sum, the move toward "product responsibility" represents an important step towards internalising environmental costs of products at the stage where it matters most: that of product design and manufacture. However, the priority given to recycling *per se* has led to some adverse economic and environmental outcomes, so that price signals should be allowed a greater role in decisions about the final destination of waste. This distortion has been compounded by the fact that industrial activities in the sector are not based on competitive market conditions.

Selected sectoral issues

Although policies for more environmentally-friendly growth need to be considered broadly in terms of the specific environmental target, widely differing

monitoring costs mean that policies might have to be designed on a sector-specific basis. For example, the need to lower nitrogen oxides (NO_x) emissions might need different policies for cars and trucks compared with those oriented to large, stationary, installations. On the other hand, sectoral policies often have a number of environmental implications which need to be considered. This section examines policies in four key sectors: electricity, coal production, transport and agriculture.

Electricity: reconciling efficient supply with environmental objectives

Liberalisation and the promotion of alternative sources of energy

The liberalisation of the electricity sector has brought lower prices and this tendency is expected to continue if competition is allowed to develop (chapter III). This development has raised two environmental policy questions: *i*) are lower electricity prices (and lower oil and raw material prices for that matter) compatible with achieving environmental goals; and *ii*) what should be the policy with respect to fuel substitution? In particular, should renewable energy sources, which are generally more expensive than gas or oil-fuelled generators, be promoted in the framework of competition in the energy market and, if so, how?

Increased competition following liberalisation places generating companies under pressure to select low cost energy sources and to make sure that the conversion of primary energy to electricity is energy efficient. Improved efficiency resulting from competition should be welcomed from an environmental and economic perspective. *Ceteris paribus*, this development has reduced the long term attractiveness of both nuclear power (an environmental policy goal) and coal to the advantage of gas. For the market to work efficiently, however, will require sound and reliable framework conditions, a point stressed by the IEA in its energy policy review of Germany, and correct price signals. Lower electricity prices could be expected to raise consumption although the effects on carbon dioxide emissions will depend on the energy mix. As noted above, in view of the Kyoto commitments the best solution would be to impose a tax or tradeable permit on carbon dioxide emissions associated with primary energy inputs rather than to tax electricity.

Electricity liberalisation includes special provisions for electricity produced from renewable sources (such as wind, solar and bio-mass) to be given access to the grid at above-market prices, the excess being treated as a network cost to be paid by all customers of the network involved. Access is limited to 5 per cent of transmissions. Integrated generation/heating systems, which are regarded as environmentally friendly because of their high rates of energy efficiency, are also given special protection, by allowing third party access to be denied under some circumstances.[111] With the energy tax reform of April 1999,

support has been further increased: taxes on electricity based on renewable energy sources will be refunded in the form of development grants, while the existing taxes on mineral oils and natural gas have been abolished for energy-efficient integrated generators. Support measures which ensure higher prices for renewable energy sources in local markets are unlikely to be effective for long in a liberalised electricity market where companies can move operations out of high cost areas. A quota scheme for energy from renewable sources has been put forward as an alternative but the question would arise as to the price which would be paid for such electricity. Moreover, at some stage preferential access is likely to prove incompatible with the liberalised European electricity market. More generally, protecting alternative sources of electricity generation is a second best policy, the first best being to tax primary energy on the basis of relative emissions. As noted above, this latter course of action would probably require agreement at the EU level.

Although appropriate pricing of primary energy inputs would place all sources of electricity generation on a more equal competitive basis (nuclear power raises other issues), thereby reducing the need for special support programmes, special issues do need to be addressed with respect to integrated generation/heating systems. Several factors appear to make such systems at present uncompetitive.[112] The capital costs are large and heating for residential purposes is only used for part of the year, at which time the energy efficiency is indeed high. For the remainder of the year it is not. Another problem appears to be that since the integrated generators are city based (S*tadtwerke*) their scale might be too small to be efficient. Indeed, this appears to be the case in cities producing only electricity. The case for an across-the-board protection of combined generators is thus questionable. An undistorted system of environmental taxation would help direct the inevitable restructuring of city generating capacity in the direction of economic efficiency and the attainment of environment targets.

As noted above, an important goal of the government is the eventual closure of nuclear power generators, which account for about 30 per cent of Germany's electricity generation. The goal is to achieve this closure gradually and by a voluntary agreement with the companies, both in order to preserve security of supply and to avoid payment of compensation to the owners. This is a political decision which will affect the attainment of environmental targets. Even if nuclear energy is replaced by gas-turbine generators, which can be brought into operation relatively quickly, these are still associated with CO_2 and other emissions. Technological advances may, of course, create future energy savings, but it will be important that unrealistic assumptions are not made about the short term capacity of the economy to realise these. A phase-out of nuclear power would thus have to be underpinned by other concrete policy measures.

The coal industry: conflicting objectives

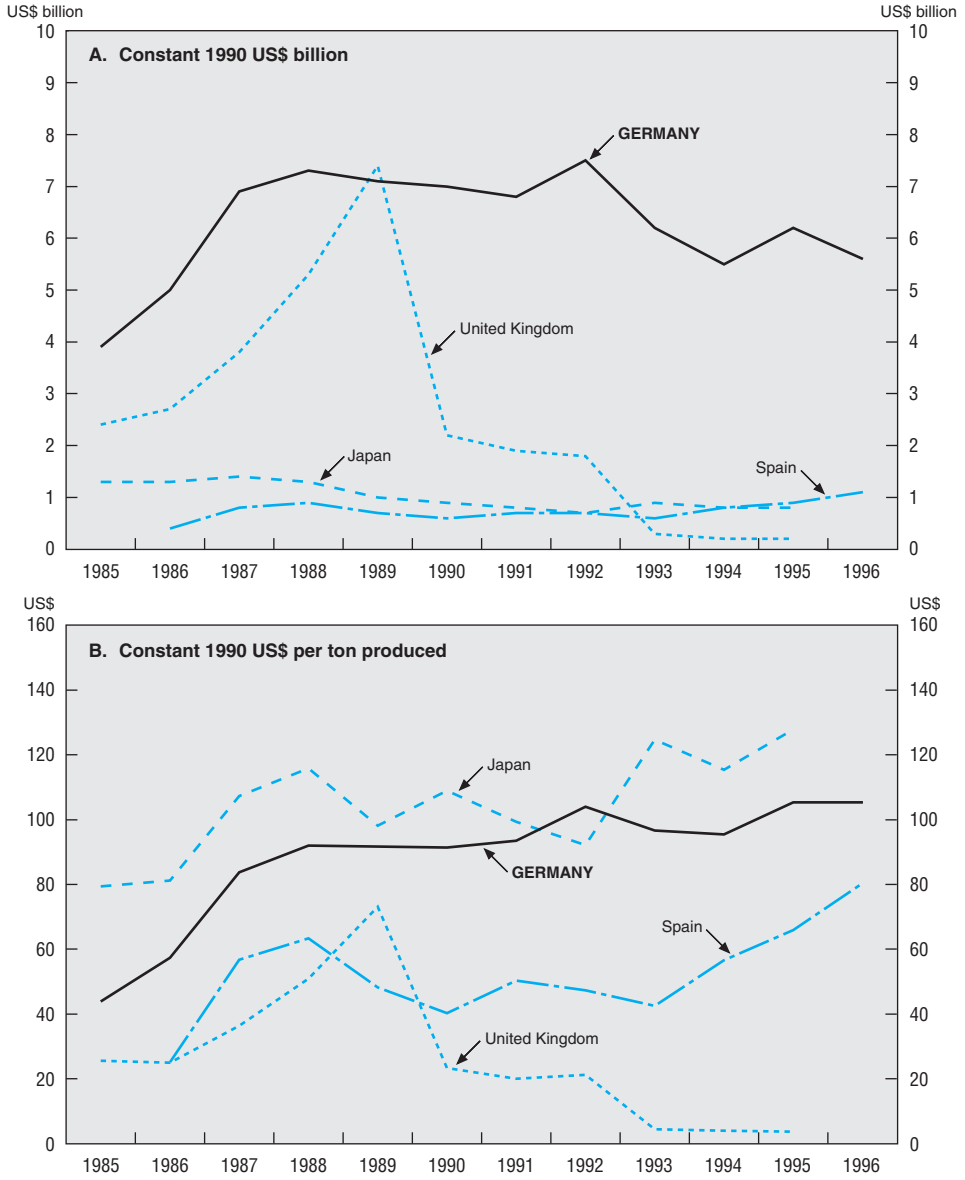
The coal industry represents an example of the conflict between narrow definitions of economic interest and wider concepts of both economic welfare and environmental objectives. Not only has hard coal continued to be subsidised – although at a declining rate – but one state government recently fought to have a new coal fired station established in its state in order to preserve jobs. Moreover, lignite production (with a high carbon dioxide emission intensity) in the new Länder is being supported until 2003 (with an option to extend to 2005) by guaranteed access to the grid at higher prices for electricity produced from this fuel. At the same time, the stated objectives of Germany's coal policy are reductions in the sales of subsidised coal for electricity production and in the steel industry. Extraction costs are not competitive with other foreign suppliers, amounting to around US$50/tonne for brown coal and US$200/tonne for hard coal,[113] far above world prices which has led in the past to subsidies. These reached a peak between 1987 and 1995 with roughly DM 12 billion per year.[114] This amounted to a subsidy per worker in coal mining of roughly DM 100 000 per year[115] and a producer subsidy equivalent of over $100 per tonne (Figure 19). In 1997 the support programme for the coal industry was changed to direct payments from the Federal government and the state Nordrhein-Westfalen.[116] Subsidies will be reduced from DM 8.9 billion in 1997 to DM 5.3 billion in 2005, although the rate of decline is not rapid (Figure 20). The new government has decided not to change this plan. The extraction path of hard coal has been largely unaffected by the different policy measures. Since the peak production of about 140 to 150 million tonnes around 1960, output has more or less continuously fallen to 46.5 million tonnes in 1997. However, the IEA's *Review of Germany*, 1998, noted that the decrease in the United Kingdom and French production had been more rapid. Capacity will be further reduced to 30 million tonnes in the year 2005. About one half of the mines will be closed between 1997 and 2005.[117] In the same period employment will fall from 85 000 to 36 000. In order to manage this process the German coal companies have merged in the "Deutsche Steinkohle AG".

Although lignite production is not subsidised and output has been reduced, two Länder have taken measures to maintain production through subsidies for the building of two new generators. In addition, under the contract for privatisation of VEAG (the supra regional utility in the new Länder) the company undertook to modernise lignite fired power plants and the building of new plants. The commitment was to invest DM 20 billion of which DM 3 billion had been invested by 1997. Access to the grid at higher prices is ensured up to 2003 and possibly to 2005.

The policy of subsidising coal extraction and of encouraging the use of coal for electricity generation contradicts the climate policy objectives of the German government since coal has the highest carbon dioxide content of all fossil

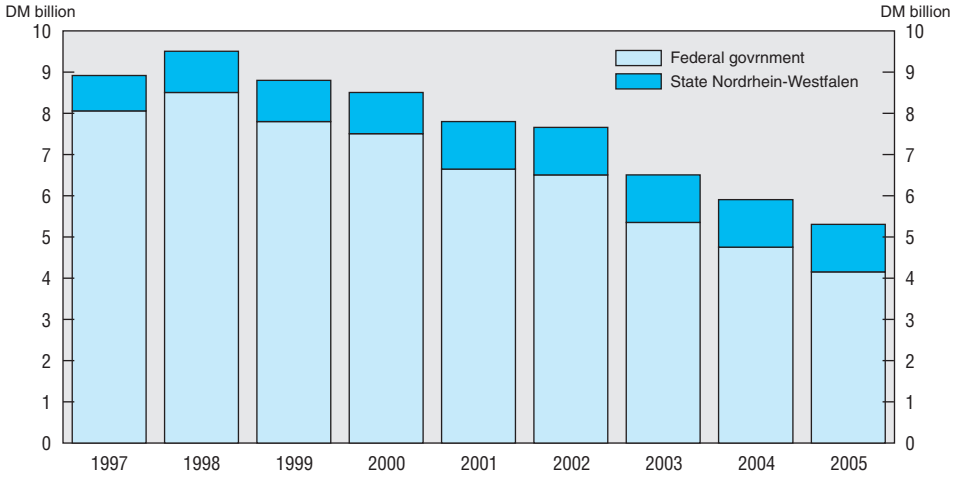
Figure 19. **Subsidies to hard coal production**

Producer subsidy equivalent (PSE)



Source: International Energy Agency (IEA).

Figure 20. **The financial aid programme to the coal industry**
Planned payments in billion of deutschemarks



Source: Storchmann/Kyron (1997).

fuels. This policy has induced a structure of power plants strongly oriented towards coal incinerators making it difficult to switch to fuels with a lower carbon content, since these power plants represent high sunk costs with a service life of several decades. Moreover, even if subsidies were to be cut, low price imported coal would extend the working lives of the plants. In the next decade several plants nevertheless need to be replaced which will place a premium on having a consistent policy framework in place. A continued orientation towards coal would direct long-term investments in a direction contradictory to climate objectives.

Traffic and transport

Traffic contributes importantly to both carbon dioxide and other air pollution emissions and is responsible for a wide range of local environmental problems such as congestion and noise. Although CO_2 emissions from the transport sector stabilised in 1996 and 1997, it is too early to say whether this will hold once overall economic activity accelerates since efficiency gains in fuel use have tended to be offset by rising transport volumes.[118] This is particularly so with respect to road transport where tonnages have increased by a quarter in the period 1990/1995, as well as in air transport (both domestic and international). A number of studies suggest that externality effects are not fully captured by cur-

rent transport pricing.[119] One estimate[120] puts the uncovered social costs of road transport in Germany at around 2½ per cent of GDP in the early 1990s, about half of which was due to air pollution and CO_2 emissions. The valuation of these latter effects is certainly too low since the shadow price of emissions has risen in line with changed environmental objectives. Even though transport charges have risen in the intervening period, the estimate is very much on the conservative side.

Transport planning in Germany is based on cost benefit analysis which includes an evaluation of environmental effects. However, according to work commissioned by the U*mweltbundesamt* environmental effects are still inadequately considered, monetary values do not correspond with recent research results and projects are neither assessed as a network nor in comparison with other transport modes.[121] Transport really needs to be considered as an integrated network with complementary modes rather than broken down into individual components for the purposes of determining which part accounts for what levels of emissions. In the absence of a network policy, discussion with respect to transport has often stressed the advantages of inter-modal substitution: switching from road transport to rail, from private to public transport, and from air to rail. Indeed, evidence does suggest widely differing energy use per unit of service.[122] However, the scope for substitution is often limited for economic and technological reasons. Although the situation is gradually improving, the railways have been inefficiently run and have not been competitive with road transport in many areas. Moreover, despite a building programme for terminals to transfer cargo from road to rail, the capacity of the lines is limited; 80 per cent of the volume is carried by only 50 per cent of the rail network so that extensive rail building would be required to raise substitution and this raises the issue of cost recovery, which for rail transport appears to be low.

Policy options are limited by institutional constraints in the case of air transport and road freight. Kerosene for air transport is not taxed at present, despite clear environmental externalities. Introduction of a tax requires unanimous consent of the EU. For road transport, most indicators point to the need to raise taxes on diesel which would affect both cars and road transport. However, fixed fees for German based transport companies are higher than in neighbouring countries so that the government has been under pressure to avoid placing domestic operators at a disadvantage. Proposals to levy a road tax on foreign trucks runs against EU law with respect to national treatment. The best method would be to move to a system of road user charges for all trucks, which is becoming technically feasible. Such a move would, however, require changes in the regulatory regime, in particular a lowering in the relatively high fixed costs levied on road transport.

Policy with respect to automobiles has varied from voluntary agreements to changes in tax treatment. German car producers have agreed to lower by 25 per

cent (2005 in comparison with 1990) average fuel consumption on cars sold in Germany while the government has been furthering development of a fuel-efficient three litre per 100 Km auto. While engine fuel consumption has improved, the weight of cars has also increased (thereby raising actual consumption) as consumers have preferred larger more comfortable and secure autos. Lower energy prices in comparison with the 1970s have led to a clear change in consumer preferences. To counter this tendency, the automobile tax was reformed in 1997 and is now based more on emissions than on car weight. Older cars, which are responsible for some 60 per cent of emissions, now pay some four times the basic rate. However, the tax differentiation is still only a fraction of that which could be expected on the basis of emissions alone and some observers judge that the tax structure favours diesel technology over petrol, which is at odds with other policy targets.[123] There is still a need to shift taxes away from the type of car to actual use. The use of private transport is also conditioned by a number of fiscal measures related to other policy objectives. The tax treatment of company cars was changed several years ago for fiscal reasons but a side effect has been a move toward smaller cars for company fleets. Another important factor has also been the deductibility of private transport to and from work from income tax. The new government has sought to widen tax exemptions to include public transport so as to make the system more mode neutral.

In sum, road pricing will need to be tightened both with respect to passenger and road transport. In practice this will mean increases in both petrol and diesel taxes with more pronounced rises in the latter. On the other hand, some fixed taxes on both cars and trucks should at the same time be reduced. With respect to railways, a balance needs to be maintained between environmental objectives and economic considerations: efficiency needs to be raised as infrastructure cost recovery is the lowest of any transport mode.

Agriculture

Production-linked support policies have led to a number of environmental problems including methane gas emissions, a high level of nitrates in ground water and increasing encroachment of natural habitats.[124] These environmental problems have been due to high concentrations of livestock in some regions, intensive land use and resort to large quantities of fertiliser and pesticides.[125] Although support policies are now less linked to output than they were in the past, and subsidies for the use of light fuels are being phased out, farmers are still facing distorted market and policy signals. Moreover, the polluter pays principle is hardly applied. Fertiliser use per hectare is limited by regulation rather than through a nitrogen-based fertiliser tax which would also affect fertiliser use up to the legal limit. Rather subsidies are often used, a case in point being the water charge which is used to compensate farmers for the limitation of fertiliser use.

Organic farming has been growing rapidly and now accounts for 1.8 per cent of arable land use. Revenues from organic farming are some 15 to 35 per cent lower than those achieved by conventional methods so that its spread has been supported not only by consumer preference but also by new subsidy programmes of the EU and the federal and state governments.

In the agricultural sector, policies to promote economic efficiency would also serve to support the achievement of environmental objectives. The current level of output-related support needs to be reduced but this will not obviate the need for selected taxes to be applied on inputs such as fertilisers. More importantly, the polluter pays principle also needs to be extended to this sector.

Assessment: realising the aim of sustainable growth

Overall, air and water emissions have been drastically reduced for many substances during the past twenty years, and the environmental quality of water resources, the soil and the air has improved in many instances. However, the reduction in emissions of many substances during the 1990s has been slower than in the 1980s and the gains which have been achieved are mainly due to developments in the new Länder: western German emissions often have been stagnating or even increasing, whereas those in the new Länder have been falling at a pace comparable to the earlier developments in the west. While the reliance on direct regulation and voluntary agreements has achieved the desired emission cuts in the case of stationary emission source this has been less the case for mobile or non-point sources. The tendency for emission reductions to slow could indicate that the inexpensive or even zero-cost options in environmental policy are about to be exhausted – at least in western Germany – and that for a number of pollutants Germany is entering the area of increasing marginal costs of abatement.

If the above analysis is correct, the reliance of environmental policy on voluntary agreements, subsidies, regulation, information and on education may not now be enough to create the appropriate incentives for further emission reductions at least cost. Since additional reductions might only be achieved at rising marginal abatement costs there will be a greater need to take an economy-wide approach when considering environmental policy, and to integrate sectoral policy more closely in such a framework. In moving ahead, attention will have to be given to defining environmental targets more clearly and to rebalancing the regulatory framework with a greater use of economic instruments. The latter are more efficient at ensuring least cost abatement provided they are broad-based. In that case, environmental policies might result in a "double dividend", in the sense of meeting labour market and environmental goals simultaneously, but a more selected approach could lead to an inefficient choice and setting of policy

instruments. The ecological tax reform of 1999 can be seen as an example of how labour-market and competitiveness issues can compromise the achievement of environmental policy targets. As a result, neither target is likely to be met effectively. In rebalancing the policy framework, a key condition will be to develop a regulatory framework which is compatible with the use of economic instruments, where these are appropriate. Proposals to unify environmental policies into a Code of Environmental Law (U*mweltgesetzbuch*) have been made by academics[126] and by a commission of the Ministry of the Environment[127] and need to be implemented.

Economic instruments may be difficult to use in some complex cases where monitoring costs are high or where the emissions are unusually complex, but in many cases they may offer advantages. For example, the environmental problems related to traffic and agriculture are difficult to control by regulatory instruments. Traffic-related emissions come from a large number of small and mobile sources which are difficult to regulate directly. Agricultural non-point pollution problems can not be directly controlled at the source. More generally, not all of the incentive structures support rational product design and development. Hence, a correction of prices through the use of economic instruments may be a superior choice. Economic instruments might also be highly effective in lowering CO_2 emissions. To this end a priority will be to set energy tax rates more rationally so as to reflect relative emission intensities. Alternatively, tradeable permits might be considered which would fit more closely with proposed international quota mechanisms. In the case of global-warming issues, however, German domestic policy with respect to CO_2 emissions cannot hope to achieve much in isolation although the demonstration effect is certainly important. Rather, policy also needs to be directed at encouraging the development of taxation and permit trading at an international or European level and at establishing project-based mechanisms. Germany is a good example of why such instruments could offer great advantages. Arguably, the investment in environmental protection in the new Länder has brought more returns in terms of improved environmental quality in Europe than if the same resources had been used in the old Länder.

A key determinant of the efficiency of potential policy measures – either of a sectoral or overall focus – is the efficiency of individual markets. Past S*urveys* have drawn attention to a number of inefficient markets, which will in turn lower the efficiency of policies directed towards the environment and might themselves serve to worsen the outcome. The agriculture sector is a good example of distorted markets leading to negative environmental affects in some regions, as are subsidies for coal production. Urban sprawl which, *inter alia*, impacts on land use, biodiversity and transport associated pollution is in part encouraged by the pattern of public support for homebuilding, and rent laws fail to give sufficient incentive to tenants and proprietors to save energy.[128] At a minimum, some form of comparison between rents gross of heating costs could be considered which

would minimise changes to the existing legal framework.[129] The electricity market has been highly regulated up to recently and although it has held energy prices high, this has scarcely been for the right reasons and innovation and energy conservation have been discouraged. A factor which has been holding back a more rational approach to policy instruments has been the unresolved issue of nuclear energy, as a reliable timetable for its phasing out still needs to be agreed. In all the above cases, the relevant markets are either practically non-existent or are still highly distorted. Effectively addressing the microeconomic reform issues involved would lead to both increased economic efficiency and a more cost-effective achievement of environmental targets.

Glossary to Chapter IV

Acidification	Caused by atmospheric emissions of sulphur dioxide and nitrogen oxides which, through "acid rain", lead to deposition of sulphur and nitrogen in the soil and water surfaces (causing damage to aquatic life and forests)
Acid rain	Describes a type of pollution which washes out of the atmosphere as dilute sulphuric and nitric acids
Anthropogenic	Describes an object or disturbance to the environment which is man-made or which follows from the emissions of harmful substances as a result of human activities
BAU	Business As Usual
Bcm	Billion cubic metres
Biomass	The total quantity of weight of organisms in a given area or volume
Biotope	The smallest subdivision of a habitat, characterised by a high degree of uniformity in its environmental conditions and in its plant and animal life
CH_4	Methane
CHP	Combined production of heat and power; sometimes, when referring to industrial CHP, the term "co-generation" is used.
CO_2	Carbon dioxide
End-of-pipe	Technology which removes pollutants from gases or liquids after the industrial process and before discharge into the environment
Eutrophication	A process of pollution that occurs when a lake or stream becomes over-rich in algae or aquatic plants due to nutrient leaching (nitrogen and phosphorous) which kill animal life by depriving it of oxygen
FCCC	Framework Convention on Climate Change
FGD	Flue-gas desulphurisation
GHG	Greenhouse Gas
GW	Gigawatt, or 1 watt $\times 10^9$
GWh	Gigawatt-hour = one gigawatt $\times$ one hour, or one watt $\times$ one hour $\times 10^9$
ha	Hectare
HFC	Hydrofluorocarbons
IEA	International Energy Agency
kW	Kilowatt, or 1 watt $\times 10^3$
kWh	Kilowatt-hour = one kilowatt $\times$ one hour, or one watt $\times$ one hour $\times 10^3$
Mcm	Million cubic metres
Mt	Million tonnes
Mtoe	Million tonnes oil equivalent; see toe
MvA	Megawatt of electricity or 1 watt $\times 10^6$

MWh	Megawatt hour
NMVOC	Non-methane volatile organic compounds
N_2O	Nitrous oxide
NO_x	Nitrogen oxides
Particulates	Tiny flecks of soot and similar substances in the atmosphere. A principal source of particulates is black exhaust from road vehicles
PFC	Perfluorocarbons
Sequestration	Term used in Kyoto Protocol. Absorption of CO_2 by a sink, often a growing forest
Sink	In this context, a place where, or a process by which, greenhouse gases are removed from the atmosphere
SO_2	Sulphur dioxide
SO_x	Sulphur oxides
TFC	Total final consumption of energy; the difference between TPES and TFC con sists of net energy losses in the production of electricity and synthetic gas, refinery use and other energy sector uses and losses
Toe	Tonne of oil equivalent, defined as 10^7 kcal
Tonne	Metric ton (10^3 kilograms)
TPES/GDP	A ratio of total primary energy supply (in Mtoe) to GDP (measured in billions of US dollars at 1990 prices and exchange rates); energy intensity
TPES	Total primary energy supply
TW	Terawatt, or 1 watt $\times$ 10^{12}
TWh	Terawatt $\times$ one hour, or one watt $\times$ one hour $\times$ 10^{12}
VOC	Volatile Organic Compound

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Notes

1. Moreover, the tabled draft law needed a number of technical corrections and additional discussions, all of which appear to have depressed the business climate. Share prices also under-performed those in other European countries.
2. On an old national accounts basis since, at the time of writing, household accounts were not available on the new base.
3. The branches are: bureau machinery and equipment for data processing; aerospace; equipment for broadcasting and message transmission; automobile construction; and appliances in optics, measurement, control engineering. See: Deutsches Institut für Wirtschaftsforschung, Berlin, Institut für Weltwirtschaft, Kiel, Institut für Wirtschaftsforschung Halle, *Gesamtwirtschaftliche und unternehmerische Anpassungsfortschritte in Ostdeutschland*, Halle 1999.
4. See: Deutsches Institut für Wirtschaftsforschung, Berlin, Institut für Weltwirtschaft, Kiel, Institut für Wirtschaftsforschung Halle, *op. cit.*
5. The switch to the euro has involved some changes to the structure and operation of the German monetary system. To avoid competitive distortions, compulsory bank reserves are now remunerated. A major change to the financial system is that commercial banks can no longer endorse bills of exchange (*Wechsel*) with the central bank for a discount. The discount facility, which was a special feature of the German refinancing system, was used extensively by credit institutions as a base for refinance. However, the consequences for liquidity management by banks is limited since the recourse to discount credit was restricted in volume terms and bills of exchange are still recognised as collateral for central bank credits.
6. However, these measures are likely to overestimate real rates as measured in terms of inflationary expectations: since the extraordinarily slow increase or even fall in prices experienced in 1998 are likely to constitute only a temporary phenomenon, inflation expectations – although low – might exceed the inflation rates measured in 1998.
7. See *Economic Outlook* 66 for a discussion of the prospects for the euro area.
8. Federal tax receipts exceeded the official projections on which the 1998 budget was based by DM 9.7 billion (¼ per cent of GDP). Reasons for unexpectedly low tax receipts in the past were considered in the 1998 OECD *Economic Survey of Germany*.
9. Local government balances in the west swung from a deficit of DM 5½ billion in 1997 to a surplus of roughly the same size, and in the east communities were able to reduce their deficit from DM 1½ billion in 1997 to DM 0.8 billion the following year. Although this outcome is partly attributable to the upswing in tax revenues, it is dominated by spending reductions.

10. No further efforts have been made to establish a domestic stability pact to prevent "excessive" deficits by one layer of government at the expense of the others.

11. In 1999 the federal government has taken on the debt of three off-budget funds – the Inherited Debt Funds (*Erblastentilgungsfonds*), the Railway Funds (*Bundeseisenbahnvermögen*) and the Coal Funds (*Ausgleichsfonds zur Sicherung des Steinkohleeinsatzes*) – totalling DM 390 billion or 10½ per cent of GDP. Since the federal government has been responsible in the past for the debt servicing of these funds, this step increases the fiscal transparency of the public sector. This change is reflected in higher interest payments in 1999. Transfers to off-budget funds are reduced accordingly. However, total transfers do not decline due to offsetting increases in transfers to the pension system and the Post Office.

12. The government's stated objectives are still in terms of the old system of national accounts with the budget projected to decline from 2 per cent this year to 1½ per cent in 2000.

13. The "golden rule", which is in the constitution, obliges new debt issues to be equal to or less than government investment. Privatisation receipts are, in this definition, counted as revenues.

14. In these simulations disturbances of the economic environment comprise real supply shocks, real private demand shocks and nominal shocks. For details see "Estimating prudent budgetary margins for 11 EU countries: a simulated SVAR model approach", OECD, *Economic Department Working Paper No. 216*.

15. Distortions arise from the business trades tax because tax breaks imply that a large share of small and medium-sized companies are exempted from the tax. Moreover, the marginal tax rate is very high.

16. The present tax system discriminates against external equity finance in favour of borrowing in that interest paid is fully deductible from the tax base, while distributed profits are not. This deficiency cannot be remedied by giving a tax preference in favour of retained profits.

17. This imbalance is even larger if social security contributions, the "solidarity" income tax surcharge and the business trades tax are taken into consideration as well.

18. A ruling of the Federal Finance Court (*Bundesfinanzhof*) of February 1999 has been widely interpreted as implying that the differential tax treatment of business and non-business income even in the present split rate system might be unconstitutional.

19. In fact, the present split rate system in combination with high and increasing social security contribution rates appear to have contributed to the increase in "apparent self employment" (*Scheinselbständigkeit*) which is the target of new provisions in social security legislation, see Chapter III below.

20. The problem in the eastern German health system is that high unemployment and lower wages than in the old states have lowered the contribution base while health expenditures have tended to equilibrate with those in the west. Per capita pharmaceutical costs are particularly high.

21. Health is essentially a Länder responsibility so that the upper house of parliament will probably need to approve the legislation.

22. The contribution base refers to the growth rate of that part of wages used to calculate contributions for the country as a whole. For the new Länder, the highest of either

the growth of the wage base in the new Länder or in Germany as a whole will be used.

23. This provision will serve to control supply both by limiting the registration of more specialists as well as by limiting any potential influx of new general practitioners from hospitals.

24. This will require that the Länder give up their bed plans.

25. To compensate the health funds for additional expenditures of some DM 8 billion per year several non-health benefits (funeral and pregnancy allowances) will be transferred to the states.

26. There are also cases where collective agreements between an employers' association and a union refer to individual companies.

27. Bundesministerium für Arbeit und Sozialordnung, *Tarifvertragliche Arbeitsbedingungen im Jahre* 1998, Bonn 1999. For a summary see: Clasen, Lothar, "Tarifentwicklung 1998 – Differenzierte Abschlüsse", *Bundesarbeitsblatt*, 3/1999, Bundesministerium für Arbeit und Sozialordnung, Bonn 1999. Detailed examples of tariff contracts in various branches are contained in: Bundesministerium für Arbeit und Sozialordnung, *Tarifvertragliche Regelungen in ausgewählten Wirtschaftszweigen*, Bonn, 1999. In 1998, 76 per cent of employees in the west, and 63 per cent in the east were employed in enterprises which were bound by tariff agreements.

28. In the metal and electrical industry, which has often been the trend setter for tariff agreements in Germany, the number of member firms in the employers' association (*Gesamtmetall*) in the west declined from 57.5 per cent in 1980 to 44 per cent in 1997. The number of workers employed within member firms in the old Länder declined from 2.95 million in 1980 to 2.08 million in 1997. In the new Länder the number declined from 944 000 in 1990 to 108 000 in 1997. The evolution is also heavily influenced by sectoral shifts and labour shedding particularly in the east. Local branches of the metal employers' association are responding to the trend to leave the organisation out of discontent with tariff agreements by introducing a new type of membership which does not imply coverage by a tariff contract.

29. Joint study by the Institute of the German Economy and the Institute for Labour market and Employment Research, 1999. The investigation is based on a firm panel of the IAB, comprising 9 000 enterprises, which is considered to be representative for the private sector of the German economy. See: Kohaut, S. And C. Schnabel, *Tarifbindung im Wandel*, iw-trends z, 1999.

30. The study found that about half of the enterprises base their wage payments on branch-wide tariff agreements even if they are not legally bound to do so. 44 per cent pay wages below tariff, and 10 per cent above tariff. See: Laschke, B., "Industrieumfragen 1998 – Dokumentation der Hauptergebnisse", *Institut für Wirtschaftsforschung Halle*, *Diskussionspapier* 86, Halle 1999.

31. The figures reflect the number of such contracts in January of the respective year.

32. The term "subsidy" is not entirely appropriate since social contributions are not risk-based but are related only to current income. If the social security system were entirely paid from tax revenues as they are in a number of countries, the issue would be one of setting low or even zero rates of income tax for low wage levels and the debate would not be conducted in terms of the granting of subsidies but in terms of progressivity of the tax code.

33. A number of simulations have been made, assuming that no social security contributions need to be paid for monthly earnings of up to DM 1 400 or DM 1 500 and that the subsidies are gradually reduced to zero at earnings of DM 2 800 or DM 3 000. For part-time employment, financial support accrues proportionately. Estimates for the number of new jobs that might be generated under this scheme range from 50 000 to 220 000. Under the assumption that 150 000 new jobs are generated and wages remain unchanged, the DIW (1999) projects the gross costs of the scheme to total around DM 14 billion. The net costs – after savings in social benefits and additional tax receipts – are projected to amount to some DM 12 billion. See also: Bender, Stefan, Bruno Kaltenborn, Helmut Rudolph, and Ulrich Walwei, "Die Diagnose stimmt, die Therapie noch nicht", IAB *Kurzbericht* No. 6, 1999; Buslei, Hermann, Viktor Steiner, Felix Brosius, Peter Jacobebbinghaus, and Ralf Martin, *Beschäftigungseffekte von Lohnsubventionen im Niedriglohnbereich*, Zentrum für Europaische Wirtschaftsforschung, Mannheim, May 1999.

34. According to a survey conducted by the Chamber of Commerce in summer 1998, amongst those enterprises affected by the legislation, 11 per cent reported they had hired employees because of the liberalisation. Deutscher Industrie und Handelstag, *Impulse für den Arbeitsmarkt*, Bonn, August 1998.

35. See OECD *Economic Surveys of Germany* (1997 and 1998) for more details.

36. Information from the establishment panel of the Institute for Employment Research of the Federal Labour Office indicates that the growth in minor employment observed in recent years reflects both additional employment and a break-up of existing jobs into several casual positions. See Institut für Arbeitsmarkt- und Berufsforschung, *The change in employment forms – empirical results and first explanatory approaches*, Nürnberg 1999.

37. The new legislation became effective in April 1999. For persons holding only one casual job, employers are required to pay 12 per cent social security contributions for the pension system and 10 per cent for the health funds. These contributions replace a 20 per cent lump sum income tax payment in the old system. There are no further obligations to pay taxes or contributions on the employee's side, but they can opt to increase their pension claims by voluntary contributions to the system. If minor employment is combined with regular employment, both the employer and the employee pay some 22 per cent social security contributions. This is a change from past practice where no cumulation for the employee was the rule. On the employer's side these payments replace the 20 per cent lump-sum income tax. In addition, the employee has to pay taxes on his combined earnings and needs a wage tax card. Under special conditions, the obligation to pay contributions to the health funds does not apply. Whereas in the past the earnings ceilings for minor employment were periodically revised upwards, in line with the development of wages, they are now frozen at DM 630 in the west and in the east. Marginal jobs are therefore bound to become less important over time. Persons considered to be apparently self-employed have to pay social security contributions as if they were in dependent employment and are considered as such if they meet two of the following four conditions: they have no employees who are liable to pay social security contributions; they have just one customer; their mode of occupation is typical for dependent employment; they do not act in the market in an entrepreneurial way. The legislation has proved too sweeping and is currently being reviewed.

38. Deutscher Industrie- und Handelstag, *Mehr Kosten, Weniger Jobs*, Bonn, 1999.

39. Joint study conducted by the Institut für Sozialforschung und Gesellschaftspolitik, Bonn, and Kienbaum Unternehmensberatung, Gummersbach; quoted in *Frankfurter Allgemeine Zeitung* 27 July 1999.

40. A somewhat higher rate of job openings has recently been observed in some fields with above-average minor employment, but this could be due to a wide range of factors.

41. As has been argued in previous *Surveys*, active labour market programmes (ALMPs) can be a valuable instrument to improve employment prospects for unemployed people with particularly disadvantageous labour market characteristics. A summary of some new ALMPs applied in Germany is given in OECD *Economic Survey of Germany* (1998).

42. The stock of persons participating in ALMPs was subsequently somewhat reduced. In May 1999 367 000 people in the east participated in work provision schemes and 147 000 in training measures.

43. This figure also contains income support for employees designed to safeguard existing employment in periods of temporary labour demand slack, *e.g.* support for temporarily reduced working hours (*Kurzarbeitergeld*) or "bad weather support" for construction workers (*Schlechtwettergeld*).

44. See OECD *Surveys of Germany*, 1997 and 1998 for descriptions of active labour market programmes. The Structural Adjustment Measures provide jobs in defined fields such as environmental, social and cultural services, monument conservation and urban renewal.

45. Fay, Robert G. (1996), "Enhancing the effectiveness of active labour market policies: evidence from programme evaluations in OECD countries", OECD *Labour Market and Occasional Papers*, Paris, No. 18. Martin, John P. (1999), "What Works among active labour market policies: evidence from OECD countries' experiences", OECD *Labour Market and Occasional Papers*, Paris, No. 35.

46. If related to the working-age population, the number of persons in training and work provision schemes is seven times higher in the east than in the west.

47. Several factors might explain the low effectiveness of active labour-market programmes. During the phase of participation, persons in ALMPs have less time to actively search for a job than recipients of unemployment benefits. Eligibility for unemployment-related income support is also effectively prolonged by participation in ALMPs. The effective lengthening of the benefit period should reduce search intensity for a regular job. This might interact with a slow downwards adjustment of reservation wages if wages paid in schemes exceed market wages. The schemes may also add little to marketable skills of the unemployed. In part this is attributable to the fact that subsidised employment in employment organisations (*Beschäftigungsgesellschaften*) are often of little relevance for market activity. Furthermore, the high subsidisation of labour may lead to an inefficient combination of capital and labour in subsidised production which does not reflect the capital endowment of working places in the market. In extreme cases this might even imply a loss of qualifications with respect to market-relevant skills. With these deficiencies, participation in a job creation scheme might give a negative signal to potential employers ("stigmatisation"). This would be less of a problem for schemes which provide market relevant training.

48. There is some evidence on crowding-out effects in gardening and landscape architecture. Results from a full census of commercial enterprises in this field indicate

that the companies changed activity patterns to escape competition from subsidised jobs.

49. Institut für Arbeitsmarkt und Berufsforschung, "Neue Bundesländer: Arbeitsmarktentwicklung und aktive Arbeitsmarktpolitik im ostdeutschen Transformationsprozeß 1990-1996", IAB-*Werkstattbericht*, No. 5, 1997.

50. Institut für Wirtschaftsforschung *et al.* (1999), *Gesamtwirtschaftliche und unternehmerische Anpassungs-fortschritte in Ostdeutschland – Neunzehnter Bericht*, Halle.

51. For example, the transmission fee is only operative for transmission of more than 100 km even though there is no ground for supposing that higher costs are involved. The high cost also does not favour development of an electricity exchange. V. Lang, "Die Rolle kommunaler Unternehmen im Strommarkt", *Wirtschaftsdienst*, No. 4, 1999.

52. To put the issue in perspective, it is important to note that although there are 900 such producers they account for only 10 per cent of electricity production although they distribute about a third of the electricity which is consumed. Three large producers, which also have stakes in the commune based generators, account for 53 per cent of the power produced and the largest six account for 80 per cent.

53. Länder have also sought to develop the cable network for broadcasting which would reduce the potential for it to be developed for telecommunications. The federal government is currently holding discussions with the Länder about this issue.

54. The telecommunications regulator can use either price cap regulations or cost comparisons in setting prices and can be concerned as much with low prices set by the old monopolist to hinder competition as by high prices to exploit a market position. The powers are thus in principle wide enough to encourage competition. See H. Lamouroux, "Price-cap Regulierung und Telekommunikationsgesetz", *Wirtschaftsdienst*, No. 5, 1999.

55. See BMU (1998*a*).

56. BMU (1997).

57. The process of making the concept of sustainability operational has proceeded along several complementary paths and is not yet complete. At the political and consensus building level, talks were started in June 1996 with 130 representatives of industry, environmental and consumer organisations, trade unions, scientists and all levels of government with a view to forming a national environmental policy plan for sustainable development. Although no consensus about the role and contents of such a plan was reached, a programme of measures was drawn up in April 1998. A number of local governments have also established their own plans (the Agenda 21 process). The present government wants to continue this debate with a view to setting concrete policy targets.

58. According to the Report to the Special Session of the United Nations General Assembly on Environment and Development in 1997 in New York this process of discussion about specific objectives has not been completed yet. BMU (1997).

59. These management rules were proposed in the 1996 Report of the German Council of Environmental Advisors (SRU, 1996) and by the Enquete Kommission of the German Parliament on the Protection of Humanity and the Environment (Deutscher Bundestag, 1998*b*).

60. Bundesregierung, IWG, (1998).

61. DIW, (1999*a*).

62. For a summary of the rates of reduction of pollutants see OECD (1993). This process has continued and the water quality in surface waters in the new Länder is improving as well.
63. Klepper *et al.* 1995*a*.
64. The environmental effect of the energy tax is considered by most observers to be small, one study reporting an optimistic scenario of at most a 2 per cent further reduction in carbon dioxide emissions. ZEW (1999). Moreover, estimates of the environmental impact from the planned increases of the energy tax rate in the period 2000 to 2003 are also small. RWI (1999*b*).
65. Jänicke and Weidner, 1997.
66. Enquete-Commission 1995, Deutscher Bundestag 1997.
67. It is not clear how externalities have been valued and whether they are consistent between studies. BMU, (1998*c*).
68. The European regulation of emissions from large combustion facilities was based on the German Ordinance, illustrating the possibility for some large countries to lead by setting tough standards.
69. The quotas for refillable containers in the packaging ordinance or the planned prohibition of sales of mineral water in one-way bottles (which has been withdrawn) are just two examples. The voluntary take-back guarantee for used cars has also been criticised by foreign car-makers as discriminatory for manufacturers not located in Germany.
70. For example, Finland had a carbon tax but to protect local electricity generators they imposed an import tax for electricity. This was ruled to be out of line with EU regulations. See OECD *Economic Survey of Finland*, 1999.
71. BMU (1997), p. 8.
72. Rennings *et al.*, 1997.
73. Two examples give an indication of the size of the task. In the lignite mining area, more than 160 open cast pits needed levelling, 38 500 dumping grounds were in need of rehabilitation and 750 deposits of industrial waste had to be treated. Up to 1996, 20 000 hectares of land had been rendered reusable and 390 million trees had been planted. From 1993 to 1997 some DM 7.1 billion and for 1998 to 2002 some DM 6 billion of funding has been made available. In the case of the uranium mining and industrial complex at Wismut some DM 13 billion has been made available for rehabilitation.
74. Employment in environmental protection is often put forward not as an indicator of cost but as an indicator of collateral benefits from environmental policy. DIW *et al.*, 1996. This is not appropriate and confuses labour market issues with questions of foregone income from environmental policy measures. In effect labour is being valued at zero opportunity cost.
75. A great deal of effort has been devoted to devising sustainability indicators and over a three-year trial to 1999, 134 indicators are being tested. These indicators cover social and institutional headings pointing to the wide definition of sustainable development which has been adopted in German political dialogue.
76. For the case of water charges and water management. See OECD 1999*c*.
77. Germany's forest area is 46 per cent privately owned and of the remainder 20 per cent is owned by local communities and 30 per cent by the Länder. State forests are

centrally managed by each federal state which also exert some influence on local forest administration.

78. BMU, (1998*d*).
79. BMU, (1998*d*).
80. Bundesregierung, IWG, (1998).
81. All these numbers are taken from Stein and Strobel (1997), Tables 84-87.
82. Bundesregierung, IWG, (1998).
83. Bundesregierung, IWG, (1998).
84. IEA, (1997), p. 146.
85. A back-of-the envelope calculation suggests the magnitudes which could be involved. With electricity production projected to account for 36 per cent of CO_2 emissions in 2005, and assuming all 30 per cent of nuclear capacity would be replaced at average levels of CO_2 emissions, the projected decline of CO_2 in comparison with 1990 would all but disappear.
86. Buttermann and Hillebrand (1998).
87. Hentrich (1999).
88. Klepper and Scholz (1998).
89. The tax rates also vary widely if they are calculated not in terms of carbon dioxide but in terms of energy units (*i.e.* joules).
90. It is not possible to be categorical at this stage since externalities associated with other emissions need also to be considered.
91. One option would have been to impose a tax on fuel rods for nuclear power plants along the lines of schemes already in force in the Netherlands, Sweden and Japan. But the externality argument for this is weak and it would not have resolved the issue of imports of nuclear power.
92. Emissions of pollutants into the air are predominantly controlled through the Federal Emission Control Act (B*undesimmissionsschutzgesetz* BImSchV) of 1974, the Ordinance on Large Combustion Facilities (G*roßfeuerungsanlagenverordnung*) of 1983, the Ordinance on Small Combustion Facilities (K*leinfeuerungsanlagenverordnung*), and the Technical Instructions on Air Pollution (TA L*uft*). These regulations control power stations, industrial plants, livestock facilities, cars, and household appliances including heating systems. They are tightened according to improvements of the best available technologies.
93. Heister *et al.* 1990.
94. I*bid*. The NO_x-study also found that the permissible air-sheds need to be quite small and will result in very thin markets for emissions permits or in locally differing emission taxes. The computational and administrative requirements for localised emission taxes could be substantial.
95. For a wider review of experience in implementing permit schemes see OECD 1999*d*.
96. SRU 1998.
97. Water quality in surface waters is regulated by the Federal Water Management Act. The federal government has only authority to establish framework legislation whereas the administration and control is given to the Länder and county authorities. An exception is the waste water charge which is part of the Federal Water Management Act and which is administered on the federal level.

98. The vehicle tax has been reformed so as to be more closely related to the estimated level of emissions.

99. SRU, (1996). The expert council for the environment in its 1996 report stated: "The waste water charge has deteriorated from an incentive charge towards a pure financing instrument and the argument for levying the charge as well as the underlying economic argument are not convincing".

100. See *e.g.* Meyer-Renschhausen (1990) or Maas (1987).

101. SRU, (1996).

102. Nuclear waste issues are discussed at length in IEA, 1998.

103. The waste management law, the Law on Circular Flow Economy and Waste Management (*Kreislaufwirtschafts- und Abfallgesetz*, KrW-/AbfG) came into force in October 1996.

104. For a practical example applied in the United Kingdom of a landfill tax oriented to these ends see David Pearce, "Cost Benefit Analysis and Environmental Policy", *Oxford Review of Economic Policy*, Vol. 14, No. 4, 1998, and articles therein.

105. Sachverständigenrat für Umweltfragen, *Umweltgutachten* 1998, Wiesbaden 1998.

106. Waste designed to be re-processed might be transported without legal restrictions over any distance, while there are legal restrictions on the transportation of waste intended to be dumped.

107. The repercussions of this inefficient arrangement became clear when households also deposited non-packaging waste of the same material in the waste containers for packaging material. In 1993, non-packaging material (including non-recyclable "residual waste") accounted for 40 per cent of the waste deposited in the bins of the voluntary Dual System. This contributed to a financial crisis of the scheme in the first half of the 1990s, since the Dual System was obliged to compensate waste processors for the collected non-packaging waste as well. See Benzler, Guido, Marianne Halstrick-Schwenk, Paul Klemmer and Klaus Löbbe, "Wettbewerbskonformität von Rücknahmeverpflichtungen im Abfallbereich", *Untersuchungen des Rheinisch-Westfälischen Instituts für Wirtschaftsforschung*, No. 17, Essen, 1995.

108. Sachverständigenrat für Umweltfragen, *Umweltgutachten* 1998, Wiesbaden 1998.

109. In fact shredder waste – waste left after all other recoverable materials have been removed – only amounts to 1.5 per cent of the domestic waste volume. It is however classified as hazardous.

110. In the Dual System, some firms did not acquire a "Green Point" licence but benefited from the fact that participating firms contributed to fulfilling the prescribed reprocessing quota collectively. To close this loophole the ordinance was revised in 1998, now requiring firms to prove that they either fulfil their disposal obligations individually or participate in a collective system.

111. OECD, *Economic Survey of Germany*, 1998.

112. IEA, (1998), pp. 71 and 119.

113. Heinloth, (1997).

114. Storchmann and Kyro, (1997).

115. Neu, (1995).

116. See OECD *Economic Survey of Germany*, 1997.

117. Beermann, (1998).

118. "CO_2 Emissionen in Deutschland: Weiterhin vom Zielpfad entfernt", DIW *Wochenbericht*, 6/99, February, (1999*a*).
119. *Efficient Transport for Europe: Policies for internalisation of External Costs*, ECMT, OECD, Paris, 1998.
120. Kageson, (1993).
121. See *Entwicklung eines Verfahrens zur Aufstellung umweltorientierter Fernverkehrskonzepte als Beitrag zur Bundesverkehrswegeplanung*, Umweltbundesamt, Berlin, March 1998.
122. "Energieverbrauch im Verkehr in Deutschland", DIW *Wochenbericht*, 10/99, March (1999*b*).
123. H.J. Luhmann, "Die neue Kfz-Steuer ist halbherzig", *Der Tagesspiegel*, 25 June 1997. The need for a greater differentiation of tax rates is reported in the same article to be based on calculations by the Bundesumweltamt.
124. The OECD's *Environmental Performance Review of Germany* in 1993 emphasised that Germany uses nitrogen fertilisers relatively intensively, concluding that from a social point of view fertiliser used per unit of area is too high.
125. *Agriculture and the environment: Issues and policies*, OECD, Paris, 1998
126. Kloepfer *et al.* (1990).
127. BMU (1998).
128. Hentrich, (1999). Some estimates point to potential energy savings of up to 90 per cent in old buildings. There is a special loan programme to encourage better insulation of accommodation if carried out as part of a renovation.
129. There are proposals to include information about heating costs when providing specifications of a house for sale.

Annex I

The effectiveness of active labour market measures in eastern Germany

Active labour market measures (ALMPs) have been used extensively in the new Länder, whereas in the old Länder they have been applied on a much more limited scale. Accordingly, most evaluation studies of the effectiveness of ALMPs in enhancing labour-market performance refer to the east.

The primary goal of ALMPs is to foster the transition from unemployment into regular employment so that studies have focused on this transition. For example, survey information on exits from participation in an ALMP over an eight-year time span in Saxony-Anhalt indicates that, on average, roughly one third of participants in training measures exited into employment (Table A1). The bulk of participants, some 60 per cent, re-entered unemployment while around 6 per cent moved to a job-creation scheme. The proportion exiting to regular employment was lower for job-creation schemes at around a fifth.

The greater effectiveness of training measures in comparison with job-creation schemes is also indicated by comparisons of re-entry frequencies into the unemployment benefits register. For a summary of findings derived from the benefits statistics see: Sachverständigenrat zur Begutachtung der gesamtwirtschaftlichen Entwicklung, *Jahresgutachten* 1995/96, Stuttgart 1995.

Descriptive approaches are, however, methodologically weak and econometric studies which compare the labour-market outcome of participants in ALMPs with those of unemployed non-participants while controlling for potential selectivity in programme participation are preferable. Table A2 summarises the findings from econometric studies for eastern Germany. They are all based of longitudinal data sets of individual labour market transitions. The methodological framework rests either on a hazard rate approach, which models conditional exit probabilities for the sample of participants and non-participants, or a

Table A1. **Exits from ALMPs in Saxony-Anhalt 1990-1997**

Per cent

From/To	Unemployment	Employment	Inactivity	Training schemes	Job creation schemes
Training schemes	60.8	31.3	2.2	. .	5.7
Job creation schemes	64.9	22.4	4.8	7.9	. .

Source: Institut für Wirtschaftsforschung, Halle, 1999.

Table A2. **Econometric studies on the effectiveness of ALMPs in eastern Germany**

Author	Time span/data set	Impact on employment	
		Job-creation schemes	
Eichler, Lechner, 1999	1992-97/LMMSA	Positive	
Hübler, 1997	1993-94/LMME	Males: Females:	Negative Insignificant
Kraus, Puhani, Steiner, 1998	1990-94/LMME	Negative	
Steiner, Kraus, 1995	1990-92/LMME	Males: Females:	Insignificant Negative
		Training schemes	
Fitzenberger, Prey, 1995, 1998	1990-94/SOEP	Outside of firm: Inside of firm:	Positive Not positive
Hübler, 1997	1993-94/LMME	Males:	Positive
Kraus, Puhani, Steiner 1997	1990-94/LMME	1990-92: 1992-94:	Negative Positive
Lechner, 1999	1990-96/SOEP	Insignificant	
Pannenberg, 1995	1990-92/SOEP	Insignificant	
Pannenberg, 1996	1990-94/SOEP	Positive	
Staat, 1997	1992-94/SOEP	Insignificant	

LMME: Labour market monitor eastern Germany (*Arbeitsmarktmonitor Ostdeutschland*).
LMMSA: Labour market monitor Saxony-Anhalt (*Arbeitsmarktmonitor Sachsen-Anhalt*).
SOEP: Socio-economic panel (*Sozio-ökonomisches Panel*).
Source: OECD.

statistical matching approach, which constructs a comparison group of non-participants and then compares average outcomes for both groups. While the results of these studies differ, they nevertheless point to modestly positive employment effects for participants in training programmes. By contrast, most studies conclude that the employment effects of participation in job-creation schemes are either insignificant or negative.

Efficiency costs

A full evaluation of ALMPs would also need to take into account potential deadweight costs and worker displacement effects (deadweight costs concern expenditures which are actually not necessary to achieve a given labour-market outcome, while displacement effects refer to the possibility that aggregate employment will not change and that there will be only a substitution between individuals). Research in other countries suggests that job-creation/work-provision schemes are associated with high deadweight costs and displacement effects (Fay, 1996; Martin, 1998). Little empirical evidence is, however, available for Germany.

Preliminary indications about deadweight costs are available from an ongoing research project of the Institute for Labour Market and Occupational Research of the Federal Labour Office (*Institut für Arbeitsmarkt- und Berufsforschung der Bundesanstalt für Arbeit*, IAB). As part of the project, 149 sampled companies – 95 in the west and 54 in the east – were asked about the impact of wage subsidies on hirings. Fourteen per cent of the sampled companies in the west and 22 per cent in the east answered that the wage subsidies they receive over-compensated for the lower productivity of the supported workers they had hired. At 32 per cent this share was highest for the structural adjustment programmes (SAM O/W) granted in the east. (The share of such schemes in all wage subsidisation programmes for enterprises in the new Länder totalled 80 per cent in December 1998.) Similarly, 28 per cent in the west and 30 per cent in the east replied that they would have hired the same worker without financial support for the person hired. In the east, the share was 27 per cent for companies receiving structural adjustment support. Another 17 per cent in the west and 22 per cent in the east answered that without support for the person hired, they would have employed another person. The latter response does not necessarily indicate the size of the displacement effect because in some cases companies would have hired another supported person.

The overall validity of such results is limited by the fact that the sample size is relatively small and labour demand responses are not assessed against those of a benchmark sample of firms receiving no aid. Nevertheless, managers are likely to understate the benefits for their companies derived from the wage subsidies. Hence, the actual share of enterprises where wage subsidies overcompensate for the lower productivity of supported workers might be substantially higher than reported. The same holds for responses to the question whether the same or another person would have been hired without the subsidies.

The conclusion appears to be that ALMPs and subsidies need to be closely targeted on groups with specific handicaps if deadweight costs are to be minimised. Worker displacement effects might nevertheless be difficult to contain.

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Annex II

Projections of emissions to 2005

In view of the detailed regulatory instruments which are a feature of German environmental policy, projecting CO_2 emissions presents a number of challenges and uncertainties. The IKARUS model (Stein and Strobel, 1997) attempts to establish a projection framework by specifying various technologies at a quite detailed sectoral level. This allows projections to be related to specific policy measures and sectors. As an optimisation model, it has also been used to calculate shadow prices for different levels of efficient emission control. Criticisms of such an approach include the macroeconomic assumptions and the way in which relative prices are allowed to impact on energy use and emission levels.

Estimates of carbon dioxide emissions derived from this model indicate that, taking account of all policy decisions made up to 1996, Germany is likely to fall short of its target emission level in 2005 by a large margin (Table A3). Subsequent policy initiatives, including the energy tax, which became effective in April 1999, have not been sufficient to alter this conclusion substantially. The environmental effect of the energy tax is considered by most observers to be small, one study reporting an optimistic scenario of at most a 2 per cent further reduction in carbon dioxide emissions (ZEW (1999)). Further announced increases for 2000 and 2001 would not seem to alter this conclusion significantly. The major contribution in terms of carbon dioxide reductions comes from activities not related to traffic. Total emissions of carbon dioxide in 2005 are expected to be 863.7 million tonnes, compared with 998.2 million tonnes in 1990. This is an expected reduction of 13.5 per cent compared with the target of 25 per cent set by the government. Private households, road traffic, and electricity demand are the sectors contributing in an accounting sense for the shortfall relative to target. With respect to other greenhouse gases, only for fluorocarbon (CFC) substitutes is an increase observable and this is expected to continue. The percentage reduction in carbon dioxide is below that expected for other important greenhouse gases, such as methane and nitrogen oxide (Table A4).

Nitrogen oxide emissions, having fallen in the 1980s and 1990s, are expected to increase again. The same is true for the volatile organic compounds. However, the projected near-term increase is even larger. Despite the significant reduction in emissions, progress has not been enough to arrest the damage and destruction of forests, although the large-scale forecast deterioration in German forests has been avoided (Box 6).

Table A3. **Projections of CO_2 emissions**

With policies in force as of 1996, in million tons of CO_2

	1990	1995	Scenario		Percentage change
			2000	2005	1990/2005
Industry	169.7	126.8	116.9	107.1	–36.9
Small users	75.7	51.9	61.6	56.5	–25.4
Households	128.4	135.2	115.9	110.7	–13.8
Traffic	184.9	196.1	223.0	224.0	21.1
Final energy users	558.8	509.9	517.5	498.4	–10.8
Electricity production	353.6	317.5	316.7	318.9	–9.8
District heating	42.9	31.7	30.0	27.5	–35.9
Other energy production	43.0	24.0	21.0	19.0	–55.8
Energy sector	439.4	373.2	367.8	365.3	–16.9
Total emissions	998.2	883.1	885.3	863.7	–13.5

Source: Stein and Strobel (1997).

Table A4. **Development of pollutant emissions**

In kilotonnes

	1970	1980	1990	1996	2005 (expected)	Change	
						Change 1990/1996	Change 1996/2005
						Per cent	
Sulphur dioxide (SO_2)	7 720	7 510	5 326	1 850	990	–65	–46
Nitrogen oxides (NO_x)	2 690	3 330	2 640	1 868	2 130	–31	+14
NM_x-VOCs	2 477	3 220	3 155	1 920	2 700	–41	+40
Dust	3 270	2 609	2 024	518	n.a.	–74	. .

Source: BMU (1998*a*), IWG (1998).

Annex III

Chronology of main economic events

1998

October

Following the federal parliamentary elections, the new federal government comes into office.

1999

January

The conversion rate of the Deutschemark to the euro is set at DM 1.95583 and the responsibility for setting monetary policy is shifted from the Bundesbank to the European Central Bank.

The first Stability Programme in accordance with the EU Stability and Growth Pact is approved by the Council of the Ministers of the EU.

The Tax Reform Law 1998 comes into force, which updates the valuation of company pension reserves to revised mortality rate tables.

The Tax Relief Law 1999 comes into force. It lowers statutory income tax rates and reduces the tax burden for households, with revenues foregone compensated by broadening the tax base for income from business activity.

New social legislation comes into operation which reverses certain laws introduced by the previous government, notably with respect to the demographic adjustment of pensions, the tightening of eligibility for invalidity pensions, relaxing dismissal protection, and reducing statutory sickness pay by enterprises. The powers of the Minister of Labour to extend wage agreements in construction to non-contracting parties are widened. To reduce youth unemployment, a public support scheme comes into operation, which aims at making available 100 000 places in training programmes or work provision schemes to young people being unemployed or threatened by unemployment.

New legislation comes into force, which reverses higher co-payments by patients and restrictions on the provision of health services introduced by the previous government.

March

The Minister of Finance, Oskar Lafontaine resigns; his successor is Hans Eichel.

April

Legislation comes into force extending the obligation to pay social security contributions to casual employment (jobs entailing a small number of hours per week and with pay not exceeding DM 630 per month in the west or DM 590 in the east), and to the "apparently self-employed" who are considered as dependent employees if certain criteria are fulfilled. Redundancy payments are no longer credited against unemployment insurance benefits (*Arbeitslosengeld*).

Taxes on energy are introduced, the proceeds of which are utilised to lower pension contribution rates by 0.8 percentage points.

Tax reform commission reports on proposals to reform taxation of business income.

The ECB reduces the repurchase rate by 50 basis points to 2.5 per cent.

May

The official tax revenue estimates for 1999 project additional revenues of DM 10 billion in comparison with the estimates of November 1998.

June

The government presents its medium-term fiscal consolidation package ("Future Programme 2000"). It projects budget savings rising from DM 30 billion in 2000 to DM 49 billion in 2003. Major spending measures comprise further reductions in personnel, reduced pension adjustments in 2000 and 2001, a curbing of social transfers and global spending restraints.

Draft health reform law presented to the parliament.

August

New legislation comes into force, opening active labour market measures to a wider group of workers with limited employment opportunities and revising certain eligibility requirements for unemployment benefits.

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Czech Republic	Denmark	Finland	France	Germany	Greece
Population												
Total	Thousands	1996	18 289	8 060	10 157	29 964	10 316	5 262	5 125	58 380	81 877	10 465
Inhabitants per sq. km	Number	1996	2	96	333	3	131	122	15	106	229	79
Net average annual increase over previous 10 years	%	1996	1.3	0.6	0.3	1.3	0	0.3	0.4	0.5	3	0.5
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	8 344	3 737 (94)	3 675 (95)	13 676	4 918	2 593	2 087	21 951	35 360	3 824 (95)
of which: Agriculture	% of TCE	1996	5.1	7.2 (94)	2.5 (95)	4.1	6.3	4	7.1	4.6	3.3	20.4 (95)
Industry	% of TCE	1996	22.5	33.2 (94)	26.7 (95)	22.8	42	27	27.6	25.9	37.5	23.2 (95)
Services	% of TCE	1996	72.4	59.6 (94)	71.4 (95)	73.1	51.7	69	65.3	69.5	59.1	56.4 (95)
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	398.9	228.7	268.2	579.2	56.2	174.9	125.1	1 536.6	2 353.5	91.2 (95)
Per capita	US$	1996	21 812	28 384	26 409	19 330	5 445	33 230	24 420	26 323	28 738	8 722 (95)
At current prices using current PPPs[3]	Bill. US$	1996	372.7	172.4	222	645.1	..	118	96.7	1 198.6	1 736.1	133.5
Per capita	US$	1996	20 376	21 395	21 856	21 529	..	22 418	18 871	20 533	21 200	12 743
Average annual volume growth over previous 5 years	%	1996	3.9	1.6	1.2	2.2	2	2.2	1.6	1.2	1.4	1.3 (95)
Gross fixed capital formation (GFCF)	% of GDP	1996	20.3	23.8	17.3	17.7	33	16.7	16.1	17.4	20.6	17 (95)
of which: Machinery and equipment	% of GDP	1996	10.2 (95)	8.8 (95)	7.5 (95)	6.6	..	7.9 (95)	6.4 (95)	7.8	7.6	7.7 (95)
Residential construction	% of GDP	1996	4.6 (95)	5.9 (95)	4.6 (95)	5.4	..	3.2 (95)	3.5 (95)	4.4	7.3	3.3 (95)
Average annual volume growth over previous 5 years	%	1996	5.6	2.1	0.3	2.2	9.4	2	–4.1	–1.5	0.2	0.5 (95)
Gross saving ratio[4]	% of GDP	1996	18	21.9	22.2	17.8	..	17.6	19.6	18.7	20	16 (95)
General government												
Current expenditure on goods and services	% of GDP	1996	17	19.8	14.5	18.7	21.5	25.2	21.9	19.4	19.8	20.8 (95)
Current disbursements[5]	% of GDP	1995	35.6	48.6	52.2	45.8	..	59.6	55.9	50.9	46.6	52.1
Current receipts	% of GDP	1995	34.9	47.4	49.9	42.7	..	58.1	52.8	46.9	45.9	45
Net official development assistance	% of GNP	1995	0.36	0.33	0.38	0.38	..	0.96	0.32	0.55	0.31	0.13
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	12 596	12 152	13 793	12 959	..	12 027	10 282	12 506	12 244	9 473
Passenger cars, per 1 000 inhabitants	Number	1994	460	433	416	466	282	312	368	430	488	199
Telephones, per 1 000 inhabitants	Number	1994	496	466	449	576	209	604	551	547	483[8]	478
Television sets, per 1 000 inhabitants	Number	1993	489	479	453	618	476	538	504	412	559	202
Doctors, per 1 000 inhabitants	Number	1995	2.2 (91)	2.7	3.7 (94)	2.2	2.9	2.9 (94)	2.8	2.9	3.4	3.9 (94)
Infant mortality per 1 000 live births	Number	1995	5.7	5.4	7.6 (94)	6.3 (94)	7.7	5.5	4	5.8 (94)	5.3	8.1
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	1.7	5.2	2.7	2.4	..	3.2	3.8	2.6	4.2	11.8
Consumer prices	%	1996	2.4	2.9	2.2	1.4	11.9	1.9	1.5	2	3.1	11.6
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	60 288	57 870	170 223[7]	202 320	21 910	51 030	40 576	288 450	521 263	11 501
As % of GDP	%	1996	15.1	25.3	63.5	34.9	39	29.2	32.4	18.8	22.1	12.9 (95)
Average annual increase over previous 5 years	%	1996	7.5	7.1	7.6	9.7	..	6.2	12.1	6.3	5.4	5.8
Imports of goods, cif*	Mill. US$	1996	61 374	67 376	160 917[7]	170 931	27 721	44 987	30 911	271 348	455 741	27 402
As % of GDP	%	1996	15.4	29.5	60	29.5	49.3	25.7	24.7	17.7	19.4	30.4 (95)
Average annual increase over previous 5 years	%	1996	9.7	5.9	5.9	7.7	..	5.6	7.3	3.9	3.3	6.6
Total official reserves[6]	Mill. SDRs	1996	10 107	15 901	11 789[7]	14 202	8 590	9 834	4 810	18 635	57 844	12 171
As ratio of average monthly imports of goods	Ratio	1996	2	2.8	0.9	1	..	2.6	1.9	0.8	1.5	5.3

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Hungary	Iceland	Ireland	Italy	Japan	Korea	Luxembourg	Mexico	Netherlands	New Zealand
Population												
Total	Thousands	1996	10 193	270	3 621	57 473	125 864	45 545	418	96 582	15 494	3 640
Inhabitants per sq. km	Number	1996	77	3	52	191	333	458	161	48	380	14
Net average annual increase over previous 10 years	%	1996	–0.3	1.1	0.2	0	0.4	1	1.3	2	0.6	1.1
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	3 605	142	1 307	20 036	64 860	20 764	212 (95)	32 385 (95)	6 983	1 688
of which: Agriculture	% of TCE	1996	8.4	9.2	10.7	7	5.5	11.6	2.8 (95)	23.5 (95)	3.9	9.5
Industry	% of TCE	1996	33	23.9	27.2	32.1	33.3	32.5	30.7 (90)	21.7 (95)	22.4	24.6
Services	% of TCE	1996	58.6	66.2	62.3	60.9	61.2	55.9	66.1 (90)	54.8 (95)	73.8	65.9
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	43.7 (95)	7.3	70.7	1 214.2	4 595.2	484.8	17	329.4	396	65.9
Per capita	US$	1996	4 273 (95)	27 076	19 525	21 127	36 509	10 644	40 791	3 411	25 511	18 093
At current prices using current PPPs[3]	Bill. US$	1996	..	6.3	68.8	1 148	2 924.5	618.5	13.5	751.1	324.5	63.6
Per capita	US$	1996	..	23 242	18 988	19 974	23 235	13 580	32 416	7 776	20 905	17 473
Average annual volume growth over previous 5 years	%	1996	–2.4 (95)	1.5	7.1	1	1.5	7.1	4.8	1.7	2.3	3.7
Gross fixed capital formation (GFCF)	% of GDP	1996	19.3 (95)	17.5	17.2	17	29.7	36.8	20.8	18	19.7	20.9
of which: Machinery and equipment	% of GDP	1996	..	6.7	5.5 (95)	8.8	10.1 (95)	13	..	8.8	9.4	10
Residential construction	% of GDP	1996	. .	3.9	4.9 (95)	4.5	5.3 (95)	7.6	..	4.7	5	5.6
Average annual volume growth over previous 5 years	%	1996	–0.9 (95)	–1.4	6	–1.4	1.3	6.9	0.2	–0.7	2.2	9.6
Gross saving ratio[4]	% of GDP	1996	..	15.6	21.7	20.5	31.4	34.2	37.5	22.7	25.7	16
General government												
Current expenditure on goods and services	% of GDP	1996	24.9 (95)	20.8	14.1	16.4	9.7	10.6	13.6	9.7[10]	14	14.4
Current disbursements[5]	% of GDP	1995	..	35.1	39.2 (94)	49.5	28.5	15.1	..	..	51.8	..
Current receipts	% of GDP	1995	..	36	39.3 (94)	44.5	32	25.1	..	..	50	..
Net official development assistance	% of GNP	1995	..	..	0.29	0.15	0.28	0.03	0.36	..	0.81	0.23
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	..	14 244	10 020	12 224	13 912	7 354	17 811	5 045	12 477	10 895
Passenger cars, per 1 000 inhabitants	Number	1994	212	434	264	517	342	115	544	91	383	457
Telephones, per 1 000 inhabitants	Number	1994	170	557	350	429	480	397	564	93	509	470
Television sets, per 1 000 inhabitants	Number	1993	427	335	301	429	618	215	261	150	491	
Doctors, per 1 000 inhabitants	Number	1995	3.4	3.9 (94)	3.4	3.0 (94)	1.7	1.6 (92)	1.8 (94)	1.1	2.2 (93)	1.6
Infant mortality per 1 000 live births	Number	1995	11	6.1	6.3	6.6 (94)	4.3	9	5.3 (94)	17 (94)	5.5	7.2 (94)
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	..	..	3.7	3.5	1.8	..	..	–1.6	2.4	1.5
Consumer prices	%	1996	23.2	2.6	2.2	4.5	0.7	5.3	2.4	19.7	2.5	2
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	15 674	1 891	48 416	250 842	411 067	129 715	..	96 000	203 953	14 316
As % of GDP	%	1996	35.9	26	68.5	20.7	8.9	26.8	..	29.1	51.5	21.7
Average annual increase over previous 5 years	%	1996	8.9	4	14.8	8.2	5.5	12.5	..	17.6	8.9	8.2
Imports of goods, cif*	Mill. US$	1996	18 105	2 032	35 763	206 904	349 149	150 340	..	89 469	184 389	14 682
As % of GDP	%	1996	41.4	27.9	50.6	17	7.6	31	..	27.2	46.6	22.3
Average annual increase over previous 5 years	%	1996	9.6	3.4	11.5	2.5	8	13.9	..	12.4	7.8	11.8
Total official reserves[6]	Mill. SDRs	1996	6 812	316	5 706	31 954	150 663	23 670	..	13 514	18 615	4 140
As ratio of average monthly imports of goods	Ratio	1996	..	1.9	1.9	1.9	5.2	..	..	1.8	1.2	3.4

* At current prices and exchange rates.
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4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Poland	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population											
Total	Thousands	1996	4 370	38 618	9 935	39 270	8 901	7 085	62 695	58 782	265 557
Inhabitants per sq. km	Number	1996	13	123	108	78	20	172	80	240	28
Net average annual increase over previous 10 years	%	1996	0.5	0.3	–0.1	0.2	0.6	0.8	2	0.3	1
Employment											
Total civilian employment (TCE)[2]	Thousands	1996	2 110	14 977	4 475	12 394	3 963	3 803	20 895	26 088	126 708
of which: Agriculture	% of TCE	1996	5.2	22.1	12.2	8.7	2.9	4.5	44.9	2	2.8
Industry	% of TCE	1996	23.4 (95)	31.7	31.4	29.7	26.1	27.7	22	27.4	23.8
Services	% of TCE	1996	71.5 (95)	46.2	56.4	61.6	71	67.4	33.1	71	73.3
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1996	157.8	117.9 (95)	103.6	584.9	251.7	294.3	181.5	1 153.4	7 388.1
Per capita	US$	1996	36 020	3 057 (95)	10 425	14 894	28 283	41 411	2 894	19 621	27 821
At current prices using current PPPs[3]	Bill. US$	1996	106.7	..	130.1	587.2	171.4	180.6	383.3	1 095.5	7 388.1
Per capita	US$	1996	24 364	..	13 100	14 954	19 258	25 402	6 114	18 636	27 821
Average annual volume growth over previous 5 years	%	1996	4.1	2.2 (95)	1.5	1.3	1	0.1	4.4	2.2	2.8
Gross fixed capital formation (GFCF)	% of GDP	1996	20.5	17.1 (95)	24.1	20.1	14.8	20.2	25	15.5	17.6
of which: Machinery and equipment	% of GDP	1996	8.4	..	11.7 (93)	6.1 (95)	7.9	9.3	11.9	7.6	8.3 (95)
Residential construction	% of GDP	1996	2.6 (94)	..	5.2 (93)	4.3 (95)	1.9	11[9]	8.4 (95)	3	4.1 (95)
Average annual volume growth over previous 5 years	%	1996	2.8	5.4 (95)	2.2	–1	–2.6	–0.8	6.9	1.3	6.9
Gross saving ratio[4]	% of GDP	1996	29.9	..	21.6	20.7	16	27.1	22.3	14.6	16.6
General government											
Current expenditure on goods and services	% of GDP	1996	20.5	16.9 (95)	18.5	16.3	26.2	14.3	11.6	21.1	15.6
Current disbursements[5]	% of GDP	1995	45.8	..	42.5 (93)	41.2	63.8	47.7	..	42.3 (94)	34.3
Current receipts	% of GDP	1995	50.9	..	39.8 (93)	37.9	57.5	53.8	..	37.2 (94)	32.1
Net official development assistance	% of GNP	1995	0.87	..	0.27	0.24	0.77	0.34	0.07	0.28	0.1
Indicators of living standards											
Private consumption per capita using current PPPs[3]	US$	1996	11 593	..	8 522	9 339	10 096	15 632	4 130	11 865	18 908
Passenger cars, per 1 000 inhabitants	Number	1994	381	186	357	351	406 (93)	451	47	372	565
Telephones, per 1 000 inhabitants	Number	1994	554	131	350	371	683	597	201	489	602
Television sets, per 1 000 inhabitants	Number	1993	427	298	190	400	470	400	176	435	816
Doctors, per 1 000 inhabitants	Number	1995	2.8	2.3	3	4.1 (93)	3.1	3.1 (94)	1.2	1.6 (94)	2.6 (94)
Infant mortality per 1 000 live births	Number	1995	4	13.6	7.4	6 (94)	4	5	46.8 (94)	6.2 (94)	8 (94)
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1996	3.2	..	..	5.8	4.8	..	..	4.9	2.7
Consumer prices	%	1996	1.9	..	5.6	4.7	2.7	2.2	81.6	2.7	2.9
Foreign trade											
Exports of goods, fob*	Mill. US$	1996	49 576	24 417	24 614	102 067	84 836	79 581	23 301	259 941	625 075
As % of GDP	%	1996	31.4	20.7	23.8	17.5	33.7	27	12.8	22.5	8.5
Average annual increase over previous 5 years	%	1996	7.8	..	8.6	11.2	9	5.3	11.1	7	8.2
Imports of goods, cif*	Mill. US$	1996	35 575	37 185	35 192	121 838	66 825	78 052	43 094	287 033	795 289
As % of GDP	%	1996	22.5	31.5	34	20.8	26.5	26.5	23.7	24.9	10.8
Average annual increase over previous 5 years	%	1996	6.9	..	6.1	5.5	6	3.2	15.1	6.5	10.3
Total official reserves[6]	Mill. SDRs	1996	18 441	12 409	11 070	40 284	13 288	26 727	11 430	27 745	44 536
As ratio of average monthly imports of goods	Ratio	1996	6.2	..	3.8	4	2.4	4.1	3.2	1.2	0.7

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